CHAPTER 05

Continuity and Differentiability

In this Chapter...

- Continuity
- Differentiability and Derivatives of Various Functions
- Derivatives of Implicit Functions
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In class XI, we have studied about the concept of limit, left hand and right hand limits. We had also learnt how to differentiate polynomial function and trigonometric functions. In this chapter, we will introduce the very important concepts of continuity, differentiability and relations between them.

Continuity

Continuity of a Function at a Point

A function $f(x)$ is said to be continuous at a point $x = a$, if the value of function at $x = a$ is equal to the limit of the function at $x = a$.

i.e. $f(x)$ is continuous at $x = a$ if

$$(\text{LHL})_{x=a} = (\text{RHL})_{x=a} = f(a)$$

A function $f(x)$ is said to be **discontinuous** at $x = a$, if it is not continuous at $x = a$, i.e.

- $\lim\limits_{x \to a} f(a)$ does not exist.
- Left hand limit and right hand limit are not equal.
- $\lim\limits_{x \to a} f(x) \neq f(a)$

EXAMPLE 1. Examine the continuity of the following functions at given points.

(i) $f(x) = x^2$ at $x = -2$

(ii) $f(x) = 2x^2 - 1$ at $x = 3$

(iii) $f(x) = |x - 5|$ at $x = 5$ [NCERT]

Sol. (i) Given, $f(x) = x^2$

At point $x = -2$, $f(-2) = (-2)^2 = 4$

Now, $\lim\limits_{x \to -2} f(x) = \lim\limits_{x \to -2} x^2 = (-2)^2 = 4$

Here, $f(-2) = \lim\limits_{x \to -2} f(x)$

Hence, $f(x)$ is continuous at $x = -2$.

(ii) Given, $f(x) = 2x^2 - 1$

At point $x = 3$, $f(3) = 2(3)^2 - 1 = 2(9) - 1 = 17$

Now, $\lim\limits_{x \to 3} f(x) = \lim\limits_{x \to 3} (2x^2 - 1)$

$= 2(3)^2 - 1 = 17$

Here, $f(3) = \lim\limits_{x \to 3} f(x)$

Hence, $f(x)$ is continuous at $x = 3$.

(iii) Given, $f(x) = |x - 5| = \begin{cases} x - 5, & \text{if } x \geq 5 \\ 5 - x, & \text{if } x < 5 \end{cases}$

At point $x = 5$, $f(5) = |5 - 5| = 0$

Now, LHL $= \lim_{x \to 5^-} f(x) = \lim_{x \to 5^-} (5 - x)$

$= \lim_{h \to 0} [5 - (5 - h)] = \lim_{h \to 0} h = 0$

[put $x = 5 - h$, when $x \to 5^-$, then $h \to 0$]

and RHL $= \lim_{x \to 5^+} f(x) = \lim_{x \to 5^+} (x - 5)$

$= \lim_{h \to 0} [(5 + h) - 5] = \lim_{h \to 0} h = 0$

[put $x = 5 + h$; when $x \to 5^+$, then $h \to 0$]

Thus, LHL = RHL $= f(5) = 0$

Hence, $f(x)$ is continuous at $x = 5$

EXAMPLE 2. Discuss the continuity of the function $f(x)$ at $x = 0$, if $f(x) = \begin{cases} 2x - 1, & x < 0 \\ 2x + 1 & x \geq 0 \end{cases}$

Sol. We have, $f(x) = \begin{cases} 2x - 1, & x < 0 \\ 2x + 1 & x \geq 0 \end{cases}$

At $x = 0$, $\lim_{x \to 0^-} f(x) = \lim_{x \to 0} (2x - 1) = 2(0) - 1 = -1$

$\lim_{x \to 0^+} f(x) = \lim_{x \to 0} (2x + 1) = 2(0) + 1 = 1$

Here, $\lim_{x \to 0^-} \neq \lim_{x \to 0^+}$

Hence, $f(x)$ is not continuous at $x = -2$.

EXAMPLE 3. Discuss the continuity of the function

$$f(x) = \begin{cases} \dfrac{x^2 - x - 6}{x + 2}, & \text{if } x \neq -2 \\ -5, & \text{if } x = -2 \end{cases} \quad \text{at } x = -2.$$

Sol. We have, $f(x) = \begin{cases} \dfrac{x^2 - x - 6}{x + 2}, & \text{if } x \neq -2 \\ -5, & \text{if } x = -2 \end{cases}$

At $x = -2$, $\lim_{x \to -2} f(x) = \lim_{x \to -2} \dfrac{x^2 - x - 6}{x + 2}$ ...(i)

Now, factorising the numerator, we get

$\lim_{x \to -2} f(x) = \lim_{x \to -2} \dfrac{(x - 3)(x + 2)}{(x + 2)}$

$= \lim_{x \to -2} (x - 3)$

$= -2 - 3 = -5$

Also, at $x = -2$, $f(x) = -5$

i.e. $f(-2) = -5$

Thus, $\lim_{x \to -2} f(x) = f(-2)$

Hence, $f(x)$ is continuous at $x = -2$.

EXAMPLE 4. Discuss the continuity of the function

$$f(x) = \begin{cases} \dfrac{\sin 5x}{x}, & \text{if } x \neq 0 \\ 5, & \text{if } x = 0 \end{cases} \quad \text{at } x = 0.$$

Sol. We have, $f(x) = \begin{cases} \dfrac{\sin 5x}{x}, & \text{if } x \neq 0 \\ 5, & \text{if } x = 0 \end{cases}$

At $x = 0$, $\lim_{x \to 0} f(x) = \lim_{x \to 0} \dfrac{\sin 5x}{x}$

On multiplying numerator and denominator by 5 and then using result $\lim_{x \to 0} \dfrac{\sin x}{x} = 1$, we get

$= \lim_{x \to 0} \dfrac{5 \sin 5x}{5x} = 5 \times 1 = 5$

Also, at $x = 0$, $f(0) = 5$

Thus, $\lim_{x \to 0} f(x) = f(0)$

Hence, $f(x)$ is continuous at $x = 0$.

EXAMPLE 5. Discuss the continuity of the function

$$f(x) = \begin{cases} \dfrac{|x|}{x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases} \quad \text{at } x = 0.$$

[NCERT]

Sol. We have, $f(x) = \begin{cases} \dfrac{|x|}{x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$

At $x = 0$, LHL $= \lim_{x \to 0^-} f(x)$

$= \lim_{x \to 0^-} \dfrac{|x|}{x} = \lim_{h \to 0} \dfrac{|0 - h|}{0 - h}$

[put $x = 0 - h$; when $x \to 0^-$, then $h \to 0$]

$= \lim_{h \to 0} \dfrac{h}{-h}$ [$\because |0 - h| = |-h| = h$]

$= \lim_{h \to 0} (-1) = -1$

and RHL $= \lim_{x \to 0^+} f(x)$

$= \lim_{x \to 0^+} \dfrac{|x|}{x} = \lim_{h \to 0} \dfrac{|0 + h|}{0 + h}$

[put $x = 0 + h$; when $x \to 0^+$, then $h \to 0$]

$= \lim_{h \to 0} \dfrac{h}{h}$ [$\because |0 + h| = |h| = h$]

$= \lim_{h \to 0} (1) = 1$

Thus, LHL $\neq$ RHL

Hence, $f(x)$ is not continuous at $x = 0$.

EXAMPLE 6. Discuss the continuity of the function $f(x)$ at $x = 1/2$, when $f(x)$ is defined as follows.

$$f(x) = \begin{cases} 1/2 + x, & \text{if } 0 \le x < 1/2 \\ 1, & \text{if } x = 1/2 \\ 3/2 + x, & \text{if } 1/2 < x \le 1 \end{cases}$$ [Delhi 2011]

Sol. The given function is

$$f(x) = \begin{cases} \frac{1}{2} + x, & \text{if } 0 \le x < \frac{1}{2} \\ 1, & \text{if } x = \frac{1}{2} \\ \frac{3}{2} + x, & \text{if } \frac{1}{2} < x \le 1 \end{cases}$$

We have to check continuity of $f(x)$ at $x = \frac{1}{2}$.

Now, $\text{LHL} = \lim\limits_{x \to \frac{1}{2}^-} f(x) = \lim\limits_{x \to \frac{1}{2}^-} \left(\frac{1}{2} + x\right)$

$= \lim\limits_{h \to 0} \left(\frac{1}{2} + \frac{1}{2} - h\right)$

$\left[\text{put } x = \frac{1}{2} - h; \text{ when } x \to \frac{1^-}{2}, h \to 0\right]$

$= \lim\limits_{h \to 0} (1 - h) = 1$ [put $h = 0$]

$\text{RHL} = \lim\limits_{x \to \frac{1}{2}^+} f(x) = \lim\limits_{x \to \frac{1}{2}^+} \left(\frac{3}{2} + x\right)$

$= \lim\limits_{h \to 0} \left(\frac{3}{2} + \frac{1}{2} + h\right)$

$\left[\text{put } x = \frac{1}{2} + h; \text{ when } x \to \frac{1^+}{2}, h \to 0\right]$

$= \lim\limits_{h \to 0} (2 + h) = 2$ [put $h = 0$]

We know that a function $f(x)$ is said to be continuous at point $x = a$, if $\text{LHL} = \text{RHL} = f(a)$.

Here, $\text{LHL} \neq \text{RHL}$

Hence, $f(x)$ is discontinuous at $x = \frac{1}{2}$.

EXAMPLE 7. Show that the function $f(x)$ given by

$$f(x) = \begin{cases} x \sin \frac{1}{x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$$

is continuous at $x = 0$. [NCERT Exemplar]

Sol. Given, $f(x) = \begin{cases} x \sin \frac{1}{x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$ and point is $x = 0$.

At $x = 0$, $f(0) = 0$

$\text{LHL} = \lim\limits_{x \to 0^-} f(x) = \lim\limits_{h \to 0} f(0 - h)$

$= \lim\limits_{h \to 0} f(-h)$

$= \lim\limits_{h \to 0} (-h) \sin\left(\frac{1}{-h}\right)$

$= \lim\limits_{h \to 0} h \sin \frac{1}{h}$ [$\because \sin(-\theta) = -\sin\theta$]

$= 0 \times$ (An oscillating value lies between -1 and 1)

$= 0$

$\text{RHL} = \lim\limits_{x \to 0^+} f(x) = \lim\limits_{h \to 0} f(0 + h)$

$= \lim\limits_{h \to 0} f(h) = \lim\limits_{h \to 0} h \sin\left(\frac{1}{h}\right)$

$= 0 \times$ (An oscillating value lies between -1 and 1)

$= 0$

Here, $\text{LHL} = f(0) = \text{RHL} = 0$

Hence, $f(x)$ is continuous at $x = 0$.

EXAMPLE 8. Show that the function

$$f(x) = \begin{cases} \frac{e^{1/x} - 1}{e^{1/x} + 1}, & \text{when } x \neq 0 \\ 0, & \text{when } x = 0 \end{cases}$$

is discontinuous at $x = 0$. [NCERT Exemplar]

Sol. We have, $\begin{cases} \frac{e^{1/x} - 1}{e^{1/x} + 1}, & \text{when } x \neq 0 \\ 0, & \text{when } x = 0 \end{cases}$

At $x = 0$, $f(0) = 0$

$\text{LHL} = \lim\limits_{x \to 0^-} f(x) = \lim\limits_{x \to 0^-} \frac{(e^{1/x} - 1)}{(e^{1/x} + 1)}$

$= \lim\limits_{h \to 0} \frac{e^{\frac{1}{0-h}} - 1}{e^{\frac{1}{0-h}} + 1} = \lim\limits_{h \to 0} \frac{e^{-1/h} - 1}{e^{-1/h} + 1}$

[put $x = 0 - h$; when $x \to 0^-$, then $h \to 0$]

$= \lim\limits_{h \to 0} \frac{\frac{1}{e^{1/h}} - 1}{\frac{1}{e^{1/h}} + 1} = \frac{0 - 1}{0 + 1} = -1$ $\left[\because \lim\limits_{h \to 0} \frac{1}{e^{1/h}} = 0\right]$

and $\text{RHL} = \lim\limits_{x \to 0^+} f(x) = \lim\limits_{x \to 0^+} \frac{e^{1/x} - 1}{e^{1/x} + 1}$

$= \lim\limits_{h \to 0} \frac{e^{\frac{1}{(0+h)}} - 1}{e^{\frac{1}{(0+h)}} + 1}$

[put $x = 0 + h$; when $x \to 0^+$, then $h \to 0$]

$= \lim\limits_{h \to 0} \frac{e^{1/h} - 1}{e^{1/h} + 1}$

On dividing numerator and denominator by $e^{1/h}$, we get

$$\text{RHL} = \lim_{h\to 0} \frac{\left(1-\frac{1}{e^{1/h}}\right)}{\left(1+\frac{1}{e^{1/h}}\right)} = \frac{1-0}{1+0} = 1 \qquad \left[\because \lim_{h\to 0} \frac{1}{e^{1/h}} = 0\right]$$

Here, LHL ≠ RHL

Hence, $f(x)$ is not continuous at $x = 0$.

Continuity of a Function in its Domain

A function $f(x)$ is said to be continuous in its domain, if it is continuous at every point in its domain.

(i) A function $f(x)$ is said to be continuous in open interval (a, b), if it is continuous for every value of x in the interval (a, b).

(ii) A function $f(x)$ is said to be continuous in closed interval $[a, b]$, if

(a) it is continuous for every value of x in the open interval (a, b).

(b) $f(x)$ is continuous at $x = a$ from right,
i.e. $\lim_{x\to a^+} f(x) = f(a)$.

(c) $f(x)$ is continuous at $x = b$ from left,
i.e. $\lim_{x\to b^-} f(x) = f(b)$.

EXAMPLE 9. Discuss the continuity of the function f given by $f(x) = x^3 + x^2 - 1$ in the set of real number.

Sol. Given, $f(x) = x^3 + x^2 - 1$.

Here, $f(x)$ is always defined in R.

Let c be any real number i.e. $c \in R$.

Then, at $x = c$, $f(c) = c^3 + c^2 - 1$

and $$\lim_{x\to c} f(x) = \lim_{x\to c}(x^3 + x^2 - 1)$$
$$= c^3 + c^2 - 1$$

Thus, $$f(c) = \lim_{x\to c} f(x)$$

So, $f(x)$ is continuous at every real number.

Hence, f is continuous for every real values of x.

EXAMPLE 10. Discuss the continuity of the function f, where f is defined by

$$f(x) = \begin{cases} -2, & \text{if } x \le -1 \\ 2x, & \text{if } -1 < x \le 1. \\ 2, & \text{if } x > 1 \end{cases}$$

[NCERT]

Sol. We have, $f(x) = \begin{cases} -2, & \text{if } x \le -1 \\ 2x, & \text{if } -1 < x \le 1. \\ 2, & \text{if } x > 1 \end{cases}$

At $x = -1$, $f(-1) = -2$

$$\text{LHL} = \lim_{x\to -1^-} f(x) = \lim_{x\to -1^-} (-2) = -2$$

$$\text{RHL} = \lim_{x\to -1^+} f(x) = \lim_{x\to -1^+} 2x = \lim_{h\to 0} 2(-1+h)$$

[put $x = -1 + h$; when $x \to -1^+$, then $h \to 0$]

$$= \lim_{h\to 0} (-2 + 2h) = -2$$

Thus, $f(-1) = \text{LHL} = \text{RHL}$

$\therefore$ $f(x)$ is continuous at $x = -1$.

At $x = 1$, $f(1) = 2$

$$\text{LHL} = \lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} 2x = \lim_{h\to 0} 2(1-h)$$

[put $x = 1 - h$; when $x \to 1^-$, then $h \to 0$]

$$= \lim_{h\to 0} (2 - 2h) = 2$$

$$\text{RHL} = \lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} 2 = 2$$

Thus, $f(1) = \text{LHL} = \text{RHL}$

So, $f(x)$ is continuous at $x = 1$.

Clearly, $f(x)$ is continuous at all real numbers other than 1 and -1.

Hence, $f(x)$ is continuous for every value of x.

Algebra of Continuous Functions

We know that continuity of a function at a point is entirely dictated by the limit of the function at that point, so algebra of continuous function is similar to algebra of limits, as given below

Theorem 1 Let f and g be two real functions continuous at a real number c, then

(i) $(f + g)$ is continuous at $x = c$.

(ii) $(f - g)$ is continuous at $x = c$.

(iii) fg is continuous at $x = c$.

(iv) $\frac{f}{g}$ is continuous at $x = c$, provided that $g(c) \neq 0$.

Theorem 2 Suppose f and g are real valued functions such that (fog) is defined at c. If g is continuous at c and if f is continuous at $g(c)$, then (fog) is continuous at c.

Other important results

There are some functions which are always continuous in their respective domain. e.g.

(i) Every constant function is continuous.

(ii) Every identity function is continuous.

(iii) Every rational function is continuous.

(iv) Every polynomial function is continuous.

(v) Modulus function $f(x) = |x|$ is continuous.

(vi) All trigonometric functions are continuous.

EXAMPLE 11. Discuss the continuity of sine function. [NCERT]

Sol. We know that $\lim_{x \to 0} \sin x = 0$ and $f(x) = \sin x$ is defined for every real number. To show $f(x)$ is continuous for every real number. Let c be a real number.

Then, $\lim_{x \to c} f(x) = \lim_{x \to c} \sin x = \lim_{h \to 0} \sin(c + h)$

[put $x = c + h$; when $x \to c$, then $h \to 0$]

$= \lim_{h \to 0} [\sin c \cos h + \cos c \sin h]$

$= \lim_{h \to 0} (\sin c \cos h) + \lim_{h \to 0} (\cos c \sin h)$

$= \sin c \lim_{h \to 0} \cos h + \cos c \lim_{h \to 0} \sin h$

$= \sin c \times 1 + \cos c \times 0$

[$\because \lim_{x \to 0} \cos x = 1$ and $\lim_{x \to 0} \sin x = 0$]

$= \sin c = f(c)$

Hence, f is a continuous function.

EXAMPLE 12. Examine that $e^{|x|}$ is a continuous function.

Sol. Let $g(x) = |x|$ and $h(x) = e^x$

Then, $hog(x) = h[g(x)] = h(|x|) = e^{|x|} = f(x)$

Since, $g(x)$ and $h(x)$ are both continuous function for all $x \in R$, so composition of $g(x)$ and $h(x)$ is also a continuous function for all $x \in R$.

Hence, $f(x) = e^{|x|}$ is a continuous function.

EXAMPLE 13. Discuss the continuity of

$$f(x) = |\sin x + \cos x| \text{ at } x = \pi.$$

Sol. Let $g(x) = \sin x + \cos x$ and $k(y) = |y|$

We shall first show that g is continuous at $x = \pi$ and k is continuous at $y = g(\pi)$.

Now, $\lim_{x \to \pi} g(x) = \lim_{x \to \pi} (\sin x + \cos x)$

$= \sin \pi + \cos \pi = -1$

Also, $g(\pi) = \sin \pi + \cos \pi = -1$

$\Rightarrow \lim_{x \to \pi} g(x) = g(\pi)$

So, g is continuous at $x = \pi$.

Now, $g(\pi) = -1$

$\therefore k[g(\pi)] = k(-1) = |-1| = 1$

Now, $\lim_{y \to (-1)^+} k(y) = \lim_{y \to (-1)^+} |y| = \lim_{h \to 0} |-1 + h| = 1$

[put $y = -1 + h$; when $y \to -1^+$, then $h \to 0$]

and $\lim_{y \to (-1)^-} k(y) = \lim_{y \to (-1)^-} |y| = \lim_{h \to 0} |-1 - h| = 1$

[put $y = -1 - h$; when $y \to -1^-$, then $h \to 0$]

$\therefore k[g(\pi)] = \lim_{y \to g(\pi)} k(y)$

This shows that k is continuous at $g(\pi)$.

Consequently, (kog) is continuous at $x = \pi$.

But $kog(x) = k[g(x)] = k(\sin x + \cos x)$

$= |\sin x + \cos x| = f(x)$

Hence, $f(x)$ is continuous at $x = \pi$.

EXAMPLE 14. Show that the function defined by $f(x) = \sin x^2$ is a continuous function. [NCERT]

Sol. Given, $f(x) = \sin x^2$

It is defined for every real number.

Suppose $g(x) = \sin x$ and $h(x) = x^2$, then $f(x) = goh(x)$

Also, both $g(x)$ and $h(x)$ are continuous functions.

So, their composition is also continuous function.

Hence, $f(x) = \sin x^2$ is a continuous function.

Problems Based on Finding Unknown Constant in Continuous Function

EXAMPLE 15. Find the value of a, so that

$$f(x) = \begin{cases} ax + 5, & \text{if } x \le 2 \\ x - 1, & \text{if } x > 2 \end{cases} \text{ is continuous at } x = 2.$$

Sol. Given, $f(x) = \begin{cases} ax + 5, & \text{if } x \le 2 \\ x - 1, & \text{if } x > 2 \end{cases}$

and at $x = 2$, $f(x)$ is continuous.

$\therefore \quad x = c = 2$

Here, a is an unknown constant.

At $x = 2$, $f(2) = a(2) + 5 = 2a + 5$

$\text{LHL} = \lim_{x \to 2^-} f(x)$

$= \lim_{x \to 2^-} (ax + 5)$

$= \lim_{h \to 0} [a(2 - h) + 5]$

$= \lim_{h \to 0} (2a - ah + 5) = 2a + 5$

$\text{RHL} = \lim_{x \to 2^+} f(x) = \lim_{x \to 2^+} (x - 1)$

$= \lim_{h \to 0} [(2 + h) - 1]$

$= \lim_{h \to 0} (1 + h) = 1$

Since, $f(x)$ is continuous at $x = 2$

At $x = 2$, LHL = RHL = $f(2)$

$\Rightarrow \quad 2a + 5 = 1 = 2a + 5$

$\therefore \quad 2a + 5 = 1$

$\Rightarrow \quad 2a = -4$

$\Rightarrow \quad a = -2$

Hence, the required value of a is -2 for which $f(x)$ is continuous at $x = 2$.

EXAMPLE 16. Find the relationship between a and b so that the function f defined by $f(x) = \begin{cases} ax + 1, & \text{if } x \le 3 \\ bx + 3, & \text{if } x > 3 \end{cases}$ is continuous at $x = 3$. **[NCERT; All India 2011]**

Sol. Given, $f(x) = \begin{cases} ax + 1, & \text{if } x \le 3 \\ bx + 3, & \text{if } x > 3 \end{cases}$

and $f(x)$ is continuous at $x = 3$.

Here, a and b are two unknowns.

Now, at $x = 3$, $f(3) = a(3) + 1 = 3a + 1$

$$\text{LHL} = \lim_{x \to 3^-} f(x) = \lim_{x \to 3^-} (ax + 1)$$

$$= \lim_{h \to 0} \{a(3 - h) + 1\}$$

[put $x = 3 - h$; when $x \to 3^-$ then $h \to 0$]

$$= \lim_{h \to 0} (3a - ah + 1) = 3a + 1$$

$$\text{and RHL} = \lim_{x \to 3^+} f(x) = \lim_{x \to 3^+} \{bx + 3\}$$

$$= \lim_{h \to 0} \{b(3 + h) + 3\}$$

[put $x = 3 + h$; when $x \to 3^+$, then $h \to 0$]

$$= \lim_{h \to 0} \{3b + bh + 3\} = 3b + 3$$

Since, $f(x)$ is continuous at $x = 3$.

At $x = 3$, $f(3) = \text{LHL} = \text{RHL}$.

i.e. $3a + 1 = 3a + 1 = 3b + 3$

$\Rightarrow$ $3a + 1 = 3b + 3 \Rightarrow 3a = 3b + 3 - 1$

$\Rightarrow$ $a = \dfrac{(3b + 2)}{3} \Rightarrow a = b + \dfrac{2}{3}$

which is the required relation between a and b.

EXAMPLE 17. Find the values of a and b such that the function defined by $f(x) = \begin{cases} 5, & \text{if } x \le 2 \\ ax + b, & \text{if } 2 < x < 10 \\ 21, & \text{if } x \ge 10 \end{cases}$ is continuous function. **[CBSE 2023, NCERT]**

Sol. Here, $f(x) = \begin{cases} 5, & \text{if } x \le 2 \\ ax + b, & \text{if } 2 < x < 10 \\ 21, & \text{if } x \ge 10 \end{cases}$

At $x = 2$, $\text{LHL} = \lim_{x \to 2^-} f(x) = \lim_{x \to 2^-} (5) = 5$

and $\text{RHL} = \lim_{x \to 2^+} f(x) = \lim_{x \to 2^+} (ax + b)$

$\therefore$ $\text{RHL} = \lim_{h \to 0} [a(2 + h) + b]$

[put $x = 2 + h$, when $x \to 2^+$, then $h \to 0$]

$$= \lim_{h \to 0} (2a + ah + b) = 2a + b$$

Also, $f(2) = 5$

Since, $f(x)$ is continuous at $x = 2$.

$\therefore$ $\text{LHL} = \text{RHL} = f(2) \Rightarrow 2a + b = 5$ ...(i)

Now, at $x = 10$,

$$\text{LHL} = \lim_{x \to 10^-} f(x) = \lim_{x \to 10^-} (ax + b)$$

$\therefore$ $\text{LHL} = \lim_{h \to 0} [a(10 - h) + b] = \lim_{h \to 0} (10a - ah + b)$

[put $x = 10 - h$; when $x \to 10^-$, then $h \to 0$]

$= 10a + b$

and $\text{RHL} = \lim_{x \to 10^+} f(x) = \lim_{x \to 10^+} (21) = 21$

Also, $f(10) = 21$. Since, $f(x)$ is continuous at $x = 10$.

$\therefore$ $\text{LHL} = \text{RHL} = f(10)$

$\Rightarrow$ $10a + b = 21$ ...(ii)

On subtracting Eq. (i) from Eq. (ii), we get

$$8a = 16 \Rightarrow a = 2$$

On putting $a = 2$ in Eq. (i), we get $2 \times 2 + b = 5 \Rightarrow b = 1$

Hence, the values of a and b are 2 and 1, respectively.

EXAMPLE 18. Find the values of p and q for which

$$f(x) = \begin{cases} \dfrac{1 - \sin^3 x}{3\cos^2 x}, & \text{if } x < \dfrac{\pi}{2} \\ p, & \text{if } x = \dfrac{\pi}{2} \\ \dfrac{q(1 - \sin x)}{(\pi - 2x)^2}, & \text{if } x > \dfrac{\pi}{2} \end{cases}$$ is continuous at $x = \dfrac{\pi}{2}$.

[Delhi 2016]

Sol. Given, $f(x) = \begin{cases} \dfrac{1 - \sin^3 x}{3\cos^2 x}, & \text{if } x < \dfrac{\pi}{2} \\ p, & \text{if } x = \dfrac{\pi}{2} \\ \dfrac{q(1 - \sin x)}{(\pi - 2x)^2}, & \text{if } x > \dfrac{\pi}{2} \end{cases}$

is continuous at $x = \dfrac{\pi}{2}$.

$\therefore$ $(\text{LHL})_{x = \pi/2} = (\text{RHL})_{x = \pi/2} = f\left(\dfrac{\pi}{2}\right)$ ...(i)

Now, $\text{LHL} = \lim_{x \to \frac{\pi}{2}^-} f(x) = \lim_{h \to 0} f\left(\dfrac{\pi}{2} - h\right)$

$\left[\text{put } x = \dfrac{\pi}{2} - h, \text{ when } x \to \dfrac{\pi^-}{2} \text{ then } h \to 0\right]$

$$= \lim_{h \to 0} \frac{1 - \sin^3\left(\frac{\pi}{2} - h\right)}{3\cos^2\left(\frac{\pi}{2} - h\right)} = \lim_{h \to 0} \frac{1 - \cos^3 h}{3\sin^2 h}$$

$\left[\because \cos\left(\dfrac{\pi}{2} - \theta\right) = \sin\theta, \sin\left(\dfrac{\pi}{2} - \theta\right) = \cos\theta\right]$

$$= \lim_{h \to 0} \frac{(1 - \cos h)(1^2 + \cos^2 h + 1 \times \cos h)}{3(1 - \cos^2 h)}$$

$$= \lim_{h\to 0} \frac{(1-\cos h)(1+\cos^2 h+\cos h)}{3(1-\cos h)(1+\cos h)}$$

$$= \lim_{h\to 0} \frac{(1+\cos^2 h+\cos h)}{3(1+\cos h)}$$

$$= \frac{1+\cos^2 0+\cos 0}{3(1+\cos 0)} = \frac{1+1+1}{3(1+1)} = \frac{3}{3\times 2} = \frac{1}{2}$$

and RHL $= \lim_{x\to \frac{\pi^+}{2}} f(x) = \lim_{h\to 0} f\left(\frac{\pi}{2}+h\right)$

$$\left[\text{put } x = \frac{\pi}{2}+h, \text{ when } x \to \frac{\pi^+}{2} \text{ then } h \to 0\right]$$

$$= \lim_{h\to 0} \frac{q\left[1-\sin\left(\frac{\pi}{2}+h\right)\right]}{\left[\pi - 2\left(\frac{\pi}{2}+h\right)\right]^2}$$

$$= \lim_{h\to 0} \frac{q(1-\cos h)}{(\pi-\pi-2h)^2} = \lim_{h\to 0} \frac{q(1-\cos h)}{4h^2}$$

$$= \lim_{h\to 0} \frac{q\left(2\sin^2\frac{h}{2}\right)}{4h^2} \quad \left[\because \cos x = 1-2\sin^2\frac{x}{2}\right]$$

$$= \lim_{h\to 0} \frac{q\sin^2\frac{h}{2}}{2\left(\frac{h}{2}\right)^2\times 4} = \frac{q}{8}\lim_{h\to 0}\left[\frac{\sin\left(\frac{h}{2}\right)}{h/2}\right]^2$$

$$= \frac{q}{8}\times 1 = \frac{q}{8} \quad \left[\because \lim_{h\to 0}\frac{\sin h}{h} = 1\right]$$

From Eq. (i), $\frac{1}{2} = \frac{q}{8} = p$

$\Rightarrow \quad \frac{1}{2} = \frac{q}{8}$ and $\frac{1}{2} = p$

$\therefore \quad q = 4$ and $p = \frac{1}{2}$

Problems Based on Finding Points of Discontinuity of a Function

EXAMPLE 19. Find the point of the function f defined by $f(x) = \begin{cases} x+2, & \text{if } x \le 1 \\ x-2, & \text{if } x > 1 \end{cases}$.

Sol. Given, $f(x) = \begin{cases} x+2, & \text{if } x \le 1 \\ x-2, & \text{if } x > 1 \end{cases}$

When $x < 1$, we have $f(x) = x+2$, which is a polynomial function, so it is continuous at each point for $x < 1$.

Also, when $x > 1$, we have $f(x) = x-2$, which is a polynomial function, so it is continuous at each point for $x > 1$.

Now, we have to check the continuity at $x = 1$.

LHL $= \lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} (x+2) = \lim_{h\to 0} (1-h+2) = 3$

[put $x = 1-h$, when $x \to 1^-$, then $h \to 0$]

and RHL $= \lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} (x-2)$

$= \lim_{h\to 0} (1+h-2) = -1$

[put $x = 1+h$; when $x \to 1^+$, then $h \to 0$]

$\because$ LHL $\ne$ RHL at $x = 1$.

So, $f(x)$ is not continuous at $x = 1$.

Hence, $x = 1$ is the only point of discontinuity of f.

EXAMPLE 20. Find all points of discontinuity of f, where f is defined by $f(x) = \begin{cases} |x|+3, & \text{if } x \le -3 \\ -2x, & \text{if } -3 < x < 3 \\ 6x+2, & \text{if } x \ge 3 \end{cases}$. [NCERT]

Sol. Given, $f(x) = \begin{cases} |x|+3, & \text{if } x \le -3 \\ -2x, & \text{if } -3 < x < 3 \\ 6x+2, & \text{if } x \ge 3 \end{cases}$

Here, we have to check the continuity only at points $x = -3$ and $x = 3$.

[$\because$ functions defined in each interval is a polynomial, so it is continuous in each interval]

At $x = -3$,

LHL $= \lim_{x\to -3^-} f(x) = \lim_{x\to -3^-} \{|x|+3\} = \lim_{h\to 0} \{|-3-h|+3\}$

[put $x = -3-h$, when $x \to -3^-$, then $h \to 0$]

$= \lim_{h\to 0} (3+h+3) = \lim_{h\to 0} (6+h) = 6$

RHL $= \lim_{x\to -3^+} f(x) = \lim_{x\to -3^+} (-2x) = \lim_{h\to 0} -2(-3+h)$

[put $x = -3+h$; when $x \to -3^+$, then $h \to 0$]

$= \lim_{h\to 0} (6-2h) = 6$ and $f(-3) = |-3|+3 = 6$

Thus, $f(-3) =$ LHL $=$ RHL.

So, $f(x)$ is continuous at $x = -3$.

At $x = 3$,

LHL $= \lim_{x\to 3^-} f(x) = \lim_{x\to 3^-} (-2x) = \lim_{h\to 0} -2(3-h)$

[put $x = 3-h$; when $x \to 3^-$, then $h \to 0$]

$= \lim_{h\to 0} (-6+2h) = -6$

RHL $= \lim_{x\to 3^+} f(x) = \lim_{x\to 3^+} (6x+2) = \lim_{h\to 0} \{6(3+h)+2\}$

[put $x = 3+h$; when $x \to 3^+$, then $h \to 0$]

$= \lim_{h\to 0} (18+6h+2) = 20$

Thus, LHL $\ne$ RHL

So, $f(x)$ is not continuous at $x = 3$.

Hence, the point of discontinuity is $x = 3$.

Topic Practice 1

1 Examine the continuity of the following functions at the given point.

(i) $f(x) = 5x - 3$ at $x = -3$ [NCERT]

(ii) $f(x) = x^2 + 5$ at $x = -1$

(iii) $f(x) = x^2 + 3x + 4$ at $x = 1$

(iv) $f(x) = x^3 + 2x^2 - 1$ at $x = 1$ [NCERT Exemplar]

(v) $f(x) = |x - 2|$ at $x = 2$

2 Is the function f defined by $f(x) = \begin{cases} x, & \text{if } x \le 1 \\ 5, & \text{if } x > 1 \end{cases}$ continuous at $x = 0$, at $x = 1$ and at $x = 2$? [NCERT]

Directions (Q.Nos. 3-14) *Discuss the continuity of the following function at the indicated point(s).*

3 $f(x) = \begin{cases} 4x + 3, & \text{if } x \ne 4 \\ 3x + 7, & \text{if } x = 4 \end{cases}$

4 $f(x) = \begin{cases} \dfrac{x^2 - 49}{x - 7}, & \text{if } x \ne 7 \\ 14, & \text{if } x = 7 \end{cases}$

5 $f(x) = \begin{cases} 5 - x, & \text{if } x \le 0 \\ x^2, & \text{if } x > 0 \end{cases}$

6 $f(x) = \begin{cases} x^3 - 2x^2 + x + 2, & \text{if } x \ne 1 \\ 2, & \text{if } x = 1 \end{cases}$

7 $f(x) = \begin{cases} \dfrac{\sin 3x}{x}, & \text{if } x \ne 0 \\ 3, & \text{if } x = 0 \end{cases}$

8 $f(x) = \begin{cases} \dfrac{x}{|x|}, & \text{if } x \ne 0 \\ 1, & \text{if } x = 0 \end{cases}$

9 $f(x) = \begin{cases} \dfrac{1 - \cos x}{x^2}, & \text{if } x \ne 0 \\ 1, & \text{if } x = 0 \end{cases}$

10 $f(x) = \begin{cases} x^2 \sin\left(\dfrac{1}{x}\right), & \text{if } x \ne 0 \\ 0, & \text{if } x = 0 \end{cases}$

11 $f(x) = \begin{cases} 2x, & \text{if } x < 2 \\ 2, & \text{if } x = 2 \\ x^2, & \text{if } x > 2 \end{cases}$

12 $f(x) = \begin{cases} \dfrac{|x - a|}{x - a}, & \text{if } x \ne a \\ 1, & \text{if } x = a \end{cases}$

13 $f(x) = \begin{cases} 2x, & \text{if } x < 0 \\ 0, & \text{if } 0 \le x \le 1 \\ 4x, & \text{if } x > 1 \end{cases}$

14 $f(x) = \begin{cases} 1, & \text{if } x \le -1 \\ -2x - 1, & \text{if } -1 < x < 0 \\ -1, & \text{if } x \ge 0 \end{cases}$

Directions (Q.Nos. 15-19) *Show that*

15 The function $f(x) = \begin{cases} \dfrac{x^2 - x - 6}{x - 3}, & \text{if } x \ne 3 \\ 5, & \text{if } x = 3 \end{cases}$ is continuous at $x = 3$.

16 The function $f(x) = \begin{cases} \dfrac{\sin x}{x} + \cos x, & \text{if } x \ne 0 \\ 2, & \text{if } x = 0 \end{cases}$ is continuous at $x = 0$.

17 The function $f(x) = \begin{cases} \dfrac{e^{1/x}}{1 + e^{1/x}}, & \text{if } x \ne 0 \\ 0, & \text{if } x = 0 \end{cases}$. [NCERT Exemplar]

18 The function $f(x) = 2x - |x|$ is continuous at $x = 0$. [CBSE 2002]

19 $f(x) = \begin{cases} |x| \cos\dfrac{1}{x}, & \text{if } x \ne 0 \\ 0, & \text{if } x = 0 \end{cases}$ is continuous at $x = 0$. [NCERT Exemplar]

20 Examine the following functions for continuity.

(i) $f(x) = \dfrac{1}{x - 5}, x \ne 5$ [NCERT]

(ii) $f(x) = \dfrac{1}{x + 3}, x \in R$

(iii) $f(x) = \dfrac{x^2 - 25}{x + 5}, x \ne -5$ [NCERT]

(iv) $f(x) = x - 5$

(v) $f(x) = x^2 - 10$

21 Discuss the continuity of the function $f(x) = \sin x - \cos x$. [NCERT]

22 Discuss the continuity of the function $f(x) = \sin x \cdot \cos x$. [NCERT Exemplar]

23 Show that the function defined by $f(x) = \cos x^2$ is a continuous function. [NCERT]

24 Examine that $\sin|x|$ is a continuous function.

25 Show that the function $f(x) = \cos|x|$ is continuous.

26 Show that tangent function is continuous in their domain.

27 Given, $f(x) = \dfrac{1}{x - 1}$. Find the point(s) of discontinuity of composite function $y = f\{f(x)\}$.

Directions (Q.Nos. 28-43) *Determine the value of constant involved in the definition such that the following functions are continuous at the indicated point.*

28 $f(x)=\begin{cases}\dfrac{(x+3)^2-36}{x-3}, & x\neq 3\\ k, & x=3\end{cases}$, at $x=3$. [All India 2017]

29 $f(x)=\begin{cases}\dfrac{x^2+3x-10}{x-2}, & x\neq 2\\ k, & x=2\end{cases}$ at $x=2$. [Delhi 2017C]

30 $f(x)=\begin{cases}\dfrac{\sin x}{x}+\cos x, & \text{if } x\neq 0\\ k, & \text{if } x=0\end{cases}$ at $x=0$. [NCERT Exemplar]

31 $f(x)=\begin{cases}\dfrac{\sin^2 ax}{x^2}, & x\neq 0\\ 1, & x=0\end{cases}$ at $x=0$. [CBSE 2023]

32 $f(x)=\begin{cases}\dfrac{1-\cos 4x}{x^2}, & x\neq 0\\ p, & x=0\end{cases}$ at $x=0$. [Delhi 2017C]

33 $f(x)=\begin{cases}\dfrac{\sin x-\cos x}{4x-\pi}, & x\neq \dfrac{\pi}{4}\\ k, & x=\dfrac{\pi}{4}\end{cases}$ at $x=\dfrac{\pi}{4}$. [Delhi 2017C]

34 $f(x)=\begin{cases}kx+1, & \text{if } x\leq \pi\\ \cos x, & \text{if } x>\pi\end{cases}$ at $x=\pi$. [Foreign 2011]

35 $f(x)=\begin{cases}\dfrac{1-\cos x}{2x^2}, & x\neq 0\\ k, & x=0\end{cases}$ at $x=0$. [CBSE 2023]

36 $f(x)=\begin{cases}\dfrac{\cos^2 x-\sin^2 x-1}{\sqrt{x^2+1}-1}, & \text{if } x\neq 0\\ k, & \text{if } x=0\end{cases}$ is $x=0$.

37 $f(x)=\begin{cases}\dfrac{\sqrt{1+kx}-\sqrt{1-kx}}{x}, & \text{if } -1\leq x<0\\ \dfrac{2x+1}{x-1}, & \text{if } 0\leq x<1\end{cases}$ at $x=0$. [All India 2013]

38 $f(x)=\begin{cases}\dfrac{2^{x+2}-16}{4^x-16}, & \text{if } x\neq 2\\ k, & \text{if } x=2\end{cases}$ at $x=2$.

39 $f(x)=\begin{cases}m(x^2-2x), & \text{if } x<0\\ \cos x, & \text{if } x\geq 0\end{cases}$ at $x=0$.

40 $f(x)=\begin{cases}\dfrac{1-\cos 4x}{x^2}, & \text{if } x<0\\ a, & \text{if } x=0\\ \dfrac{\sqrt{x}}{\sqrt{16+\sqrt{x}}-4}, & \text{if } x>0\end{cases}$ at $x=0$? [NCERT Exemplar; All India 2011C]

41 $f(x)=\begin{cases}2x-1, & \text{if } x<2\\ k, & \text{if } x=2\\ x+1, & \text{if } x>2\end{cases}$.

42 $f(x)=\begin{cases}\lambda(x^2-2x), & \text{if } x\leq 0\\ 4x+1, & \text{if } x>0\end{cases}$ at $x=0$ and $x=1$? [Foreign 2011]

43 $f(x)=\begin{cases}\dfrac{x}{|x|+2x^2}, & \text{if } x\neq 0\\ k, & \text{if } x=0\end{cases}$ at $x=0$.

Directions (Q.Nos. 44-47) *Determine the values of a and b for which the functions $f(x)$ is continuous.*

44 $f(x)=\begin{cases}2, & \text{if } x\leq 3\\ ax+b, & \text{if } 3<x<5\\ 9, & \text{if } x\geq 5\end{cases}$.

45 $f(x)=\begin{cases}x+2, & \text{if } x\leq 2\\ ax+b, & \text{if } 2<x<5\\ 3x-2, & \text{if } x\geq 5\end{cases}$. [CBSE 2010]

46 $f(x)=\begin{cases}\dfrac{x-4}{|x-4|}+a, & \text{if } x<4\\ a+b, & \text{if } x=4\\ \dfrac{x-4}{|x-4|}+b, & \text{if } x>4\end{cases}$ at $x=4$. [NCERT Exemplar]

47 $f(x)=\begin{cases}\dfrac{\sin(a+1)x+2\sin x}{x}, & x<0\\ 2, & x=0\\ \dfrac{\sqrt{1+bx}-1}{x}, & x>0\end{cases}$. [All India 2016]

Directions (Q.Nos. 48-52) *Find the points of discontinuity of the following functions.*

48 $f(x)=\begin{cases}2x+3, & \text{if } x\leq 2\\ 2x-3, & \text{if } x>2\end{cases}$.

49 $f(x)=\begin{cases}x^3-3, & \text{if } x\leq 2\\ x^2+1, & \text{if } x>2\end{cases}$.

50 $f(x)=\begin{cases}\dfrac{1}{2}-x, & \text{if } 0\leq x<\dfrac{1}{2}\\ 1, & \text{if } x=\dfrac{1}{2}\\ \dfrac{3}{2}-x, & \text{if } \dfrac{1}{2}<x\leq 1\end{cases}$.

51 $f(x)=\begin{cases}2x+3, & \text{if } -3\leq x<-2\\ x+1, & \text{if } -2\leq x<0\\ x+2, & \text{if } 0\leq x\leq 1\end{cases}$.

52 $f(x)=|x|-|x+1|$. [NCERT]

Differentiability and Derivatives of Various Functions

Differentiability (Derivability)

A function $f(x)$ is said to be differentiable at a point $x = a$, if Left hand derivative at $x = a$ equals to Right hand derivative at $x = a$, provided both the derivatives exist.

i.e. LHD (at $x = a$) = RHD (at $x = a$),

where, Right hand derivative,

$$Rf'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

and Left hand derivative,

$$Lf'(a) = \lim_{h \to 0} \frac{f(a-h) - f(a)}{-h}$$

If $Rf'(a) \neq Lf'(a)$, the function is not differentiable at $x = a$.

EXAMPLE 1. Discuss the differentiability of $f(x) = x^2$ at $x = 1$.

Sol. We have, $f(x) = x^2$

Test for differentiability at $x = 1$.

$$\text{LHD} = \lim_{h \to 0} \frac{f(1-h) - f(1)}{-h} = \lim_{h \to 0} \frac{(1-h)^2 - 1}{-h}$$

$$= \lim_{h \to 0} \frac{1 + h^2 - 2h - 1}{-h} = \lim_{h \to 0} (-h + 2) = 2$$

$$\text{RHD} = \lim_{h \to 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \to 0} \frac{(1+h)^2 - 1}{h}$$

$$= \lim_{h \to 0} \frac{h^2 + 2h + 1 - 1}{h} = \lim_{h \to 0} (h + 2) = 2$$

$\because$ LHD = RHD = 2

Hence, $f(x)$ is differentiable at $x = 1$.

EXAMPLE 2. Prove that $f(x) = |x|$ is not differentiable at $x = 0$.

Sol. We have, $f(x) = |x|$

Test for differentiability at $x = 0$,

$$\text{LHD} = Lf'(0) = \lim_{h \to 0} \frac{f(0-h) - f(0)}{-h}$$

$$= \lim_{h \to 0} \frac{f(-h) - f(0)}{-h}$$

$$= \lim_{h \to 0} \frac{|-h| - |0|}{-h} \qquad \{\because f(x) = |x|\}$$

$$= \lim_{h \to 0} \frac{h}{-h} = \lim_{h \to 0} (-1) = -1$$

$$\text{and RHD} = Rf'(0) = \lim_{h \to 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \to 0} \frac{f(h) - f(0)}{h}$$

$$= \lim_{h \to 0} \frac{|h| - |0|}{h} = \lim_{h \to 0} \left(\frac{h}{h}\right) = \lim_{h \to 0} (1) = 1$$

$\because$ LHD $\neq$ RHD

Hence, f is not differentiable at $x = 0$. **Hence proved.**

EXAMPLE 3. Show that $f(x) = \begin{cases} \dfrac{\sin x^2}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ is differentiable at $x = 0$ and find $f'(0)$.

Sol. We have, $f(x) = \begin{cases} \dfrac{\sin x^2}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

Test for differentiability at $x = 0$

$$\text{LHD} = Lf'(0) = \lim_{h \to 0} \frac{f(0-h) - f(0)}{-h} = \lim_{h \to 0} \frac{f(-h) - f(0)}{-h}$$

$$= \lim_{h \to 0} \frac{\dfrac{\sin(-h)^2}{-h} - 0}{-h} = \lim_{h \to 0} \frac{\sin h^2}{h^2} = 1 \left[\because \lim_{x \to 0} \frac{\sin x}{x}\right]$$

$$\text{and RHD} = Rf'(0) = \frac{\lim_{h \to 0} f(0+h) - f(0)}{h} = \lim_{h \to 0} \frac{f(h) - f(0)}{h}$$

$$= \lim_{h \to 0} \frac{\dfrac{\sin h^2}{h} - 0}{h} = \lim_{h \to 0} \frac{\sin h^2}{h^2} = 1$$

Since, LHD = RHD = 1

Therefore, $f(x)$ is differentiable at $x = 0$ and $f'(0) = 1$.

EXAMPLE 4. Examine the following function $f(x)$ for differentiability at $x = 2$.

$$f(x) = \begin{cases} 4x^2 - 3x, \text{ if } x < 2 \\ 3x + 4, \text{ if } x \geq 2 \end{cases}$$

Sol. We have, $f(x) = \begin{cases} 4x^2 - 3x, \text{ if } x < 2 \\ 3x + 4, \text{ if } x \geq 2 \end{cases}$

Test for differentiability at $x = 2$,

$$\text{LHD} = \lim_{h \to 0} \frac{f(2-h) - f(2)}{-h}$$

$$= \lim_{h \to 0} \frac{\{4(2-h)^2 - 3(2-h)\} - \{3(2) + 4\}}{-h}$$

$$= \lim_{h \to 0} \frac{\{4(4 - 4h + h^2) - 6 + 3h\} - \{6 + 4\}}{-h}$$

$$= \lim_{h \to 0} \frac{16 - 16h + 4h^2 - 6 + 3h - 10}{-h}$$

$$= \lim_{h \to 0} \frac{4h^2 - 13h}{-h} = \lim_{h \to 0} (-4h + 13) = 13$$

$$\text{RHD} = \lim_{h\to 0}\frac{f(2+h)-f(2)}{h}$$
$$= \lim_{h\to 0}\frac{3(2+h)+4-(3\times 2+4)}{h}$$
$$= \lim_{h\to 0}\frac{(6+3h+4-10)}{h}$$
$$= \lim_{h\to 0}\left(\frac{3h}{h}\right) = \lim_{h\to 0}(3) = 3$$

Since, LHD ≠ RHD

Hence, f is not differentiable at $x = 2$.

Relation between Continuity and Differentiability

If a function is differentiable at a point c, then it is also continuous at that point. So, every differentiable function is continuous, but every continuous function need not to be differentiable function.

EXAMPLE 5. Show that the function $f(x) = \begin{cases} 1+x, \text{ if } x \le 2 \\ 5-x, \text{ if } x > 2 \end{cases}$ is continuous but not differentiable at $x = 2$.

Sol. We have, $f(x) = \begin{cases} 1+x, \text{ if } x \le 2 \\ 5-x, \text{ if } x > 2 \end{cases}$

For continuity at $x = 2$

$$\text{LHL} = \lim_{x\to 2^-} f(x) = \lim_{h\to 0} f(2-h)$$

[put $x = 2-h$ when $x \to 2^-$ then $h \to 0$]

$$= \lim_{h\to 0}(1+(2-h))$$
$$= \lim_{h\to 0}(3-h) = 3$$
$$\text{RHL} = \lim_{x\to 2^+} f(x) = \lim_{h\to 0} f(2+h)$$

[put $x = 2+h$ when $x \to 2^+$ then $h \to 0$]

$$= \lim_{h\to 0}(5-(2+h))$$
$$= \lim_{h\to 0}(3-h) = 3$$

Also, $f(2) = 1 + 2 = 3$

Here, LHL = RHL = $f(2)$

Hence, $f(x)$ is continuous at $x = 2$

Test for differentiability at $x = 2$

$$Lf'(2) = \lim_{h\to 0}\frac{f(2-h)-f(2)}{-h}$$
$$= \lim_{h\to 0}\frac{(1+(2-h))-(1+2)}{-h}$$
$$= \lim_{h\to 0}\frac{(3-h)-3}{-h}$$
$$= \lim_{h\to 0}\left(\frac{h}{h}\right) = 1$$

and
$$Rf'(2) = \lim_{h\to 0}\frac{f(2+h)-f(2)}{h}$$
$$= \lim_{h\to 0}\frac{(5-(2+h))-(1+2)}{h}$$
$$= \lim_{h\to 0}\frac{(3-h)-3}{h}$$
$$= \lim_{h\to 0}\frac{-h}{h} = -1$$

$\because Lf'(2) \ne Rf'(2)$

Hence, $f(x)$ is not differentiable at $x = 2$.

Hence proved.

EXAMPLE 6. Show that the function $f(x) = |x-3|, x \in R$, is continuous but not differentiable at $x = 3$. **[Delhi 2013]**

Sol. Given, $f(x) = |x-3|$

First, we check the continuity of $f(x)$ at $x = 3$.

Here, $$\text{LHL} = \lim_{x\to 3^-}|x-3| = \lim_{h\to 0}|3-h-3|$$

[put $x = 3-h$ when $x \to 3^-$, then $h \to 0$]

$$= \lim_{h\to 0}|-h| = 0$$
$$\text{RHL} = \lim_{x\to 3^+}|x-3| = \lim_{h\to 0}|3+h-3| = \lim_{h\to 0}|h| = 0$$

[put $x = 3+h$ when $x \to 3^+$, then $h \to 0$]

and $f(3) = |3-3| = 0$

Thus, LHL = RHL = $f(3)$

Hence, f is continuous at $x = 3$.

Now, let us check the differentiability of $f(x)$ at $x = 3$.

$$\text{LHD} = f'(3^-) = \lim_{h\to 0}\frac{f(3-h)-f(3)}{-h}$$

$$\left[\because Lf'(a) = \lim_{h\to 0}\frac{f(a-h)-f(a)}{-h}\right]$$

$$= \lim_{h\to 0}\frac{|3-h-3|-|3-3|}{-h}$$
$$= \lim_{h\to 0}\frac{|-h|}{-h} = \lim_{h\to 0}\frac{h}{-h} = -1$$

[$\because |-x| = x$, if $x > 0$]

$$\text{RHD} = f'(3^+) = \lim_{h\to 0}\frac{f(3+h)-f(3)}{h}$$

$$\left[\because Rf'(a) = \lim_{h\to 0}\frac{f(a+h)-f(a)}{h}\right]$$

$$= \lim_{h\to 0}\frac{|3+h-3|-|3-3|}{h}$$
$$= \lim_{h\to 0}\frac{|h|}{h} = \lim_{h\to 0}\frac{h}{h} = 1 \quad [\because |x| = x, \text{ if } x > 0]$$

Since, LHD ≠ RHD at $x = 3$.

So, f is not differentiable. **Hence proved.**

Differentiation

The process of finding derivative of a function is called differentiation. The derivative of a function is also called the **differential coefficient** of the function.

Derivatives of some standard functions are given below

(i) $\frac{d}{dx}(\text{constant}) = 0$

(ii) $\frac{d}{dx}(x^n) = nx^{n-1}$

(iii) $\frac{d}{dx}(cx^n) = cnx^{n-1}$

(iv) $\frac{d}{dx}(\sin x) = \cos x$

(v) $\frac{d}{dx}(\cos x) = -\sin x$

(vi) $\frac{d}{dx}(\tan x) = \sec^2 x$

(vii) $\frac{d}{dx}(\text{cosec } x) = -\text{cosec } x \cot x$

(viii) $\frac{d}{dx}(\sec x) = \sec x \tan x$

(ix) $\frac{d}{dx}(\cot x) = -\text{cosec}^2 x$

(x) $\frac{d}{dx}(e^x) = e^x$

(xi) $\frac{d}{dx}(a^x) = a^x \log_e a, a > 0$

(xii) $\frac{d}{dx}(\log_e x) = \frac{1}{x}, x > 0$

(xiii) $\frac{d}{dx}(\log_a x) = \frac{1}{x \log_e a}, a > 0, a \neq 1$

Algebra of Derivatives

(i) **Sum and Difference Rule** Let $y = f(x) \pm g(x)$. Then, its derivative is written as

$$\frac{dy}{dx} = \frac{d}{dx} f(x) \pm \frac{d}{dx} g(x).$$

(ii) **Product Rule** Let $y = f(x) \cdot g(x)$. Then, its derivative is written as

$$\frac{dy}{dx} = \left[\frac{d}{dx} f(x)\right] g(x) + \left[\frac{d}{dx} g(x)\right] f(x).$$

(iii) **Quotient Rule** Let $y = \frac{f(x)}{g(x)}; g(x) \neq 0$.

Then, its derivative is written as

$$\frac{dy}{dx} = \frac{g(x) \times \frac{d}{dx}[f(x)] - f(x) \times \frac{d}{dx}[g(x)]}{[g(x)]^2}$$

EXAMPLE 7. Find the derivative of $x^4 + 3x^{5/3} + 10x + 7$.

Sol. Let $y = x^4 + 3x^{5/3} + 10x + 7$

On differentiating both sides, w.r.t. x, we get

$$\frac{dy}{dx} = \frac{d}{dx}(x^4 + 3x^{5/3} + 10x + 7)$$

$$= \frac{d}{dx}(x^4) + \frac{d}{dx}(3x^{5/3}) + \frac{d}{dx}(10x) + \frac{d}{dx}(7)$$

[using sum and difference rule]

$$= 4x^{4-1} + 3\left(\frac{5}{3}\right)x^{\frac{5}{3}-1} + 10x^{1-1} + 0$$

$$\left[\because \frac{d}{dx}x^n = nx^{n-1}; \frac{d}{dx}(cx^n) = cnx^{n-1} \text{ and } \frac{d}{dx}(\text{constant}) = 0\right]$$

$$= 4x^3 + 5x^{2/3} + 10$$

EXAMPLE 8. Find the differential coefficient of $(7x^5 + 9)(px^{7/2} + q)$.

Sol. Let $y = (7x^5 + 9)(px^{7/2} + q)$

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = (7x^5 + 9)\frac{d}{dx}(px^{7/2} + q) + (px^{7/2} + q)\frac{d}{dx}(7x^5 + 9)$$

[using Product rule]

$$= (7x^5 + 9)\left(p \cdot \frac{7}{2}x^{\frac{7}{2}-1} + 0\right) + (px^{7/2} + q)(7 \cdot 5x^{5-1} + 0)$$

$$= \frac{7p}{2}x^{5/2}(7x^5 + 9) + 35x^4(px^{7/2} + q)$$

EXAMPLE 9. If $y = (2 + 3\sin x)(3 - 2\cos x)$, then find $\frac{dy}{dx}$.

Sol. We have, $y = (2 + 3\sin x)(3 - 2\cos x)$

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = \frac{d}{dx}[(2 + 3\sin x)(3 - 2\cos x)]$$

$$= (2+3\sin x)\frac{d}{dx}(3-2\cos x) + (3 - 2\cos x)\frac{d}{dx}(2+3\sin x)$$

[using Product rule]

$$= (2 + 3\sin x)(0 + 2\sin x) + (3 - 2\cos x)(0 + 3\cos x)$$
$$= (2 + 3\sin x)(2\sin x) + (3 - 2\cos x)(3\cos x)$$
$$= 4\sin x + 6\sin^2 x + 9\cos x - 6\cos^2 x$$
$$= 4\sin x + 9\cos x + 6\sin^2 x - 6\cos^2 x$$
$$= 4\sin x + 9\cos x - 6(\cos^2 x - \sin^2 x)$$
$$= 4\sin x + 9\cos x - 6\cos 2x$$
$$[\because \cos^2 x - \sin^2 x = \cos 2x]$$

EXAMPLE 10. Differentiate $\frac{\sec x - 1}{\sec x + 1}$ w.r.t. x.

Sol. Let $y = \frac{\sec x - 1}{\sec x + 1}$

$$\Rightarrow y = \frac{\left(\frac{1}{\cos x} - 1\right)}{\left(\frac{1}{\cos x} + 1\right)} \qquad \left[\because \sec x = \frac{1}{\cos x}\right]$$

$$\Rightarrow y = \frac{1 - \cos x}{1 + \cos x}$$

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = \frac{\left[(1+\cos x)\frac{d}{dx}(1-\cos x) - (1-\cos x)\frac{d}{dx}(1+\cos x)\right]}{(1 + \cos x)^2}$$

[using quotient rule]

$$= \frac{[(1 + \cos x)(0 + \sin x) - (1 - \cos x)(0 - \sin x)]}{(1 + \cos x)^2}$$
$$= \frac{[(1 + \cos x)\sin x - (1 - \cos x)(-\sin x)]}{(1 + \cos x)^2}$$
$$= \frac{[\sin x + \cos x \cdot \sin x + \sin x - \cos x \cdot \sin x]}{(1 + \cos x)^2}$$
$$= \frac{2\sin x}{(1 + \cos x)^2}$$

Derivatives of Composite Functions

Derivative of a composite function can be found out by chain rule.

Chain Rule

Chain rule is applied when the given function is the function of function, i.e. a function is in the form of $fog(x)$ or $f[g(x)]$.

Let $y = f(u)$ and $u = f(x)$, then by using chain rule, we may write $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$, when $\frac{dy}{du}$ and $\frac{du}{dx}$ both exist.

Extension of Chain Rule

Extension of chain rule is applied when the given function (say y) is a composite of three functions.

e.g.

If $y = f(u)$, $u = g(v)$ and $v = h(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx} = f'(u) \cdot g'(v) \cdot h'(x)$$

i.e. $\frac{d}{dx}[f(g(h(x)))] = f'[g(h(x))] \cdot g'(h(x)) \cdot h'(x)$.

EXAMPLE 11. Differentiate the following function with respect to x.

(i) $\sin(x^2 + 5)$ [NCERT]

(ii) $\tan\sqrt{x}$ (iii) $\sin 5x \cdot \cos 7x$

Sol. (i) Let $y = \sin u$, where $u = x^2 + 5$. Using chain rule,

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \cos u \cdot \frac{du}{dx} = \cos(x^2 + 5)\frac{d}{dx}(x^2 + 5)$$
$$= \cos(x^2 + 5) \cdot (2x)$$
$$= 2x \cdot \cos(x^2 + 5)$$

(ii) Let $y = \tan u$, where $u = \sqrt{x}$ [using chain rule]

$$\therefore \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \sec^2 u \frac{du}{dx}$$
$$= \sec^2\sqrt{x}\,\frac{d}{dx}(\sqrt{x})$$
$$= \sec^2\sqrt{x} \cdot \frac{1}{2}x^{-1/2}$$
$$= \frac{\sec^2\sqrt{x}}{2\sqrt{x}}$$

(iii) Let $y = \sin 5x \cdot \cos 7x = \frac{1}{2}[2\sin 5x \cdot \cos 7x]$

$$= \frac{1}{2}[\sin(5x + 7x) + \sin(5x - 7x)]$$
$$[\because 2\sin A\cos B = \sin(A + B) + \sin(A - B)]$$
$$= \frac{1}{2}[\sin 12x - \sin 2x]$$

$$\Rightarrow y = \frac{1}{2}\sin 12x - \frac{1}{2}\sin 2x$$

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = \frac{1}{2} \cdot \frac{d}{dx}(\sin 12x) - \frac{1}{2} \cdot \frac{d}{dx}(\sin 2x)$$

[by difference rule]

$$= \frac{1}{2}\cos 12x \frac{d}{dx}(12x) - \frac{1}{2}\cos 2x \frac{d}{dx}(2x)$$
$$= \frac{1}{2}\cos 12x \cdot 12 - \frac{1}{2}\cos 2x \cdot 2$$
$$= 6\cos 12x - \cos 2x$$

EXAMPLE 12. Differentiate the following with respect to x.

(i) $\sin^2(3x+1)$ (ii) $\sqrt{\sin\sqrt{x}}$

(iii) $2\sqrt{\cot(x^2)}$ [NCERT]

Sol. (i) Let $y = \sin^2(3x+1)$

Here, y is a composite of three functions.

Let $y = t^2$, where $t = \sin u$ and $u = 3x + 1$.

Then, $\frac{dy}{dx} = \frac{dy}{dt}\cdot\frac{dt}{du}\cdot\frac{du}{dx}$ [by chain rule]

$= \frac{d}{dt}(t^2)\cdot\frac{d}{du}(\sin u)\cdot\frac{d}{dx}(3x+1)$

$= 2t(\cos u)\cdot 3 = 6t\cos u$

$= 6\sin u\cos u$ [$\because t = \sin u$]

$= 3\sin 2u$ [$\because 2\sin\theta\cos\theta = \sin 2\theta$]

$= 3\sin 2(3x+1)$

$= 3\sin(6x+2)$ [put $u = 3x+1$]

(ii) Let $y = \sqrt{\sin\sqrt{x}} = (\sin\sqrt{x})^{1/2}$

On differentiating both sides w.r.t. x, we get

$\frac{dy}{dx} = \frac{d}{dx}(\sin\sqrt{x})^{1/2} = \frac{1}{2}(\sin\sqrt{x})^{\frac{1}{2}-1}\frac{d}{dx}(\sin\sqrt{x})$

[using chain rule]

$= \frac{1}{2}(\sin\sqrt{x})^{-1/2}(\cos\sqrt{x})\frac{d}{dx}(\sqrt{x})$

[using chain rule]

$= \frac{1}{2\sqrt{\sin\sqrt{x}}}\cdot(\cos\sqrt{x})\cdot\frac{1}{2\sqrt{x}} = \frac{\cos\sqrt{x}}{4\sqrt{x\sin\sqrt{x}}}$

(iii) Let $y = 2\sqrt{\cot(x^2)}$.

On differentiating both sides w.r.t. x, we get

$\frac{dy}{dx} = \frac{d}{dx}2(\cot x^2)^{1/2} = 2\cdot\frac{1}{2}\{\cot(x^2)\}^{\frac{1}{2}-1}\cdot\frac{d}{dx}\cot(x^2)$

$\left[\text{using chain rule } \frac{d}{dx}[fg(x)] = f'\{g(x)\}\frac{d}{dx}g(x)\right]$

$= \frac{1}{\sqrt{\cot(x^2)}}[-\operatorname{cosec}^2 x^2]\frac{d}{dx}(x^2)$

[using chain rule]

$= -\frac{\operatorname{cosec}^2(x^2)\,2x}{\sqrt{\cot(x^2)}}$

$= \frac{-2x\operatorname{cosec}^2(x^2)}{\sqrt{\cot(x^2)}}$

Topic Practice 2

1 Show that

(i) $f(x) = x^2$ is differentiable at $x = 2$.

(ii) $f(x) = x^{2/3}$ is not differentiable at $x = 0$.

2 Show that the function f given by $f(x) = |x-1|$, $x \in R$ is not differentiable at $x = 1$. [NCERT]

3 Examine the differentiability of the function

$f(x) = \begin{cases} x[x], & \text{if } 0 \le x < 2 \\ (x-1)x, & \text{if } 2 \le x < 3 \end{cases}$ at $x = 2$.

[NCERT Exemplar]

4 Prove that the greatest integer function defined by $f(x) = [x], 0 < x < 3$ is not differentiable at $x = 1$.

5 Show that the function $f(x) = \begin{cases} x-1, \text{ if } x < 2 \\ 2x-3, \text{ if } x \ge 2 \end{cases}$ is not differentiable at $x = 2$.

6 Show that the function $f(x) = |x-1| + |x+1|, \forall$ $x \in R$ is not differentiable at the points $x = -1$ and $x = 1$. [All India 2015]

7 Show that a function $f(x)$ is defined as follows

$$f(x) = \begin{cases} x^2\sin\frac{1}{x}, & \text{if } x \ne 0 \\ 0, & \text{if } x = 0 \end{cases}$$

is differentiable at $x = 0$.

8 Let $f(x) = x|x|, \forall x \in R$. Discuss the derivability of $f(x)$ at $x = 0$. [NCERT Exemplar]

9 Show that the function $f(x) = |x-5|$ is continuous but not differentiable at $x = 5$.

[Delhi 2013; NCERT Exemplar]

10 Find the value of a and b, if the function f defined by $f(x) = \begin{cases} x^2 + 3x + a, & x \le 1 \\ bx + 2, & x > 1 \end{cases}$ is differentiable at $x = 1$.

[CBSE 2016]

11 Differentiate the following functions with respect to x.

(i) $x^{20} + 30$

(ii) $7x^6 + \frac{5}{4}x^8$

(iii) $-3x^{-5} + \frac{1}{\sqrt{x}}$

(iv) $(ax^2 + b)(px^{5/3} + q)$

(v) $\cos x\cdot\sin x$

(vi) $(\tan x + x)(x^2 - \cot x)$

(vii) $\frac{1}{ax+b}$

(viii) $\frac{x+3}{x^2+1}$

(ix) $\frac{x+\sin x}{x+\cos x}$

(x) $x^2\sec x+\frac{x^2}{(1+\sin x)}$

12 Differentiate the following function with respect to x.

(i) $\tan(x^2+5)$ (ii) $\tan(\sin x)$

(iii) $\sin(\cos x^2)$ (iv) $\cos(\sin x^3)$

(v) $\sqrt{\cos(1+x^2)}$ (vi) $\operatorname{cosec}(\sqrt{x^2+2})$

(vii) $\sec\left(\frac{5}{x}\right)$

13 If $f(x)=\sin 2x-\cos 2x$, then find $f'\left(\frac{\pi}{6}\right)$.

[All India 2017C]

14 If $f(x)=|\cos x|$, then find $f'\left(\frac{3\pi}{4}\right)$.

[NCERT Exemplar]

15 If $y=x\tan x+\sec x$, then find the value of $\frac{dy}{dx}$ at $x=\frac{\pi}{4}$.

16 If $y=\sin^3\sqrt{ax^2+bx+c}$, then find $\frac{dy}{dx}$.

17 If $y=\left[x+\sqrt{x^2+a^2}\right]^n$, then prove that $\frac{dy}{dx}=\frac{ny}{\sqrt{x^2+a^2}}$.

18 If $y=\sqrt{\frac{1-\sin 2x}{1+\sin 2x}}$, then show that $\frac{dy}{dx}+\sec^2\left(\frac{\pi}{4}-x\right)=0, x\in\left[0,\frac{\pi}{4}\right]$.

Hints & Solutions

1. (i) Similar as Example 1.
(ii) Similar as Example 1.

2. Similar as Example 2.

3. Given, $f(x)=\begin{cases}x[x], & \text{if } 0\le x<2\\ (x-1)x & \text{if } 2\le x<3\end{cases}$

At $x=2$, $\text{LHD}=\lim_{h\to 0}\frac{f(2-h)-f(2)}{2-h-2}$

$=\lim_{h\to 0}\frac{(2-h)[2-h]-(2-1)2}{-h}$

$[\because [a-h]=[a-1]$, where a is any positive number$]$

$=\lim_{h\to 0}\frac{(2-h)(1)-2}{-h}=\lim_{h\to 0}\frac{-h}{-h}=1$

$\text{RHD}=\lim_{h\to 0}\frac{f(2+h)-f(2)}{2+h-2}$

$=\lim_{h\to 0}\frac{(2+h-1)(2+h)-(2-1)\cdot 2}{h}$

$=\lim_{h\to 0}\frac{(1+h)(2+h)-2}{h}$

$=\lim_{h\to 0}\frac{h(h+3)}{h}=3$

$\because$ LHD $\neq$ RHD

Hence, $f(x)$ is not differentiable at $x=2$.

4. Here, $f(x)=[x], 0<x<3$

For differentiability at $x=1$,

$\text{LHD}=\lim_{h\to 0}\frac{f(1-h)-f(1)}{-h}$

$=\lim_{h\to 0}\frac{[1-h]-[1]}{-h}$ $[\because [1-h]=0$ and $[1]=1]$

$=\lim_{h\to 0}\frac{0-1}{-h}=\lim_{h\to 0}\frac{1}{h}$, which does not exist.

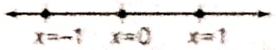
Hence, $f(x)$ is not differentiable at $x=1$.

5. Similar as Example 5.

6. **Hint** Here, given $f(x)=|x-1|+|x+1|$

Put $x-1=0$ and $x+1=0$

$\Rightarrow x=1$ and $x=-1$



***Case* I** When $x<-1$

In this case, $x-1<0$ and $x+1<0$

$\therefore |x-1|=-(x-1)$ and $|x+1|=-(x+1)$

$\therefore f(x)=-(x-1)+\{-(x+1)\}=-x+1-x-1$

$\Rightarrow f(x)=-2x$

***Case* II** When $-1\le x<1$

In this case, $x-1<0$ and $x+1\ge 0$

$\therefore |x-1|=-(x-1)$ and $|x+1|=x+1$

$\therefore f(x)=-(x-1)+(x+1)=-x+1+x+1=2$

***Case* III** When $x\ge 1$

In this case, $x-1\ge 0$ and $x+1\ge 0$

$\therefore |x-1|=x-1$ and $|x+1|=x+1$

$\therefore f(x)=(x-1)+(x+1)=2x$

Thus, $f(x)=\begin{cases}-2x, & \text{if } x<-1\\ 2, & \text{if } -1\le x<1\\ 2x, & \text{if } x\ge 1\end{cases}$

Now, examine the differentiability of the function $f(x)$, at $x=-1$ and $x=1$.

Derivatives of Implicit Functions

A function of the form $f(x, y) = 0$, where it is not possible to express y directly as a function of x in the form of $y = \phi(x)$, is said to be an **implicit function** of x and y.

e.g. $x^2 + y^2 = xy$, $\cos(x + y) = \log(xy)$

Following steps are used to find $\frac{dy}{dx}$ of implicit functions.

I. Write the given equation in x and y and then differentiate both sides of the given equation w.r.t. x.

II. Take all terms involving $\frac{dy}{dx}$ on LHS and transfer the remaining terms on RHS to get equation in the form of $\frac{dy}{dx} f(x, y) = g(x, y)$.

III. Now, $\frac{dy}{dx} = \frac{g(x, y)}{f(x, y)}$.

EXAMPLE 1. If $x^3 + y^3 = 3axy$, then find $\frac{dy}{dx}$. **[CBSE 2023]**

Sol. Given, $x^3 + y^3 = 3axy$

On differentiating both sides w.r.t. x, we get

$$3x^2 + 3y^2\frac{dy}{dx} = 3a\frac{d}{dx}(xy)$$

$$\Rightarrow \quad 3x^2 + 3y^2\frac{dy}{dx} = 3a\left(1 \cdot y + x\frac{dy}{dx}\right)$$

[by product rule of derivative]

$$\Rightarrow \quad 3x^2 + 3y^2\frac{dy}{dx} = 3ay + 3ax\frac{dy}{dx}$$

$$\Rightarrow \quad 3y^2\frac{dy}{dx} - 3ax\frac{dy}{dx} = 3ay - 3x^2$$

[taking terms involving dy/dx in LHS]

$$\Rightarrow \quad 3(y^2 - ax)\frac{dy}{dx} = 3(ay - x^2)$$

$$\Rightarrow \quad (y^2 - ax)\frac{dy}{dx} = ay - x^2$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{ay - x^2}{y^2 - ax}$$

EXAMPLE 2. Find $\frac{dy}{dx}$ for the equation

$$x^3 + y^3 = \sin(x + y).$$

Sol. We have, $x^3 + y^3 = \sin(x + y)$

On differentiating both sides w.r.t. x, we get

$$3x^2 + 3y^2 \cdot \frac{dy}{dx} = \cos(x + y)\frac{d}{dx}(x + y)$$

[by chain rule of derivative]

$$\Rightarrow \quad 3x^2 + 3y^2\frac{dy}{dx} = \cos(x + y)\left(1 + \frac{dy}{dx}\right)$$

$$\Rightarrow \quad 3x^2 + 3y^2\frac{dy}{dx} = \cos(x + y) + \cos(x + y) \cdot \frac{dy}{dx}$$

$$\Rightarrow \quad 3y^2\frac{dy}{dx} - \cos(x + y)\frac{dy}{dx} = \cos(x + y) - 3x^2$$

$\left[\text{taking terms involving } \frac{dy}{dx} \text{ in LHS}\right]$

$$\Rightarrow \quad \frac{dy}{dx}[3y^2 - \cos(x + y)] = \cos(x + y) - 3x^2$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{\cos(x + y) - 3x^2}{3y^2 - \cos(x + y)}$$

EXAMPLE 3. If $y = x\sin y$, then prove that

$$x\frac{dy}{dx} = \frac{y}{1 - x\cos y}.$$

Sol. Given, $y = x\sin y$

On differentiating both sides w.r.t. x, we get

$$\frac{d(y)}{dx} = \frac{d}{dx}(x\sin y)$$

$$\Rightarrow \quad \frac{dy}{dx} = x \cdot \cos y\frac{dy}{dx} + \sin y$$

$$\Rightarrow \quad \frac{dy}{dx} - x\cos y\frac{dy}{dx} = \sin y$$

$$\Rightarrow \quad \frac{dy}{dx}(1 - x\cos y) = \sin y$$

$$\Rightarrow \quad \frac{dy}{dx}(1 - x\cos y) = \frac{y}{x} \qquad \text{[from Eq. (i)]}$$

$$\Rightarrow \quad x\frac{dy}{dx} = \frac{y}{1 - x\cos y} \qquad \textbf{Hence proved.}$$

EXAMPLE 4. If $\sin y = x\cos(a + y)$, then show that $\frac{dy}{dx} = \frac{\cos^2(a + y)}{\cos a}$. Also, show that $\frac{dy}{dx} = \cos a$, when $x = 0$.

Sol. We have, $\sin y = x\cos(a + y)$

$$\Rightarrow \quad x = \frac{\sin y}{\cos(a + y)}$$

On differentiating both sides w.r.t. y, we get

$$\frac{dx}{dy}=\frac{\left[\cos(a+y)\frac{d}{dy}\sin y-\sin y\frac{d}{dy}\cos(a+y)\right]}{[\cos(a+y)]^2}$$

[by quotient rule of derivative]

$$=\frac{\cos(a+y)\cos y-\sin y[-\sin(a+y)\cdot 1]}{\cos^2(a+y)}$$

$$=\frac{\cos(a+y)\cos y+\sin y\sin(a+y)}{\cos^2(a+y)}$$

$$=\frac{\cos(a+y-y)}{\cos^2(a+y)}$$

$[\because \cos A\cos B+\sin A\sin B=\cos(A-B)]$

$$=\frac{\cos a}{\cos^2(a+y)}$$

$$\Rightarrow \frac{dy}{dx}=\frac{\cos^2(a+y)}{\cos a}$$

Given, $\sin y=x\cos(a+y)$

When $x=0$, then

$$\sin y=0 \Rightarrow y=n\pi, n\in I.$$

$\therefore$ At $x=0$,

$$\frac{dy}{dx}=\frac{\cos^2(a+n\pi)}{\cos a}=\frac{\cos^2 a}{\cos a}=\cos a$$

$[\because \cos^2(n\pi+\theta)=\cos^2\theta]$

Hence proved.

EXAMPLE 5. If $x+y=t+\frac{1}{t}$ and $x^3+y^3=t^3+\frac{1}{t^3}$, then prove that $\frac{dy}{dx}=-\frac{1}{x^2}$.

Sol. Given, $(x+y)=t+\frac{1}{t}$

On cubing both sides, we get

$$(x+y)^3=\left(t+\frac{1}{t}\right)^3$$

$$\Rightarrow x^3+y^3+3xy(x+y)=t^3+\frac{1}{t^3}+3\left(t+\frac{1}{t}\right)$$

$[\because (a+b)^3=a^3+b^3+3ab(a+b)]$

$$\Rightarrow x^3+y^3+3xy(x+y)=x^3+y^3+3(x+y)$$

$$\Rightarrow 3xy(x+y)=3(x+y)$$

$$\left[\because x^3+y^3=t^3+\frac{1}{t^3} \text{ and } (x+y)=t+\frac{1}{t}\right]$$

$$\Rightarrow xy=1 \quad \ldots(i)$$

On differentiating both sides w.r.t. x, we get

$$x\cdot\frac{dy}{dx}+y\cdot\frac{d}{dx}(x)=\frac{d}{dx}\cdot 1$$

[product rule of derivative]

$$\Rightarrow x\cdot\frac{dy}{dx}+y=0\Rightarrow x\frac{dy}{dx}=-y$$

$$\Rightarrow x\frac{dy}{dx}=-\frac{1}{x}$$ [from Eq. (i)]

$$\Rightarrow \frac{dy}{dx}=-\frac{1}{x^2}$$ **Hence proved.**

Topic Practice 3

Directions (Q.Nos. 1-12) *Find* $\frac{dy}{dx}$ *w.r.t. x in each of the following.*

1 $xy=a^2$

2 $x^2+y^2-4xy=5$

3 $y=\tan(x+y)$ [NCERT Exemplar]

4 $xy+y^2=\tan x+y$ [NCERT]

5 $\sec(x+y)=xy$ [NCERT Exemplar]

6 $ax^2+by^2=\cos y$

7 $x^3+x^2y+xy^2+y^3=81$ [NCERT]

8 $(x^2+y^2)^2=xy$ [CBSE 2023; NCERT Exemplar]

9 $\sin(x+y)=x^2+y^2$

10 $y=\sqrt{\sin x+y}$

11 $x+y=\sin(xy)$

12 $\sin(xy)+\frac{x}{y}=x^2-y$. [NCERT Exemplar]

13 Find dy/dx at $x=1, y=\pi/4$, if $\sin^2 y+\cos xy=K$. [Delhi 2017]

14 If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, then show that $\frac{dy}{dx} \cdot \frac{dx}{dy} = 1$ [NCERT Exemplar]

15 If $x\sqrt{1+y} + y\sqrt{1+x} = 0$ for $-1 < x, y < 1$, then prove that $\frac{dy}{dx} = \frac{-1}{(x+1)^2}$.

[NCERT, Foreign 2012; Delhi 2011]

16 If $\cos y = x\cos(a + y)$, where $\cos a \neq \pm 1$, then prove that $\frac{dy}{dx} = \frac{\cos^2(a+y)}{\sin a}$. [NCERT; Foreign 2014]

17 If $\tan(x + y) + \tan(x - y) = 1$, then prove that $\frac{\sec^2(x+y) + \sec^2(x-y)}{\sec^2(x-y) - \sec^2(x+y)}$.

Hints & Solutions

1. Similar as Example 1. **Ans.** $-y/x$

2. Similar as Example 1. **Ans.** $\frac{2y - x}{y - 2x}$

3. Similar as Example 2. **Ans.** $-\operatorname{cosec}^2(x + y)$

4. Similar as Example 2. **Ans.** $\frac{\sec^2 x - y}{(x + 2y - 1)}$

5. Similar as Example 2. **Ans.** $\frac{y - \sec(x+y)\tan(x+y)}{\sec(x+y)\tan(x+y) - x}$

6. We have, $ax^2 + by^2 = \cos y$

On differentiating both sides w.r.t. x, we get

$$2ax + 2by\frac{dy}{dx} = -\sin y\frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx}(2by + \sin y) = -2ax$$

$$\Rightarrow \frac{dy}{dx} = \frac{-2ax}{2by + \sin y}$$

7. We have, $x^3 + x^2y + xy^2 + y^3 = 81$

On differentiating both sides w.r.t. x, we get

$$3x^2 + x^2\frac{dy}{dx} + y\cdot 2x + x\cdot 2y\frac{dy}{dx} + y^2\cdot 1 + 3y^2\frac{dy}{dx} = 0$$

$$\Rightarrow x^2\frac{dy}{dx} + 2xy\frac{dy}{dx} + 3y^2\frac{dy}{dx} = -3x^2 - 2xy - y^2$$

$$\Rightarrow \frac{dy}{dx}(x^2 + 2xy + 3y^2) = -3x^2 - 2xy - y^2$$

$$\frac{dy}{dx} = \frac{-(3x^2 + 2xy + y^2)}{x^2 + 2xy + 3y^2}$$

8. Similar as Example 1. **Ans.** $\frac{y - 4x^3 - 4xy^2}{4yx^2 + 4y^3 - x}$

9. We have, $\sin(x + y) = x^2 + y^2$.

On differentiating both sides w.r.t. x, we get

$$\cos(x + y)\frac{d}{dx}(x + y) = 2x + 2y\frac{dy}{dx}$$

$$\Rightarrow \cos(x + y)\left(1 + \frac{dy}{dx}\right) = 2x + 2y\frac{dy}{dx}$$

$$\Rightarrow \cos(x + y) + \cos(x + y)\frac{dy}{dx} = 2x + 2y\frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx}[\cos(x + y) - 2y] = 2x - \cos(x + y)$$

$$\therefore \frac{dy}{dx} = \frac{2x - \cos(x + y)}{\cos(x + y) - 2y}$$

10. We have, $y = (\sin x + y)^{1/2}$

$$\therefore \frac{dy}{dx} = \frac{1}{2}(\sin x + y)^{-1/2}\cdot\frac{d}{dx}(\sin x + y)$$

[by chain rule of derivative]

$$\Rightarrow \frac{dy}{dx} = \frac{1}{2}\cdot\frac{1}{(\sin x + y)^{1/2}}\cdot\left(\cos x + \frac{dy}{dx}\right)$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{2y}\left(\cos x + \frac{dy}{dx}\right) \quad [\because (\sin x + y)^{1/2} = y]$$

$$\Rightarrow \frac{dy}{dx}\left(1 - \frac{1}{2y}\right) = \frac{\cos x}{2y}$$

$$\therefore \frac{dy}{dx} = \frac{\cos x}{2y}\cdot\frac{2y}{2y - 1} = \frac{\cos x}{2y - 1}$$

11. Given, $x + y = \sin(xy)$

On differentiating both sides, w.r.t. x, we get

$$1 + \frac{dy}{dx} = \cos(xy)\frac{d}{dx}(xy)$$

[by product rule of derivative]

$$= \cos(xy)\left(x\frac{dy}{dx} + y\right)$$

$$\Rightarrow 1 + \frac{dy}{dx} = x\cos(xy)\frac{dy}{dx} + y\cos(xy)$$

$$\Rightarrow \frac{dy}{dx}[1 - x\cos(xy)] = y\cos(xy) - 1$$

$$\Rightarrow \frac{dy}{dx} = \frac{y\cos(xy) - 1}{1 - x\cos(xy)}$$

12. Similar as Example 2.

Ans. $\dfrac{2xy^2 - y^3\cos(xy) - y}{xy^2\cos(xy) - x + y^2}$

13. We have, $\sin^2 y + \cos xy = K$

On differentiating both sides w.r.t. x, we get

$$\frac{d}{dx}(\sin^2 y + \cos xy) = \frac{d}{dx}(K)$$

$$\Rightarrow \quad \frac{d}{dx}(\sin^2 y) + \frac{d}{dx}(\cos xy) = 0$$

$$\Rightarrow \quad 2\sin y\cos y\frac{dy}{dx} + (-\sin xy)\frac{d}{dx}(xy) = 0$$

$$\Rightarrow \quad \sin 2y\frac{dy}{dx} - \sin xy\left(x\frac{dy}{dx} + y\cdot 1\right) = 0$$

$$\Rightarrow \quad \sin 2y\frac{dy}{dx} - x\sin xy\frac{dy}{dx} = y\sin xy$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{y\sin(xy)}{\sin 2y - x\sin(xy)}$$

Now, at $x = 1$, $y = \dfrac{\pi}{4}$.

$$\therefore \left(\frac{dy}{dx}\right)_{(1,\pi/4)} = \frac{\frac{\pi}{4}\sin\left(1.\frac{\pi}{4}\right)}{\sin\left(2.\frac{\pi}{4}\right) - 1\sin\left(1.\frac{\pi}{4}\right)}$$

$$= \frac{\frac{\pi}{4}\sin\left(\frac{\pi}{4}\right)}{\sin\left(\frac{\pi}{2}\right) - \sin\left(\frac{\pi}{4}\right)} = \frac{\frac{\pi}{4}\left(\frac{1}{\sqrt{2}}\right)}{1 - \frac{1}{\sqrt{2}}}$$

$$= \frac{\frac{\pi}{4\sqrt{2}}}{(\sqrt{2}-1)/\sqrt{2}} = \frac{\pi}{4\sqrt{2}} \times \frac{\sqrt{2}}{(\sqrt{2}-1)}$$

$$= \frac{\pi}{4(\sqrt{2}-1)}$$

14. We have, $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ ...(i)

On differentiating both sides w.r.t. x, we get

$$\frac{d}{dx}(ax^2) + \frac{d}{dx}(2hxy) + \frac{d}{dx}(by^2) + \frac{d}{dx}(2gx) + \frac{d}{dx}(2fy) + \frac{d}{dx}(c) = 0$$

$$\Rightarrow \quad 2ax + 2h\left(x\cdot\frac{dy}{dx} + y\cdot 1\right) + b\cdot 2y\frac{dy}{dx} + 2g + 2f\frac{dy}{dx} + 0 = 0$$

$$\Rightarrow \quad \frac{dy}{dx}[2hx + 2by + 2f] = -2ax - 2hy - 2g$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{-2(ax + hy + g)}{2(hx + by + f)} = \frac{-(ax + hy + g)}{(hx + by + f)} \qquad \text{...(ii)}$$

Now, differentiating both sides of Eq. (i) w.r.t. y, we get

$$\frac{d}{dy}(ax^2) + \frac{d}{dy}(2hxy) + \frac{d}{dy}(by^2) + \frac{d}{dy}(2gx) + \frac{d}{dy}(2fy) + \frac{d}{dy}(c) = 0$$

$$\Rightarrow \quad a\cdot 2x\cdot\frac{dx}{dy} + 2h\cdot\left(x\cdot 1 + y\cdot\frac{d}{dy}x\right) + b\cdot 2y + 2g\cdot\frac{dx}{dy} + 2f + 0 = 0$$

$$\Rightarrow \quad \frac{dx}{dy}[2ax + 2hy + 2g] = -2hx - 2by - 2f$$

$$\Rightarrow \quad \frac{dx}{dy} = \frac{-2(hx + by + f)}{2(ax + hy + g)} = \frac{-(hx + by + f)}{(ax + hy + g)} \qquad \text{...(iii)}$$

$$\therefore \quad \frac{dy}{dx}\cdot\frac{dx}{dy} = \frac{-(ax + hy + g)}{(hx + by + f)}\cdot\frac{-(hx + by + f)}{(ax + hy + g)}$$

[using Eqs. (ii) and (iii)]

$$= 1 = \text{RHS}$$

Hence proved.

15. We have, $x\sqrt{1+y} + y\sqrt{1+x} = 0$

$$\Rightarrow \quad x\sqrt{1+y} = -y\sqrt{1+x}$$

On squaring both sides, we get

$$x^2(1+y) = y^2(1+x)$$

$$\Rightarrow \quad x^2 - y^2 + x^2y - y^2x = 0$$

$$\Rightarrow \quad (x+y)(x-y) + xy(x-y) = 0$$

$$\Rightarrow \quad (x-y)(x+y+xy) = 0$$

But $x \neq y$

$$\therefore \quad x + y + xy = 0$$

$$\Rightarrow \quad y(1+x) = -x$$

$$\Rightarrow \quad y = \frac{-x - 1 + 1}{1+x}$$

$$\Rightarrow \quad y = -1 + \frac{1}{1+x}$$

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = \frac{-1}{(1+x)^2}$$

Hence proved.

16. Similar as Example 4.

Hint $x = \dfrac{\cos y}{\cos(a+y)}$, find $\dfrac{dx}{dy}$.

17. We have, $\tan(x+y) + \tan(x-y) = 1$

On differentiating both sides, w.r.t. x, we get

$$\sec^2(x+y)\left(1 + \frac{dy}{dx}\right) + \sec^2(x-y)\left(1 - \frac{dy}{dx}\right) = 0$$

$$\Rightarrow \frac{dy}{dx}[\sec^2(x+y) - \sec^2(x-y)] + \sec^2(x+y) + \sec^2(x-y)$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{\sec^2(x+y) + \sec^2(x-y)}{\sec^2(x-y) - \sec^2(x+y)}$$

Derivatives of Inverse Trigonometric Functions

Derivatives of inverse trigonometric functions are given below

- $\frac{d}{dx}(\sin^{-1}x) = \frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}(\cos^{-1}x) = \frac{-1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}(\tan^{-1}x) = \frac{1}{1+x^2}$
- $\frac{d}{dx}(\cot^{-1}x) = \frac{-1}{1+x^2}$
- $\frac{d}{dx}(\sec^{-1}x) = \frac{1}{x\sqrt{x^2-1}}$
- $\frac{d}{dx}(\text{cosec}^{-1}x) = \frac{-1}{x\sqrt{x^2-1}}$

EXAMPLE 1. Differentiate the following with respect to x.

(i) $\sin^{-1}7x$ (ii) $\cos^{-1}(5x^2+4)$

(iii) $\tan^{-1}(\cot x)$

Sol. (i) Let $y = \sin^{-1}7x$

Put $z = 7x$. Then, $y = \sin^{-1}z$

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = \frac{d}{dx}(\sin^{-1}z) = \frac{d}{dz}(\sin^{-1}z)\frac{dz}{dx} \quad \text{[by chain rule]}$$

$$= \frac{1}{\sqrt{1-z^2}}\frac{d}{dx}(7x) = \frac{7}{\sqrt{1-(7x)^2}} = \frac{7}{\sqrt{1-49x^2}}$$

$[\because z = 7x]$

(ii) Let $y = \cos^{-1}(5x^2+4)$

Put $z = 5x^2+4$

Then, $y = \cos^{-1}z$

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = \frac{d}{dx}(\cos^{-1}z) = \frac{d}{dz}(\cos^{-1}z)\frac{dz}{dx}$$

[by chain rule]

$$= \frac{-1}{\sqrt{1-z^2}}\frac{d}{dx}(5x^2+4) \qquad [\text{put } z = 5x^2+4]$$

$$= -\frac{1}{\sqrt{1-(5x^2+4)^2}} \times [5\times 2x + 0]$$

$[\text{put } z = 5x^2+4]$

$$= -\frac{10x}{\sqrt{1-(5x^2+4)^2}}$$

(iii) Let $y = \tan^{-1}(\cot x)$

Put $z = \cot x$. Then, $y = \tan^{-1}z$

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = \frac{d}{dx}(\tan^{-1}z) = \frac{d}{dz}(\tan^{-1}z)\frac{dz}{dx}$$

$$= \frac{-1}{1+z^2}\frac{d}{dx}(\cot x) \qquad \text{[by chain rule]}$$

$$= \frac{-\text{cosec}^2 x}{1+\cot^2 x} \qquad [\because z = \cot x]$$

$$= \frac{-\text{cosec}^2 x}{\text{cosec}^2 x} = -1$$

EXAMPLE 2. Differentiate each of the following functions w.r.t.x

(i) $\tan^{-1}\left(\frac{x}{2}\right) + \tan^{-1}\left(\frac{2}{x}\right)$ (ii) $\cot^{-1}\left(\frac{1-x}{1+x}\right)$

Sol. (i) Let $y = \tan^{-1}\frac{x}{2} + \tan^{-1}\frac{2}{x}$

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = \frac{d}{dx}\left(\tan^{-1}\frac{x}{2} + \tan^{-1}\frac{2}{x}\right)$$

$$= \frac{d}{dx}\left(\tan^{-1}\frac{x}{2}\right) + \frac{d}{dx}\left(\tan^{-1}\frac{2}{x}\right)$$

$$= \frac{1}{1+\left(\frac{x}{2}\right)^2}\frac{d}{dx}\left(\frac{x}{2}\right) + \frac{1}{1+\left(\frac{2}{x}\right)^2}\frac{d}{dx}\left(\frac{2}{x}\right)$$

$\left[\text{by chain rule and } \frac{d}{dx}(\tan^{-1}x) = \frac{1}{1+x^2}\right]$

$$= \frac{1}{1+\frac{x^2}{4}}\left(\frac{1}{2}\right) + \frac{1}{1+\frac{4}{x^2}}\left(\frac{-2}{x^2}\right)$$

$$= \frac{2}{4+x^2} - \frac{2}{x^2+4} = 0$$

(ii) Let $y = \cot^{-1}\left(\frac{1-x}{1+x}\right)$

Put $\frac{1-x}{1+x} = u$ ...(i)

Then, $y = \cot^{-1}u$ ...(ii)

Thus, y is a composite function, so we use chain rule of derivative.

i.e. $\frac{dy}{dx} = \frac{dy}{du}\cdot\frac{du}{dx}$ ...(iii)

Now, differentiating Eq. (i) w.r.t. x, we get

$$\frac{du}{dx}=\frac{d}{dx}\left(\frac{1-x}{1+x}\right)$$

$$=\frac{(1+x)(-1)-(1-x)(1)}{(1+x)^2} \quad \text{[by quotient rule]}$$

$$=\frac{-1-x-1+x}{(1+x)^2}=\frac{-2}{(1+x)^2}$$

On differentiating both sides of Eq. (ii) w.r.t. u, we get

$$\frac{dy}{du}=\frac{-1}{1+u^2}=\frac{-1}{1+\left(\frac{1-x}{1+x}\right)^2} \quad \text{[from Eq. (i)]}$$

$$\Rightarrow \quad \frac{dy}{du}=\frac{-(1+x)^2}{(1+x)^2+(1-x)^2}=\frac{-(1+x)^2}{2(1+x^2)}$$

On putting the values of $\frac{du}{dx}$ and $\frac{dy}{du}$ in Eq. (iii), we get

$$\frac{dy}{dx}=\frac{-(1+x)^2}{2(1+x^2)}\cdot\frac{-2}{(1+x)^2} \Rightarrow \frac{dy}{dx}=\frac{1}{1+x^2}$$

EXAMPLE 3. Show that

$$\frac{d}{dx}\left[\frac{x}{2}\sqrt{a^2-x^2}+\frac{a^2}{2}\sin^{-1}\frac{x}{a}\right]=\sqrt{a^2-x^2}.$$

Sol. We have, LHS $=\frac{d}{dx}\left[\frac{x}{2}\cdot\sqrt{a^2-x^2}+\frac{a^2}{2}\cdot\sin^{-1}\frac{x}{a}\right]$

$$=\frac{d}{dx}\left[\frac{x}{2}\cdot\sqrt{a^2-x^2}\right]+\frac{a^2}{2}\cdot\frac{d}{dx}\left[\sin^{-1}\frac{x}{a}\right]$$

$$=\frac{x}{2}\cdot\frac{d}{dx}(\sqrt{a^2-x^2})+(\sqrt{a^2-x^2})\cdot\frac{d}{dx}\left(\frac{x}{2}\right)$$

$$+\frac{a^2}{2}\cdot\frac{1}{\sqrt{1-\frac{x^2}{a^2}}}\cdot\frac{1}{a} \quad \text{[by product rule]}$$

$$=\frac{x}{2}\cdot\frac{1}{2}(a^2-x^2)^{-1/2}(-2x)+(\sqrt{a^2-x^2})$$

$$\cdot\frac{1}{2}+\frac{a^2}{2\sqrt{a^2-x^2}}$$

$$=\frac{-x^2}{2\sqrt{a^2-x^2}}+\frac{\sqrt{a^2-x^2}}{2}+\frac{a^2}{2\sqrt{a^2-x^2}}$$

$$=\frac{-x^2+(a^2-x^2)+a^2}{2\sqrt{a^2-x^2}}$$

$$=\frac{2(a^2-x^2)}{2\sqrt{a^2-x^2}}=\sqrt{a^2-x^2}=\text{RHS}$$

Hence, $\frac{d}{dx}\left[\frac{x}{2}\cdot\sqrt{a^2-x^2}+\frac{a^2}{2}\cdot\sin^{-1}\frac{x}{a}\right]=\sqrt{a^2-x^2}.$

Differentiation of Inverse Trigonometric Functions by Trigonometrical Transformations

Sometimes differentiation of inverse trigonometric functions becomes difficult by chain rule, then we can differentiate them easily by using trigonometrical transformation.

For this, we can use the following trigonometric and inverse trigonometric formulae

- $\sin 2x=2\sin x\cos x$
- $\cos 2x=\cos^2 x-\sin^2 x=2\cos^2 x-1=1-2\sin^2 x$
- $\sin 2x=\frac{2\tan x}{1+\tan^2 x}$
- $\cos 2x=\frac{1-\tan^2 x}{1+\tan^2 x}$
- $\tan 2x=\frac{2\tan x}{1-\tan^2 x}$
- $\tan(x+y)=\frac{\tan x+\tan y}{1-\tan x\tan y}$
- $\tan(x-y)=\frac{\tan x-\tan y}{1+\tan x\tan y}$
- $\sin^{-1}(-x)=-\sin^{-1}x$,
- $\cos^{-1}(-x)=\pi-\cos^{-1}x$, $\tan^{-1}(-x)=-\tan^{-1}x$
- $\sin^{-1}x+\cos^{-1}x=\frac{\pi}{2}$, $\tan^{-1}x+\cot^{-1}x=\frac{\pi}{2}$, $\sec^{-1}x+\operatorname{cosec}^{-1}x=\frac{\pi}{2}$
- $\sin^{-1}(\sqrt{1-x^2})=\cos^{-1}x$, $\cos^{-1}(\sqrt{1-x^2})=\sin^{-1}x$
- $2\tan^{-1}x=\sin^{-1}\left(\frac{2x}{1+x^2}\right)=\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ $=\tan^{-1}\left(\frac{2x}{1-x^2}\right)$
- $\tan^{-1}x+\tan^{-1}y=\tan^{-1}\left(\frac{x+y}{1-xy}\right)$, $xy<1$
- $\tan^{-1}x-\tan^{-1}y=\tan^{-1}\left(\frac{x-y}{1+xy}\right)$, $xy>-1$

Following are some substitution useful in finding derivatives.

	Expressions	Substitution
1.	$\sqrt{a^2-x^2}$	$x = a\sin\theta$ or $x = a\cos\theta$
2.	$\sqrt{a^2+x^2}$ or a^2+x^2	$x = a\tan\theta$ or $x = a\cot\theta$
3.	$\sqrt{x^2-a^2}$	$x = a\sec\theta$ or $x = a\operatorname{cosec}\theta$
4.	$\sqrt{\frac{a-x}{a+x}}$ or $\sqrt{\frac{a+x}{a-x}}$	$x = a\cos 2\theta$
5.	$\sqrt{\frac{x}{a-x}}$ or $\sqrt{\frac{a-x}{x}}$	$x = a\sin^2\theta$
6.	$\sqrt{\frac{x}{a+x}}$ or $\sqrt{\frac{a+x}{x}}$	$x = a\tan^2\theta$
7.	$\sqrt{\frac{x}{x-a}}$ or $\sqrt{\frac{x-a}{x}}$	$x = a\sec^2\theta$
8.	$\sqrt{\frac{a^2-x^2}{a^2+x^2}}$ or $\sqrt{\frac{a^2+x^2}{a^2-x^2}}$	$x^2 = a^2\cos^2\theta$

EXAMPLE 4. Differentiale each of the following w.r.t. x.

(i) $\tan^{-1}\left(\frac{1+\cos x}{\sin x}\right)$ **[CBSE 2018]**

(ii) $\tan^{-1}(\sec x + \tan x)$

Sol. (i) Let $y = \tan^{-1}\left(\frac{1+\cos x}{\sin x}\right) = \tan^{-1}\left(\frac{2\cos^2\frac{x}{2}}{2\sin\frac{x}{2}\cos\frac{x}{2}}\right)$

$\left[\because 1+\cos A = 2\cos^2\frac{A}{2} \text{ and } \sin A = 2\sin\frac{A}{2}\cos\frac{A}{2}\right]$

$= \tan^{-1}\left(\frac{\cos\frac{x}{2}}{\sin\frac{x}{2}}\right) = \tan^{-1}\left(\cot\frac{x}{2}\right)$

$= \frac{\pi}{2} - \cot^{-1}\left(\cot\frac{x}{2}\right) \quad \left[\because \tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}\right]$

$= \frac{\pi}{2} - \frac{x}{2} \quad [\because \cot^{-1}(\cot x) = x]$

Now, on differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = 0 - \frac{1}{2} = -\frac{1}{2}$$

(ii) Let $y = \tan^{-1}(\sec x + \tan x) = \tan^{-1}\left(\frac{1}{\cos x} + \frac{\sin x}{\cos x}\right)$

$\Rightarrow y = \tan^{-1}\left(\frac{1+\sin x}{\cos x}\right) = \tan^{-1}\left(\frac{1-\cos\left(\frac{\pi}{2}+x\right)}{\sin\left(\frac{\pi}{2}+x\right)}\right)$

$\left[\because \cos\left(\frac{\pi}{2}+x\right) = -\sin x \text{ and } \sin\left(\frac{\pi}{2}+x\right) = \cos x\right]$

$\Rightarrow y = \tan^{-1}\left[\frac{2\sin^2\left(\frac{\pi}{4}+\frac{x}{2}\right)}{2\sin\left(\frac{\pi}{4}+\frac{x}{2}\right)\cdot\cos\left(\frac{\pi}{4}+\frac{x}{2}\right)}\right]$

$\left[\because \sin x = 2\sin\frac{x}{2}\cos\frac{x}{2}\right]$

$\Rightarrow y = \tan^{-1}\left[\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)\right] \Rightarrow y = \frac{\pi}{4}+\frac{x}{2}$

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = 0 + \frac{1}{2} \Rightarrow \frac{dy}{dx} = \frac{1}{2}$$

EXAMPLE 5. Differentiate w.r.t. x

(i) $\sin^{-1}\left(\frac{1-x^2}{1+x^2}\right), 0 < x < 1$ **[NCERT]**

(ii) $\sec^{-1}\left(\frac{1}{2x^2-1}\right), 0 < x < \frac{1}{\sqrt{2}}$ **[NCERT]**

Sol. (i) Let $y = \sin^{-1}\left(\frac{1-x^2}{1+x^2}\right)$

Putting $x = \tan\theta \Rightarrow \theta = \tan^{-1}x$

$\therefore \quad y = \sin^{-1}\left(\frac{1-\tan^2\theta}{1+\tan^2\theta}\right)$

$\Rightarrow \quad y = \sin^{-1}(\cos 2\theta) = \sin^{-1}\left[\sin\left(\frac{\pi}{2}-2\theta\right)\right]$

$\Rightarrow \quad y = \frac{\pi}{2} - 2\theta \quad [\because \sin^{-1}(\sin\theta) = \theta]$

$\Rightarrow \quad y = \frac{\pi}{2} - 2\tan^{-1}x$

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{\pi}{2} - 2\tan^{-1}x\right) = \frac{d}{dx}\left(\frac{\pi}{2}\right) - 2\frac{d}{dx}(\tan^{-1}x)$$

$$= 0 - 2\cdot\frac{1}{1+x^2} = \frac{-2}{1+x^2}$$

(ii) Let $y = \sec^{-1}\left(\frac{1}{2x^2-1}\right)$

Putting $x = \cos\theta \Rightarrow \theta = \cos^{-1}x$

$$\therefore\ y = \sec^{-1}\left(\frac{1}{2\cos^2\theta - 1}\right) \Rightarrow y = \sec^{-1}\left(\frac{1}{\cos 2\theta}\right)$$

$[\because \cos 2\theta = 2\cos^2\theta - 1]$

$$= \sec^{-1}(\sec 2\theta) = 2\theta = 2\cos^{-1} x$$

On differentiating both sides w.r.t. x

$$\frac{dy}{dx} = \frac{d}{dx}(2\cos^{-1} x)$$

$$= 2\frac{d}{dx}(\cos^{-1} x) = 2\left(\frac{-1}{\sqrt{1-x^2}}\right) = \frac{-2}{\sqrt{1-x^2}}$$

EXAMPLE 6. If $y = \sec^{-1}\left(\frac{x^2+1}{x^2-1}\right)$, then find $\frac{dy}{dx}$.

Sol. We have, $y = \sec^{-1}\left(\frac{x^2+1}{x^2-1}\right)$.

Here, putting $x = \tan\theta$, we get

$$y = \sec^{-1}\left(\frac{\tan^2\theta + 1}{\tan^2\theta - 1}\right)$$

$$\Rightarrow \quad y = \cos^{-1}\left(\frac{\tan^2\theta - 1}{\tan^2\theta + 1}\right) \quad \left[\because \sec^{-1}\theta = \cos^{-1}\left(\frac{1}{\theta}\right)\right]$$

$$\Rightarrow \quad y = \cos^{-1}\left[-\left(\frac{1-\tan^2\theta}{1+\tan^2\theta}\right)\right]$$

$$\Rightarrow \quad y = \cos^{-1}[-\cos 2\theta] \quad \left[\because \cos 2\theta = \frac{1-\tan^2\theta}{1+\tan^2\theta}\right]$$

$$\Rightarrow \quad y = \pi - \cos^{-1}(\cos 2\theta) \quad [\because \cos^{-1}(-x) = \pi - \cos^{-1} x]$$

$$\Rightarrow \quad y = \pi - 2\theta \quad [\because \cos^{-1}(\cos x) = x]$$

$$\Rightarrow \quad y = \pi - 2\tan^{-1} x \quad [\because \tan\theta = x \Rightarrow \theta = \tan^{-1} x] \quad ...(i)$$

On differentiating both sides of Eq. (i) w.r.t. x, we get

$$\frac{dy}{dx} = 0 - \frac{2}{1+x^2} \Rightarrow \frac{dy}{dx} = \frac{-2}{1+x^2}$$

EXAMPLE 7. If $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$, then prove that $\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$.

Sol. We have, $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$.

On putting $x = \sin\theta$ and $y = \sin\phi$, it becomes

$$\cos\theta + \cos\phi = a(\sin\theta - \sin\phi)$$

$$\Rightarrow 2\cos\left(\frac{\theta+\phi}{2}\right)\cdot\cos\left(\frac{\theta-\phi}{2}\right) = a2\cos\left(\frac{\theta+\phi}{2}\right)\cdot\sin\left(\frac{\theta-\phi}{2}\right)$$

$$\Rightarrow \quad \cot\frac{\theta-\phi}{2} = a \Rightarrow \theta - \phi = 2\cot^{-1} a$$

$$\Rightarrow \quad \sin^{-1} x - \sin^{-1} y = 2\cot^{-1} a$$

On differentiating both sides w.r.t. x, we get

$$\frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-y^2}}\frac{dy}{dx} = 0$$

Hence, $\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$

EXAMPLE 8. Find $\frac{dy}{dx}$, if $y = \sin^{-1}\left(\frac{5x + 12\sqrt{1-x^2}}{13}\right)$.

Sol. Given, $y = \sin^{-1}\left(\frac{5x + 12\sqrt{1-x^2}}{13}\right)$.

Put $x = \cos\theta$, then $\sqrt{1-x^2} = \sin\theta$.

$$\therefore y = \sin^{-1}\left(\frac{5\cos\theta + 12\sin\theta}{13}\right) = \sin^{-1}\left(\frac{12}{13}\sin\theta + \frac{5}{13}\cos\theta\right) \quad ...(i)$$

Let $\frac{12}{13} = \cos\alpha$, then $\frac{5}{13} = \sin\alpha$ and $\tan\alpha = \frac{5}{12}$

$$\therefore \quad y = \sin^{-1}(\cos\alpha\sin\theta + \sin\alpha\cos\theta)$$

$$= \sin^{-1}(\sin\theta\cos\alpha + \cos\theta\sin\alpha)$$

$$= \sin^{-1}\{\sin(\theta+\alpha)\} = \theta + \alpha$$

$$\Rightarrow \quad y = \cos^{-1} x + \tan^{-1}\left(\frac{5}{12}\right)$$

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = \frac{-1}{\sqrt{1-x^2}} + 0 = \frac{-1}{\sqrt{1-x^2}}$$

EXAMPLE 9. Differentiate $\tan^{-1}\left(\frac{a\cos x - b\sin x}{b\cos x + a\sin x}\right)$ $-\frac{\pi}{2} < x < \frac{\pi}{2}$ and $\frac{a}{b}\tan x > -1$. [NCERT Exemplar]

Sol. Let $y = \tan^{-1}\left(\frac{a\cos x - b\sin x}{b\cos x + a\sin x}\right)$

$$\Rightarrow \quad y = \tan^{-1}\left[\frac{\left(\frac{a\cos x - b\sin x}{b\cos x}\right)}{\left(\frac{b\cos x + a\sin x}{b\cos x}\right)}\right]$$

[dividing the numerator and denominator by $b\cos x$]

$$\Rightarrow \quad y = \tan^{-1}\left[\frac{\left(\frac{a}{b} - \tan x\right)}{\left(1 + \frac{a}{b}\tan x\right)}\right]$$

Put $\frac{a}{b} = \tan\theta \Rightarrow \theta = \tan^{-1}\left(\frac{a}{b}\right)$

Then, $y = \tan^{-1}\left[\frac{\tan\theta - \tan x}{1 + \tan\theta\cdot\tan x}\right]$

$$\Rightarrow \quad y = \tan^{-1}[\tan(\theta - x)] \quad \left[\because \tan(x-y) = \frac{\tan x - \tan y}{1 + \tan x\cdot\tan y}\right]$$

$$\Rightarrow \quad y = \theta - x \quad [\because \tan^{-1}(\tan\theta) = \theta]$$

$$\Rightarrow \quad y = \tan^{-1}\left(\frac{a}{b}\right) - x \quad \left[\because \theta = \tan^{-1}\left(\frac{a}{b}\right)\right]$$

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = 0 - 1 = -1$$

Topic Practice 4

1 Differentiate the following functions w.r.t. x.

(i) $\cos^{-1} 5x$ (ii) $\tan^{-1}\sqrt{x}$

(iii) $\sin^{-1}\left(\frac{x}{a}\right)$ (iv) $\sin^{-1}(x\sqrt{x})$

(v) $\cos^{-1} x^4$ (vi) $\cos^{-1}(\sin x)$

(vii) $\sin^{-1}(3x+2)$ (viii) $\cos^{-1}(\sqrt{\cos x})$

(ix) $[\sqrt{1-x^2}\sin^{-1} x - x]$ (x) $\sqrt{\cot^{-1}\sqrt{x}}$

(xi) $x\sec^{-1} x$

Directions (Q. Nos. 2-20) *Differentiate each of the following with respect to x.* [NCERT]

2 $\cot^{-1}\left(\frac{1-x}{1+x}\right)$.

3 $\sin^{-1}(2x\sqrt{1-x^2}), -\frac{1}{\sqrt{2}} < x < \frac{1}{\sqrt{2}}$.

4 $\cos^{-1}(4x^3 - 3x), -1 < x < 1$.

5 $\cos^{-1}\sqrt{\frac{1+\cos x}{2}}$.

6 $\sin^{-1}\left(\sqrt{\frac{1+x}{2}}\right), x \neq 0$.

7 $\cos^{-1}\left(\frac{\sin x + \cos x}{\sqrt{2}}\right), \frac{-\pi}{4} < x < \frac{\pi}{4}$.

8 $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$. [NCERT]

9 $\sin^{-1}(6x\sqrt{1-9x^2}), -\frac{1}{3\sqrt{2}} < x < \frac{1}{3\sqrt{2}}$. [Delhi 2017]

10 $\sin^{-1}(2x\sqrt{1-x^2}), -\frac{1}{\sqrt{2}} < x < \frac{1}{\sqrt{2}}$. [NCERT]

11 $\tan^{-1}\left(\frac{\sqrt{1+a^2x^2}-1}{ax}\right)$.

12 $\tan^{-1}(\sqrt{1+x^2}+x)$.

13 $\sin^{-1}\left(\frac{\sqrt{1+x}-\sqrt{1-x}}{2}\right)$.

14 $\tan^{-1}\left(\frac{\cos x + \sin x}{\cos x - \sin x}\right)$.

15 $\tan^{-1}\left(\frac{\cos x}{1+\sin x}\right)$.

16 $\cos^{-1}\left(\frac{3\cos x - 4\sin x}{5}\right)$.

17 $\tan^{-1}\left(\frac{\sqrt{1+x}-\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}}\right)$.

18 $\tan^{-1}\left[\frac{\sqrt{1+x^2}+\sqrt{1-x^2}}{\sqrt{1+x^2}-\sqrt{1-x^2}}\right], -1 < x < 1$.

19 $\tan^{-1}\left(\frac{3a^2x - x^3}{a^3 - 3ax^2}\right), -\frac{1}{\sqrt{3}} < \frac{x}{a} < \frac{1}{\sqrt{3}}$. [NCERT Exemplar]

20 $\frac{dy}{dx}$, $y = \sin^{-1}\left[\frac{6x - 4\sqrt{1-4x^2}}{5}\right]$. [All India 2016]

21 If the derivative of $\tan^{-1}(a+bx)$ takes the value $\frac{dy}{dx} = 1$ at $x = 0$, then prove that $b = 1 + a^2$.

22 If $y = \sin^{-1}[x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2}]$ and $0 < x < 1$, then find dy/dx. [All India 2014C]

23 If $y = \frac{x\sin^{-1} x}{\sqrt{1-x^2}}$, then prove that

$$\frac{dy}{dx} = \frac{x(\sqrt{1-x^2}) + \sin^{-1} x}{(1-x^2)^{\frac{3}{2}}}.$$

24 $\sqrt{1-x^4} + \sqrt{1-y^4} = a(x^2 - y^2)$, then prove that

$$\frac{dy}{dx} = \frac{x}{y}\sqrt{\frac{1-y^4}{1-x^4}}.$$ [All India 2016C]

Hints & Solutions

1. (i) Let $y = \cos^{-1}(5x)$

$$\Rightarrow \frac{dy}{dx} = \frac{-1}{\sqrt{1-(5x)^2}} \frac{d}{dx}(5x) = \frac{-5}{\sqrt{1-25x^2}}$$

(ii) Similar as Example 1 (iii). **Ans.** $\dfrac{1}{2\sqrt{x}\,(1+x)}$

(iii) Similar as Example 1 (i). **Ans.** $\dfrac{1}{\sqrt{a^2-x^2}}$

(iv) Similar Example 1 (i). **Ans.** $\dfrac{3}{2}\dfrac{\sqrt{x}}{\sqrt{1-x^3}}$

(v) Similar as Example 1 (ii). **Ans.** $\dfrac{-4x^3}{\sqrt{1-x^8}}$

(vi) Similar as Example 1 (ii). **Ans.** -1

(vii) Similar as Example 1(i). **Ans.** $\dfrac{3}{\sqrt{1-(3x+2)^2}}$

(viii) Similar as Example 1 (ii). **Ans.** $\dfrac{1}{2}\sqrt{\sec x + 1}$

(ix) Let $y = \sqrt{1-x^2}\,\sin^{-1}x - x$

$$\frac{dy}{dx} = \sqrt{1-x^2}\frac{d}{dx}(\sin^{-1}x) + \sin^{-1}x\frac{d}{dx}(\sqrt{1-x^2}) - 1$$

$$= \sqrt{1-x^2}\cdot\frac{1}{(\sqrt{1-x^2})} + \sin^{-1}x\cdot\frac{1}{2\sqrt{1-x^2}}(-2x) - 1$$

$$= 1 - \frac{x\sin^{-1}x}{\sqrt{1-x^2}} - 1 = \frac{-x\sin^{-1}x}{\sqrt{1-x^2}}$$

(x) Let $y = \sqrt{\cot^{-1}\sqrt{x}}$

$$\therefore \frac{d}{dx} = \frac{1}{2\sqrt{\cot^{-1}\sqrt{x}}}\frac{d}{dx}(\cot^{-1}\sqrt{x})$$

$$= \frac{1}{2\sqrt{\cot^{-1}\sqrt{x}}}\left(\frac{-1}{1+(\sqrt{x})^2}\right)\frac{d}{dx}(\sqrt{x})$$

[by chain rule]

$$= \frac{1}{2\sqrt{\cot^{-1}\sqrt{x}}}\left(\frac{-1}{1+x}\right)\left(\frac{1}{2\sqrt{x}}\right)$$

$$= \frac{-1}{4(\sqrt{\cos^{-1}\sqrt{x}})(1+x)\sqrt{x}}$$

(xi) Let $y = x\sec^{-1}x$

$$\therefore \frac{dy}{dx} = x\frac{d}{dx}(\sec^{-1}x) + (\sec^{-1}x)\frac{d}{dx}(x)$$

[by product rule]

$$= x\cdot\frac{1}{x\sqrt{x^2-1}} + \sec^{-1}x = \frac{1}{\sqrt{x^2-1}} + \sec^{-1}x$$

2. Let $y = \cot^{-1}\left(\dfrac{1-x}{1+x}\right)$

Putting $x = \tan\theta$, we get

$$y = \cot^{-1}\left(\frac{1-\tan\theta}{1+\tan\theta}\right) = \cot^{-1}\left\{\tan\left(\frac{\pi}{4}-\theta\right)\right\}$$

$$= \cot^{-1}\left[\cot\left\{\frac{\pi}{2}-\left(\frac{\pi}{4}-\theta\right)\right\}\right] = \cot^{-1}\left[\cot\left(\frac{\pi}{4}+\theta\right)\right] = \frac{\pi}{4}+\theta$$

$$\Rightarrow y = \frac{\pi}{4} + \tan^{-1}x \Rightarrow \frac{dy}{dx} = 0 + \frac{1}{1+x^2} = \frac{1}{1+x^2}$$

3. Let $y = \sin^{-1}(2x\sqrt{1-x^2})$

Putting $x = \sin\theta \Rightarrow \theta = \sin^{-1}x$

$$\Rightarrow y = \sin^{-1}(2\sin\theta\sqrt{1-\sin^2\theta})$$

$$= \sin^{-1}(2\sin\theta\cdot\cos\theta)$$

$$= \sin^{-1}(\sin 2\theta) = 2\theta = 2\sin^{-1}x$$

$$\therefore \frac{dy}{dx} = \frac{d}{dx}(2\sin^{-1}x) = \frac{2}{\sqrt{1-x^2}}$$

4. Let $y = \cos^{-1}(4x^3 - 3x)$

Putting $x = \cos\theta \Rightarrow \theta = \cos^{-1}x$

$$y = \cos^{-1}(4\cos^3\theta - 3\cos\theta)$$

$$= \cos^{-1}(\cos 3\theta) = 3\theta = 3\cos^{-1}x$$

$$\therefore \frac{dy}{dx} = \frac{d}{dx}(3\cos^{-1}x) = \frac{-3}{\sqrt{1-x^2}}$$

5. Let $y = \cos^{-1}\sqrt{\dfrac{1+\cos x}{2}}$

$$= \cos^{-1}\sqrt{\frac{2\cos^2\frac{x}{2}}{2}} \qquad \left[\because 1+\cos\theta = 2\cos^2\frac{\theta}{2}\right]$$

$$= \cos^{-1}\sqrt{\cos^2\frac{x}{2}} = \cos^{-1}\left[\cos\left(\frac{x}{2}\right)\right]$$

$$\Rightarrow y = \frac{x}{2} \qquad [\because \cos^{-1}(\cos\theta) = \theta]$$

On differentiating both sides w.r.t. x, we get $\dfrac{dy}{dx} = \dfrac{1}{2}$

6. Let $y = \sin^{-1}\left(\sqrt{\dfrac{1+x}{2}}\right)$

Putting $x = \cos\theta \Rightarrow \theta = \cos^{-1}x$

$$y = \sin^{-1}\left(\sqrt{\frac{1+\cos\theta}{2}}\right) = \sin^{-1}\left(\sqrt{\frac{2\cos^2\frac{\theta}{2}}{2}}\right)$$

$$= \sin^{-1}\left(\cos\frac{\theta}{2}\right) = \sin^{-1}\left[\sin^{-1}\left(\frac{\pi}{2}-\frac{\theta}{2}\right)\right] = \frac{\pi}{2}-\frac{\theta}{2}$$

$$\therefore \frac{dy}{dx} = \frac{d}{dx}\left(\frac{\pi}{2} - \frac{1}{2}\cos^{-1}x\right) = \frac{1}{2\sqrt{1-x^2}}$$

7. **Hint** $\cos^{-1}\dfrac{(\sin x + \cos x)}{\sqrt{2}} = \cos^{-1}\left[\cos\left(\dfrac{\pi}{4}-x\right)\right]$

$$= \left(\frac{\pi}{4}-x\right) \quad \textbf{Ans. } -1$$

Differentiation of Exponential and Logarithmic Functions

Exponential Function

The function defined as $y = f(x) = b^x$ with positive base $b > 0, b \neq 1$ is called the exponential function.

- If the base of an exponential function is e i.e. $b = e$, then it is called **natural exponential** function.
- If the base of an exponential function is 10 i.e. $b = 10$, then it is called the **common exponential function.**

Logarithmic Function

If b is any number such that $b > 0$ and $b \neq 1$ and $x > 0$, then $y = \log_b x$ is equivalent to $b^y = x$.

We usually read as 'log base b of x'.

Basic properties of logarithms are given below:

(i) $\log_a (mn) = \log_a m + \log_a n$, $m > 0, n > 0, a > 0, a \neq 1$

(ii) $\log_a x = \log_a e \log_e x, a > 0, a \neq 1, x > 0$

(iii) $\log_a \frac{m}{n} = \log_a m - \log_a n, m > 0, n > 0, a > 0, a \neq 1$

(iv) $\log_a m^n = n \log_a m, m > 0, a > 0, a \neq 1$

(v) $e^{\log_e x} = x, x > 0$

(vi) $\log_a x = \frac{\log x}{\log a}, a > 0, a \neq 1, x > 0$

(vii) $\log_n m = \frac{1}{\log_m n}$, $m, n > 0$ and $m, n \neq 1$

(viii) $\log_{10} 10 = 1, \log_e e = 1$ and $\log_a 1 = 0, a > 0, a \neq 1$

Important Theorems

Theorem 1 The derivative of e^x w.r.t. x is e^x

i.e. $\frac{d}{dx}(e^x) = e^x$.

Theorem 2 The derivative of $\log x$ w.r.t. x is $\frac{1}{x}$

i.e. $\frac{d}{dx}(\log x) = \frac{1}{x}$.

EXAMPLE 1. Differentiate the following w.r.t. x.

(i) e^{x^3} (ii) $\log (\cos e^x)$ [NCERT]

Sol. (i) Let $y = e^{x^3}$

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = \frac{d}{dx} e^{x^3} \frac{d}{dx}(x^3) \qquad \text{[by chain rule]}$$

$$= e^{x^3} \cdot 3x^2 = 3x^2 e^{x^3}$$

(ii) Let $y = \log (\cos e^x)$, it is a composite function.

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = \frac{1}{\cos e^x} \cdot \frac{d}{dx}(\cos e^x)$$

$$\left[\text{by chain rule and } \frac{d}{dx}(\log t) = \frac{1}{t}\right]$$

$$= \frac{1}{\cos e^x}(-\sin e^x) \frac{d}{dx}(e^x)$$

[again by chain rule]

$$= -\tan e^x \times e^x = -e^x \tan e^x$$

EXAMPLE 2. Differentiate each of the following w.r.t. x.

(i) $e^{\sqrt{\cot x}}$ (ii) $\log \tan \frac{x}{2}$

Sol. (i) Let $y = e^{\sqrt{\cot x}}$

Taking log on both sides, we get

$$\log y = \sqrt{\cot x} \cdot \log e = \sqrt{\cot x} \qquad [\because \log_e e = 1]$$

On differentiating both sides w.r.t. x, we get

$$\frac{1}{y}\frac{dy}{dx} = \frac{1}{2\sqrt{\cot x}} \frac{d}{dx}(\cot x)$$

$$\Rightarrow \frac{1}{y} \cdot \frac{dy}{dx} = \frac{1}{2\sqrt{\cot x}} \times (-\operatorname{cosec}^2 x)$$

$$\Rightarrow \frac{dy}{dx} = y\left[-\frac{\operatorname{cosec}^2 x}{2\sqrt{\cot x}}\right]$$

$$\Rightarrow \frac{dy}{dx} = -e^{\sqrt{\cot x}} \frac{\operatorname{cosec}^2 x}{2\sqrt{\cot x}}$$

(ii) Let $y = \log \tan \frac{x}{2}$

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = \frac{1}{\tan \frac{x}{2}} \frac{d}{dx}\left(\tan \frac{x}{2}\right)$$

$$= \frac{1}{\tan \frac{x}{2}} \times \sec^2 \frac{x}{2} \times \frac{d}{dx}\left(\frac{x}{2}\right)$$

$$= \frac{\cos\frac{x}{2}}{\sin\frac{x}{2}} \times \frac{1}{\cos^2\frac{x}{2}} \times \frac{1}{2} = \frac{1}{2\sin\frac{x}{2}\cos\frac{x}{2}}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{1}{\sin x} = \text{cosec } x$$

EXAMPLE 3. Differentiate $y = \log_{\cos x} \sin x$ w.r.t. x.

Sol. Given $y = \log_{\cos x} \sin x = \dfrac{\log_e \sin x}{\log_e \cos x}$ $\left[\because \log_a b = \dfrac{\log b}{\log a}\right]$

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = \frac{\log_e \cos x \frac{d}{dx}(\log_e \sin x) - \log_e \sin x \frac{d}{dx}(\log_e \cos x)}{(\log_e \cos x)^2}$$

[by quotient rule]

$$= \frac{\log_e \cos x \frac{1}{\sin x}\frac{d}{dx}(\sin x) - \log_e \sin x \frac{1}{\cos x}\frac{d}{dx}(\cos x)}{(\log_e \cos x)^2}$$

$$= \frac{\log_e \cos x \times \frac{\cos x}{\sin x} - \log_e \sin x \times \frac{(-\sin x)}{\cos x}}{(\log_e \cos x)^2}$$

$$= \frac{\cot x \log_e \cos x + \tan x \log_e \sin x}{(\log_e \cos x)^2}$$

Logarithmic Differentiation

When the given function is in the form $[f(x)]^{g(x)}$, where $f(x)$ and $g(x)$ are functions of x.

Then, we take logarithm both sides and use properties of logarithm to simplify it and then differentiate it. This process is known as **logarithmic differentiation.**

Let $\quad y = [f(x)]^{g(x)}$

Taking logarithm (to base e), we get

$$\log y = g(x) \log (f(x))$$

Using chain rule, we get

$$\frac{dy}{dx} = [f(x)]^{g(x)} \left[\frac{g(x)}{f(x)} f'(x) + g'(x) \log f(x)\right]$$

Here, functions should always be positive otherwise their logarithms are not defined.

EXAMPLE 4. Differentiate each of the following functions w.r.t. x

(i) a^x, where a is a positive constant [NCERT]

(ii) $x^{\frac{1}{x}}$

Sol. (i) Let $y = a^x$

Taking log on both sides, we get

$$\log y = \log a^x \Rightarrow \log y = x \log a$$

On differentiating both sides w.r.t. x, we get

$$\frac{1}{y}\frac{dy}{dx} = \log a \Rightarrow \frac{dy}{dx} = y \log a$$

Thus, $\dfrac{d}{dx}(a^x) = a^x \log a$

(ii) Let $\quad y = x^{\frac{1}{x}}$

Taking log on both sides, we get

$$\log y = \log x^{\frac{1}{x}}$$

$$\Rightarrow \quad \log y = \frac{1}{x}\log x = \frac{\log x}{x}$$

On differentiating both sides w.r.t. x, we get

$$\frac{1}{y}\frac{dy}{dx} = \frac{x\frac{d}{dx}(\log x) - \log x \frac{d}{dx}(x)}{x^2}$$

$$\Rightarrow \quad \frac{1}{y}\frac{dy}{dx} = \frac{x \cdot \frac{1}{x} - \log x \cdot 1}{x^2}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{y(1 - \log x)}{x^2}$$

Thus, $\dfrac{d}{dx}x^{\frac{1}{x}} = \dfrac{x^{\frac{1}{x}}(1 - \log x)}{x^2}$

EXAMPLE 5. If $y = \log\sqrt{\dfrac{1+\sin^2 x}{1-\sin x}}$, then find $\dfrac{dy}{dx}$. [CBSE 2023]

Sol. Given, $y = \log\sqrt{\dfrac{1+\sin^2 x}{1-\sin x}} \Rightarrow y = \log\left(\dfrac{1+\sin^2 x}{1-\sin x}\right)^{1/2}$

$$\Rightarrow \quad y = \frac{1}{2}\log\left(\frac{1+\sin^2 x}{1-\sin x}\right) \qquad [\because \log m^n = n\log m]$$

$$\Rightarrow \quad y = \frac{1}{2}[\log(1+\sin^2 x) - \log(1-\sin x)]$$

$$\left[\because \log\left(\frac{m}{n}\right) = \log m - \log n\right]$$

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = \frac{1}{2}\left[\frac{1}{1+\sin^2 x} \times (0 + 2\sin x \cdot \cos x) - \frac{1}{1-\sin x}(-\cos x)\right]$$

$$= \frac{1}{2}\left[\frac{\sin 2x}{1+\sin^2 x} + \frac{\cos x}{1-\sin x}\right]$$

EXAMPLE 6. Differentiate $\sqrt{\frac{(x-3)(x^2+4)}{3x^2+4x+5}}$ with respect to x. [NCERT]

Sol. Let given function be $y = \sqrt{\frac{(x-3)(x^2+4)}{3x^2+4x+5}}$

$$\Rightarrow \quad y = \left\{\frac{(x-3)(x^2+4)}{3x^2+4x+5}\right\}^{1/2}$$

Taking log on both sides, we get

$$\log y = \log\left\{\frac{(x-3)(x^2+4)}{3x^2+4x+5}\right\}^{1/2}$$

$$= \frac{1}{2}\log\left\{\frac{(x-3)(x^2+4)}{3x^2+4x+5}\right\} \quad [\because \log m^n = n\log m]$$

$$= \frac{1}{2}[\log(x-3)(x^2+4) - \log(3x^2+4x+5)]$$

$$[\because \log(m/n) = \log m - \log n]$$

$$= \frac{1}{2}[\log(x-3) + \log(x^2+4) - \log(3x^2+4x+5)] \quad ...(i)$$

$$[\because \log mn = \log m + \log n]$$

On differentiating both sides of Eq. (i) w.r.t. x, we get

$$\frac{1}{y}\frac{dy}{dx} = \frac{1}{2}\left[\frac{1}{(x-3)}\frac{d}{dx}(x-3) + \frac{1}{(x^2+4)}\frac{d}{dx}(x^2+4) - \frac{1}{(3x^2+4x+5)}\frac{d}{dx}(3x^2+4x+5)\right]$$

[by chain rule]

$$= \frac{1}{2}\left[\frac{1}{x-3} + \frac{2x}{x^2+4} - \frac{6x+4}{3x^2+4x+5}\right]$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{y}{2}\left[\frac{1}{x-3} + \frac{2x}{x^2+4} - \frac{6x+4}{3x^2+4x+5}\right]$$

$$= \frac{1}{2}\sqrt{\frac{(x-3)(x^2+4)}{3x^2+4x+5}}\left[\frac{1}{x-3} + \frac{2x}{x^2+4} - \frac{6x+4}{3x^2+4x+5}\right]$$

EXAMPLE 7. Differentiate each of the following functions w.r.t. x

(i) $x^{\sin^{-1}x}$ (ii) $(\sin x)^{\log x}$

Sol. (i) Let $y = x^{\sin^{-1}x}$

Taking logarithm on both sides, we get

$$\log y = \log x^{\sin^{-1}x}$$

$$\Rightarrow \quad \log y = \sin^{-1}x \log x \quad [\because \log m^n = n\log m]$$

On differentiating both sides w.r.t. x, we get

$$\frac{1}{y}\frac{dy}{dx} = \sin^{-1}x \cdot \frac{1}{x} + \frac{1}{\sqrt{1-x^2}}\log x$$

[using product rule]

$$\Rightarrow \frac{dy}{dx} = y\left(\frac{\sin^{-1}x}{x} + \frac{\log x}{\sqrt{1-x^2}}\right)$$

$$\Rightarrow \frac{dy}{dx} = x^{\sin^{-1}x}\left(\frac{\sin^{-1}x}{x} + \frac{\log x}{\sqrt{1-x^2}}\right) \quad [\because y = x^{\sin^{-1}x}]$$

(ii) Let $y = (\sin x)^{\log x}$

Taking logarithm on both sides, we get

$$\log y = \log(\sin x)^{\log x}$$

$$\Rightarrow \quad \log y = \log x \log(\sin x)$$

On differentiating both sides w.r.t. x, we get

$$\frac{1}{y}\frac{dy}{dx} = \log x \cdot \frac{d}{dx}(\log(\sin x)) + \log(\sin x)\cdot\frac{d}{dx}(\log x)$$

$$= \log x \cdot \frac{\cos x}{\sin x} + \log(\sin x)\cdot\frac{1}{x}$$

$$\Rightarrow \frac{dy}{dx} = y\left[\log x \cdot \cot x + \frac{\log(\sin x)}{x}\right]$$

$$= (\sin x)^{\log x}\left[\log x \cdot \cot x + \frac{\log(\sin x)}{x}\right]$$

EXAMPLE 8. If $y = (\cos x)^x + \sin^{-1}\sqrt{3x}$, find $\frac{dy}{dx}$. **[All India 2017C]**

Sol. Let $y = (\cos x)^x + \sin^{-1}\sqrt{3x}$

Again, let $u = (\cos x)^x$, $v = \sin^{-1}\sqrt{3x}$

$\therefore \quad y = u + v$

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx} \quad ...(i)$$

Now, $u = (\cos x)^x$

Taking log on both sides, we get

$$\log u = \log(\cos x)^x = x\log(\cos x)$$

On differentiating both sides w.r.t. x, we get

$$\frac{1}{u}\frac{du}{dx} = x\frac{d}{dx}\log(\cos x) + \log(\cos x)\frac{d}{dx}(x)$$

$$= \frac{x}{\cos x}(-\sin x) + \log(\cos x)$$

$$\Rightarrow \quad \frac{du}{dx} = u[-x\tan x + \log\cos x]$$

$$\Rightarrow \quad \frac{du}{dx} = (\cos x)^x[-x\tan x + \log\cos x]$$

Again, $v = \sin^{-1}\sqrt{3x}$

$$\frac{dv}{dx} = \frac{1}{\sqrt{1-(\sqrt{3x})^2}}\frac{d}{dx}[(3x)^{1/2}]$$

$$\Rightarrow \quad \frac{dv}{dx} = \frac{1}{\sqrt{1-3x}}\frac{1}{2}(3x)^{-1/2}\times 3$$

$$\Rightarrow \quad \frac{dv}{dx} = \frac{1}{\sqrt{1-3x}}\times\frac{3}{2\sqrt{3x}}$$

$$\Rightarrow \quad \frac{dv}{dx} = \frac{3}{2\sqrt{3x}\sqrt{1-3x}} \Rightarrow \frac{dv}{dx} = \frac{\sqrt{3}}{2\sqrt{x-3x^2}}$$

On putting the values of $\frac{du}{dx}$ and $\frac{dv}{dx}$ in Eq. (i), we get

$$\frac{dy}{dx} = (\cos x)^x [-x \tan x + \log \cos x] + \frac{\sqrt{3}}{2\sqrt{x-3x^2}}$$

EXAMPLE 9. If $y = x^{x^x}$, then find $\frac{dy}{dx}$.

Sol. Given, $y = x^{x^x}$ ...(i)

Taking log on both sides, we get

$$\log y = \log x^{x^x} \quad \text{...(ii)}$$

$$\Rightarrow \quad \log y = x^x \log x \qquad [\because \log m^n = n \log m]$$

Again, taking log on both sides, we get

$$\log \log y = \log (x^x \log x)$$

$$\Rightarrow \quad \log \log y = \log x^x + \log \log x$$

$$[\because \log (mn) = \log m + \log n]$$

$$\Rightarrow \quad \log \log y = x \log x + \log \log x$$

On differentiating both sides w.r.t. x, we get

$$\frac{1}{\log y} \cdot \frac{d}{dx}(\log y) = \left[x \frac{d}{dx}(\log x) + \log x \frac{d}{dx}(x)\right] + \left[\frac{1}{\log x} \cdot \frac{d}{dx}(\log x)\right]$$

[by product rule and chain rule]

$$\Rightarrow \quad \frac{1}{\log y} \cdot \frac{1}{y} \cdot \frac{dy}{dx} = \left[x \cdot \frac{1}{x} + \log x \cdot 1\right] + \left[\frac{1}{\log x} \cdot \frac{1}{x}\right]$$

$$\Rightarrow \quad \frac{1}{y \log y} \cdot \frac{dy}{dx} = 1 + \log x + \frac{1}{x \log x}$$

$$\Rightarrow \quad \frac{dy}{dx} = y \log y \left[1 + \log x + \frac{1}{x \log x}\right]$$

$$= x^{x^x} \log x^{x^x} \left[1 + \log x + \frac{1}{x \log x}\right]$$

[from Eqs. (i) and (ii)]

Topic Practice 5

1 Differentiate the following function with respect to x.

(i) e^{-x} (ii) e^{4x}

(iii) e^{ax^2+b} (iv) $e^{\sqrt{3x}}$

(v) $e^{\sin x}$ (vi) $e^{\cos 3x}$

(vii) $e^{\sin^{-1} x}$ (viii) $x \cos x + e^x$

(ix) $xe^x \sin x$ (x) $\log (\tan x)$

(xi) $\log\left(\cos \frac{x}{2}\right)$ (xii) $\sin(\log x)$

→ (xiii) $\log\left(x + \frac{1}{x}\right)$ (xiv) $\log (x + \sqrt{1+x^2})$

(xv) $\frac{\log x}{x}$

(xvi) $\log(\log x), x > 1$ [NCERT]

(xvii) $\log[\log(\log x^5)]$

2 Differentiate each of the following functions with respect to x.

(i) x^x (ii) $\left(\frac{1}{x}\right)^x$

(iii) $x^{\sqrt{x}}$ (iv) $x^{\cos^{-1} x}$

(v) $x^{\tan^{-1} x}$ (vi) $(\tan x)^{\log x}$

(vii) $(\log x)^{\sin x}$ (viii) $(\cos x)^{\log x}$

(ix) $\sin(x^x)$ (x) $(\sin^{-1} x)^x$

(xi) $(\cos x)^{\cos x}$

Directions (Q. Nos. 3-10) *Find* $\frac{dy}{dx}$ *w.r.t.* x.

3 $y^x = x^y$ [NCERT]

4 $(\cos x)^y = (\cos y)^x$ [Delhi 2012]

5 $x^y + y^x = a^b$ [All India 2017]

6 $y = (x+3)^2 (x+4)^3 (x+5)^4$. [NCERT]

7 $y = \cos x \cos 2x \cos 3x$ [NCERT]

8 $y = (\log x)^x + x^{\log x}$. [Delhi 2013]

9 $y = x^{\sin x} + (\sin x)^{\cos x}$. [NCERT; All India 2016]

10 $y = x^{\cot x} + \frac{2x^2 - 3}{x^2 + x + 2}$. [All India 2012]

Directions (Q.Nos. 11-16) *Prove that*

11 If $\log \sqrt{x^2 + y^2} = \tan^{-1}\left(\frac{x}{y}\right)$, then $\frac{dy}{dx} = \frac{y-x}{y+x}$.

12 If $xy = e^{x-y}$, then prove that $\frac{dy}{dx} = \frac{y(x-1)}{x(y+1)}$ [CBSE 2023]

13 If $(x-y)\cdot e^{\frac{x}{x-y}} = a$, then $y\frac{dy}{dx} + x = 2y$. [Delhi 2014C]

14 If $x^p y^q = (x+y)^{p+q}$, then $\frac{dy}{dx} = \frac{y}{x}$ [Foreign 2014]

15 If $e^x + e^y = e^{x+y}$, then $\frac{dy}{dx} + \frac{e^x(e^y-1)}{e^y(e^x-1)} = 0$.

Or If $e^x + e^y = e^{x+y}$, then $\frac{dy}{dx} + e^{y-x} = 0$. [Foreign 2014]

16 If $y = \frac{ax^2}{(x-a)(x-b)(x-c)} + \frac{bx}{(x-b)(x-c)} + \frac{c}{(x-c)} + 1$, then $\frac{y'}{y} = \frac{1}{x}\left[\frac{a}{a-x} + \frac{b}{b-x} + \frac{c}{c-x}\right]$.

17 Find the derivative of the function given by $f(x) = (1+x)(1+x^2)(1+x^4)(1+x^8)$ and hence find $f'(1)$. [NCERT]

Hints & Solutions

1. (i) Similar as Example 1 (i). **Ans.** $-e^{-x}$

(ii) Similar as Example 1 (i). **Ans.** $4e^{4x}$

(iii) Similar as Example 1 (i). **Ans.** $2ax\,e^{ax^2+b}$

(iv) Similar as Example 1 (i). **Ans.** $\frac{3e^{\sqrt{3x}}}{2\sqrt{3x}}$

(v) Similar as Example 1 (i). **Ans.** $\cos x\, e^{\sin x}$

(vi) Similar as Example 1 (i). **Ans.** $-3\sin 3x\, e^{\cos 3x}$

(vii) Let $y = e^{\sin^{-1}x}$

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = e^{\sin^{-1}x}\frac{d}{dx}(\sin^{-1}x)$$

$$= e^{\sin^{-1}x} \times \frac{1}{\sqrt{1-x^2}} = \frac{e^{\sin^{-1}x}}{\sqrt{1-x^2}}$$

(viii) We have, $f(x) = x\cos x + e^x$

On differentiating both sides w.r.t. x, we get

$$f'(x) = \frac{d}{dx}(x\cos x) + \frac{d}{dx}(e^x)$$

$$\Rightarrow f'(x) = x\frac{d}{dx}(\cos x) + \cos x\frac{d}{dx}(x) + \frac{d}{dx}(e^x)$$

$$= -x\sin x + \cos x + e^x$$

(ix) **Hint** Use product rule of derivative.

Ans. $xe^x\cos x + (x+1)e^x\sin x$

(x) Similar as Example 2 (ii). **Ans.** $\operatorname{cosec} x \cdot \sec x$

(xi) Similar as Example 2 (ii). **Ans.** $-\frac{1}{2}\tan\frac{x}{2}$

(xii) $\frac{d}{dx}[\sin(\log x)] = \cos(\log x)\frac{d}{dx}(\log x) = \frac{\cos(\log x)}{x}$

(xiii) Similar as Example 2 (ii). **Ans.** $\frac{x^2-1}{x(x^2+1)}$

(xiv) Similar as Example 2 (ii). **Ans.** $\frac{1}{\sqrt{1+x^2}}$

(xv) We have, $y = \frac{\log x}{x}$

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = \frac{\left(x\frac{d}{dx}(\log x) - \log x\frac{d}{dx}(x)\right)}{x^2}$$

[by quotient rule of derivative]

$$= \frac{\left(x\times\frac{1}{x} - \log x\right)}{x^2} = \frac{1-\log x}{x^2}$$

(xvi) Similar as Example 2 (ii). **Ans.** $\frac{1}{x\log x}$

(xvii) Let $y = \log[\log(\log x^5)]$

On differentiating both sides, we get

$$\frac{dy}{dx} = \frac{1}{\log(\log x^5)} \times \frac{1}{\log x^5} \times \frac{1}{x^5} \times 5x^4$$

$$\Rightarrow \frac{dy}{dx} = \frac{5}{x[\log(\log x^5)]\cdot\log x^5}$$

2. (i) Similar as Example 4 (ii).

Ans. $x^x(1+\log x)$

(ii) Similar as Example 4 (ii). **Ans.** $-\left(\frac{1}{x}\right)^x(1+\log x)$

(iii) Similar as Example 4 (ii). **Ans.** $x^{\sqrt{x}}\left(\frac{2+\log x}{2\sqrt{x}}\right)$

(iv) Similar as Example 7 (i).

Ans. $x^{\cos^{-1}x}\left[\frac{\cos^{-1}x}{x} - \frac{\log x}{\sqrt{1-x^2}}\right]$

(v) Similar as Example 7 (i).

Ans. $x^{\tan^{-1}x}\left(\frac{\tan^{-1}x}{x}+\frac{\log x}{1+x^2}\right)$

(vi) Similar as Example 7 (ii).

Ans. $(\tan x)^{\log x}\left[\frac{\log x}{\sin x\cos x}+\frac{\log(\tan x)}{x}\right]$

(vii) Similar as Example 7 (ii).

Ans. $(\log x)^{\sin x}\left[\frac{\sin x}{x\log x}+\cos x\cdot\log(\log x)\right]$

(viii) Similar as Example 7(ii).

Ans. $(\cos x)^{\log x}\left[\frac{\log(\cos x)}{x}-(\tan x)(\log x)\right]$

(ix) Similar as Example 7 (ii). **Ans.** $x^x(1+\log x)\cdot\cos(x^x)$

(x) Similar as Example 7 (ii).

Ans. $(\sin^{-1}x)^x\left[\log\sin^{-1}x+\frac{x}{\sin^{-1}x}\cdot\frac{1}{\sqrt{1-x^2}}\right]$

(xi) Similar as Example 7 (ii).

Ans. $(\cos x)^{\cos x}\,[(-\sin x)(1+\log\cos x)]$

3. We have, $y^x = x^y$

Taking log on both sides, we get

$$\log y^x = \log x^y$$

$$\Rightarrow \quad x\log y = y\log x$$

On differentiating both sides w.r.t. x, we get

$$x\frac{d}{dx}(\log y)+\log y\frac{d}{dx}(x)=y\frac{d}{dx}(\log x)+\log x\frac{d}{dx}(y)$$

[by product rule of derivative]

$$\Rightarrow \quad x\frac{1}{y}\cdot\frac{dy}{dx}+\log y = y\times\frac{1}{x}+\log x\times\frac{dy}{dx}$$

$$\Rightarrow \quad \frac{x}{y}\frac{dy}{dx}-\log x\frac{dy}{dx}=\frac{y}{x}-\log y$$

$$\Rightarrow \quad \frac{dy}{dx}\left[\frac{x}{y}-\log x\right]=\frac{y}{x}-\log y$$

$$\Rightarrow \quad \frac{dy}{dx}\left[\frac{x-y\log x}{y}\right]=\left[\frac{y-x\log y}{x}\right]$$

$$\Rightarrow \quad \frac{dy}{dx}=\frac{y}{x}\left[\frac{y-x\log y}{x-y\log x}\right]$$

4. Given, $(\cos x)^y = (\cos y)^x$

Taking log on both sides, we get

$$\log(\cos x)^y = \log(\cos y)^x$$

$$\therefore \quad y\log(\cos x) = x\log(\cos y)$$

On differentiating both sides w.r.t. x, we get

$$y\cdot\frac{d}{dx}\log(\cos x)+\log(\cos x)\cdot\frac{d}{dx}(y)$$

$$=x\frac{d}{dx}\log(\cos y)+\log(\cos y)\frac{d}{dx}(x)$$

$$\left[\because \frac{d}{dx}(uv)=u\frac{dv}{dx}+v\frac{du}{dx}\right]$$

$$\Rightarrow \quad y\cdot\frac{1}{\cos x}\frac{d}{dx}(\cos x)+\log(\cos x)\frac{dy}{dx}$$

$$=x\cdot\frac{1}{\cos y}\frac{d}{dx}(\cos y)+\log(\cos y)\cdot 1$$

$$\Rightarrow \quad y\cdot\frac{1}{\cos x}(-\sin x)+\log(\cos x)\cdot\frac{dy}{dx}$$

$$=x\frac{1}{\cos y}(-\sin y)\cdot\frac{dy}{dx}+\log(\cos y)\cdot 1$$

$$\Rightarrow \quad -y\tan x+\log(\cos x)\frac{dy}{dx}=-x\tan y\frac{dy}{dx}+\log(\cos y)$$

$$\Rightarrow \quad [x\tan y+\log(\cos x)]\frac{dy}{dx}=\log(\cos y)+y\tan x$$

$$\therefore \quad \frac{dy}{dx}=\frac{\log(\cos y)+y\tan x}{x\tan y+\log(\cos x)}$$

5. Given, $x^y + y^x = a^b$ ...(i)

Let $x^y = v$ and $y^x = u$ ...(ii)

On putting these values in Eq. (i), we get

$$v+u=a^b$$

On differentiating both sides w.r.t. x, we get

$$\frac{dv}{dx}+\frac{du}{dx}=0 \qquad \text{...(iii)}$$

Now, $x^y = v$ [from Eq. (ii)]

Taking log on both sides, we get

$$\log x^y = \log v \quad\Rightarrow\quad y\log x = \log v$$

On differentiating both sides w.r.t. x, we get

$$y\cdot\frac{1}{x}+\log x\cdot\frac{dy}{dx}=\frac{1}{v}\frac{dv}{dx}$$

$$\Rightarrow \quad v\left(\frac{y}{x}+\log x\cdot\frac{dy}{dx}\right)=\frac{dv}{dx}$$

$$\Rightarrow \quad \frac{dv}{dx}=x^y\left(\frac{y}{x}+\log x\cdot\frac{dy}{dx}\right) \qquad \text{...(iv)}$$

[from Eq. (ii)]

and $y^x = u$ [from Eq. (ii)]

Taking log on both sides, we get

$$\log y^x = \log u$$

$$\Rightarrow \quad x\log y = \log u$$

On differentiating both sides w.r.t. x, we get

$$x\cdot\frac{1}{y}\frac{dy}{dx}+1\cdot\log y=\frac{1}{u}\frac{du}{dx}$$

$$\Rightarrow \quad \frac{x}{y}\frac{dy}{dx}+\log y=\frac{1}{u}\frac{du}{dx}$$

$$\Rightarrow \frac{dy}{dx}\left[\frac{qx+qy-py-qy}{y(x+y)}\right]=\frac{px+qx-px-py}{(x+y)x}$$

$$\Rightarrow \frac{dy}{dx}\left[\frac{qx-py}{y(x+y)}\right]=\frac{(qx-py)}{x(x+y)}$$

$$\Rightarrow \frac{dy}{dx}=\frac{(qx-py)}{(x+y)x}\cdot\frac{y(x+y)}{(qx-py)}\Rightarrow\frac{dy}{dx}=\frac{y}{x}$$

Hence proved.

15. We have, $e^x+e^y=e^{x+y}$.

On differentiating both sides w.r.t. x, we get

$$e^x+e^y\frac{dy}{dx}=e^{x+y}\frac{d}{dx}(x+y)$$

[by chain rule of derivative]

$$\Rightarrow e^x+e^y\frac{dy}{dx}=e^{x+y}\left[1+\frac{dy}{dx}\right]$$

$$\Rightarrow e^x+e^y\frac{dy}{dx}=e^{x+y}+e^{x+y}\frac{dy}{dx}$$

$$\Rightarrow e^y\frac{dy}{dx}-e^{x+y}\frac{dy}{dx}=e^{x+y}-e^x$$

$$\Rightarrow -e^y\frac{dy}{dx}(e^x-1)=e^x(e^y-1)$$

$$\Rightarrow \frac{dy}{dx}=\frac{-e^x(e^y-1)}{e^y(e^x-1)}$$

$$\Rightarrow \frac{dy}{dx}+\frac{e^x(e^y-1)}{e^y(e^x-1)}=0$$

Or $\frac{dy}{dx}=\frac{-e^{x+y}+e^x}{e^{x+y}-e^y}=\frac{-e^x-e^y+e^x}{e^x+e^y-e^y}$ [given]

$$\Rightarrow \frac{dy}{dx}=\frac{-e^y}{e^x}=-e^{y-x}$$

$$\Rightarrow \frac{dy}{dx}+e^{y-x}=0$$

Hence proved.

16. We have,

$$y=\frac{ax^2}{(x-a)(x-b)(x-c)}+\frac{bx}{(x-b)(x-c)}+\frac{c}{(x-c)}+1$$

$$=\frac{ax^2}{(x-a)(x-b)(x-c)}+\frac{bx}{(x-b)(x-c)}+\frac{c+x-c}{x-c}$$

$$=\frac{ax^2}{(x-a)(x-b)(x-c)}+\frac{bx}{(x-b)(x-c)}+\frac{x}{x-c}$$

$$=\frac{ax^2}{(x-a)(x-b)(x-c)}+\frac{bx+x(x-b)}{(x-b)(x-c)}$$

$$=\frac{ax^2}{(x-a)(x-b)(x-c)}+\frac{bx+x^2-xb}{(x-b)(x-c)}$$

$$=\frac{ax^2}{(x-a)(x-b)(x-c)}+\frac{x^2}{(x-b)(x-c)}$$

$$=\frac{ax^2+x^2(x-a)}{(x-a)(x-b)(x-c)}$$

$$=\frac{ax^2+x^3-ax^2}{(x-a)(x-b)(x-c)}$$

$$\Rightarrow y=\frac{x^3}{(x-a)(x-b)(x-c)}$$

Taking log on both sides, we get

$$\log y=\log x^3-[\log(x-a)+\log(x-b)+\log(x-c)]$$

$$\Rightarrow \log y=3\log x-\log(x-a)-\log(x-b)-\log(x-c)$$

On differentiating both sides w.r.t. x, we get

$$\frac{1}{y}\cdot\frac{dy}{dx}=3\times\frac{1}{x}-\frac{1}{x-a}-\frac{1}{x-b}-\frac{1}{x-c}$$

$$\Rightarrow \frac{y'}{y}=\frac{3}{x}-\frac{1}{x-a}-\frac{1}{x-b}-\frac{1}{x-c}$$

$$=\frac{1}{x}+\frac{1}{x}+\frac{1}{x}-\frac{1}{x-a}-\frac{1}{x-b}-\frac{1}{x-c}$$

$$=\left[\frac{1}{x}-\frac{1}{x-a}\right]+\left[\frac{1}{x}-\frac{1}{x-b}\right]+\left[\frac{1}{x}-\frac{1}{x-c}\right]$$

$$=\left[\frac{x-a-x}{x(x-a)}\right]+\left[\frac{x-b-x}{x(x-b)}\right]+\left[\frac{x-c-x}{x(x-c)}\right]$$

$$=-\frac{a}{x(x-a)}-\frac{b}{x(x-b)}-\frac{c}{x(x-c)}$$

$$\Rightarrow \frac{y'}{y}=\frac{1}{x}\left[\frac{a}{(a-x)}+\frac{b}{(b-x)}+\frac{c}{c-x}\right]$$

Hence proved.

17. Given, $f(x)=(1+x)(1+x^2)(1+x^4)(1+x^8)$

Taking log on both sides, we get

$$\log\{f(x)\}=\log\{(1+x)(1+x^2)(1+x^4)(1+x^8)\}$$

$$\Rightarrow \log\{f(x)\}=\log(1+x)+\log(1+x^2)+\log(1+x^4)+\log(1+x^8)$$

On differentiating both sides w.r.t. x, we get

$$\frac{d}{dx}\log\{f(x)\}=\frac{d}{dx}\log(1+x)+\frac{d}{dx}\log(1+x^2)+\frac{d}{dx}\log(1+x^4)+\frac{d}{dx}\log(1+x^8)$$

$$\frac{1}{f(x)}f'(x)=\frac{1}{1+x}(0+1)+\frac{1}{1+x^2}(0+2x)+\frac{1}{1+x^4}(0+4x^3)+\frac{1}{1+x^8}(0+8x^7)$$

$$\Rightarrow f'(x)=f(x)\left\{\frac{1}{1+x}+\frac{2x}{1+x^2}+\frac{4x^3}{1+x^4}+\frac{8x^7}{1+x^8}\right\}$$

$$=(1+x)(1+x^2)(1+x^4)(1+x^8)\left\{\frac{1}{1+x}+\frac{2x}{1+x^2}+\frac{4x^3}{1+x^4}+\frac{8x^7}{1+x^8}\right\}$$

$$\therefore f'(1)=(1+1)(1+1)(1+1)(1+1)\left\{\frac{1}{1+1}+\frac{2}{1+1}+\frac{4}{1+1}+\frac{8}{1+1}\right\}$$

$$=2^4\left\{\frac{1}{2}+1+2+4\right\}=16\left(\frac{15}{2}\right)=120$$

Differentiation of Parametric Functions

Sometimes, x and y are given as functions of one another variable, say $x = f(t)$, $y = g(t)$ are two functions and t is a variable. In such case, x and y are called **parametric functions** and t is called the **parameter.**

Let $x = f(t)$ and $y = g(t)$, where t is a parameter, then

$$\frac{dy}{dx} = \frac{\left(\frac{dy}{dt}\right)}{\left(\frac{dx}{dt}\right)}, \text{ whenever } \frac{dx}{dt} \neq 0$$

$\frac{dy}{dx}$ is expressed in terms of parameter only without directly involving the main variables x and y.

EXAMPLE 1. Find $\frac{dy}{dx}$, if $x = a \log t$ and $y = b \sin t$.

Sol. We have, $x = a \log t$ and $y = b \sin t$.

On differentiating both sides w.r.t. t, we get

$$\frac{dx}{dt} = \frac{a}{t} \text{ and } \frac{dy}{dt} = b \cos t$$

Now, $$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{b \cos t}{a/t} = \frac{bt \cos t}{a}$$

$\therefore$ $$\frac{dy}{dx} = \frac{b}{a} t \cos t$$

EXAMPLE 2. If $x = \cos\theta - \cos 2\theta$ and $y = \sin\theta - \sin 2\theta$, then find $\frac{dy}{dx}$. **[NCERT]**

Sol. We have, $x = \cos\theta - \cos 2\theta$

On differentiating both sides w.r.t. θ, we get

$$\frac{dx}{d\theta} = -\sin\theta + \sin 2\theta \cdot 2 \Rightarrow \frac{dx}{d\theta} = 2\sin 2\theta - \sin\theta$$

and $y = \sin\theta - \sin 2\theta$

On differentiating both sides w.r.t. θ, we get

$$\frac{dy}{d\theta} = \cos\theta - 2\cos 2\theta$$

Now, $$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{\cos\theta - 2\cos 2\theta}{2\sin 2\theta - \sin\theta}$$

EXAMPLE 3. If $x = a\left(\cos t + \log \tan \frac{t}{2}\right)$ and $y = a \sin t$, then find $\frac{dy}{dx}$ at $t = \frac{\pi}{4}$. **[NCERT]**

Sol. We have, $x = a\left[\cos t + \log\left(\tan \frac{t}{2}\right)\right]$

On differentiating both sides w.r.t. t, we get

$$\frac{dx}{dt} = a\left(-\sin t + \frac{1}{\tan t/2} \cdot \sec^2 \frac{t}{2} \cdot \frac{1}{2}\right) \left[\because \frac{d}{dx}\{\log(x)\} = \frac{1}{x}\right]$$

$$= a\left(-\sin t + \frac{1}{2 \sin t/2 \cos t/2}\right)$$

$$= a\left(-\sin t + \frac{1}{\sin t}\right) \left[\because \sin t = 2 \sin \frac{t}{2} \cos \frac{t}{2}\right]$$

$$= a\left(\frac{1 - \sin^2 t}{\sin t}\right)$$

$\Rightarrow$ $$\frac{dx}{dt} = \frac{a \cos^2 t}{\sin t} \quad [\because 1 - \sin^2 t = \cos^2 t] \ldots(i)$$

Also, $y = a \sin t$

On differentiating both sides w.r.t. t, we get

$$\frac{dy}{dt} = a \cos t \quad \ldots(ii)$$

Now, $$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{a \cos t}{\left(\frac{a \cos^2 t}{\sin t}\right)} \quad \text{[from Eqs. (i) and (ii)]}$$

$\therefore$ $$\frac{dy}{dx} = \frac{\sin t}{\cos t} = \tan t \text{ and } \left(\frac{dy}{dx}\right)_{t=\frac{\pi}{4}} = \tan \frac{\pi}{4} = 1$$

EXAMPLE 4. If $x = ae^t (\sin t + \cos t)$ and $y = ae^t (\sin t - \cos t)$, then prove that $\frac{dy}{dx} = \frac{x + y}{x - y}$. **[All India 2015C]**

Sol. Given, $x = ae^t (\sin t + \cos t)$

On differentiating x w.r.t. t, we get

$$\frac{dx}{dt} = ae^t \frac{d}{dt}(\sin t + \cos t) + \frac{d}{dt}(ae^t) \cdot (\sin t + \cos t)$$

$\Rightarrow$ $$\frac{dx}{dt} = ae^t (\cos t - \sin t) + ae^t (\sin t + \cos t)$$

$\Rightarrow$ $$\frac{dx}{dt} = -ae^t (\sin t - \cos t) + ae^t (\sin t + \cos t)$$

$\Rightarrow$ $$\frac{dx}{dt} = -y + x$$

Also, $y = ae^t (\sin t - \cos t)$

On differentiating y both sides w.r.t. t, we get

$$\frac{dy}{dt} = ae^t \frac{d}{dt}(\sin t - \cos t) + \frac{d}{dt}(ae^t) \cdot (\sin t - \cos t)$$

$\Rightarrow$ $$\frac{dy}{dt} = ae^t (\cos t + \sin t) + ae^t (\sin t - \cos t)$$

$\Rightarrow$ $$\frac{dy}{dt} = ae^t (\sin t + \cos t) + ae^t (\sin t - \cos t) = x + y$$

Now, $$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{x + y}{-y + x} = \frac{x + y}{x - y}$$ **Hence proved.**

EXAMPLE 5. If $x = a\left(t + \frac{1}{t}\right)$ and $y = a\left(t - \frac{1}{t}\right)$, then prove that $\frac{dy}{dx} = \frac{x}{y}$. [Delhi 2016C]

Sol. We have, $x = a\left(t + \frac{1}{t}\right)$ ...(i)

and $y = a\left(t - \frac{1}{t}\right)$ ...(ii)

Now, $\frac{dx}{dt} = a\left(1 - \frac{1}{t^2}\right) = a\left(\frac{t^2 - 1}{t^2}\right)$ ...(iii)

and $\frac{dy}{dt} = a\left(1 + \frac{1}{t^2}\right) = a\left(\frac{t^2 + 1}{t^2}\right)$ ...(iv)

$\therefore \quad \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{a\left(\frac{t^2+1}{t^2}\right)}{a\left(\frac{t^2-1}{t^2}\right)}$ [using Eqs. (iii) and (iv)]

$= \frac{a(t^2+1)}{t^2} \cdot \frac{t^2}{a(t^2-1)} = \frac{t^2+1}{t^2-1}$ ...(v)

Now, consider $\frac{x}{y} = \frac{a\left(\frac{t^2+1}{t}\right)}{a\left(\frac{t^2-1}{t}\right)} = \frac{t^2+1}{t^2-1}$ ...(vi)

From Eqs. (v) and (vi), we get

$\frac{dy}{dx} = \frac{x}{y}$ **Hence proved.**

EXAMPLE 6. If $x = \sqrt{a^{\sin^{-1}t}}$, $y = \sqrt{a^{\cos^{-1}t}}$, then show that $\frac{dy}{dx} = -\frac{y}{x}$. [NCERT]

Sol. Given, $x = \sqrt{a^{\sin^{-1}t}}$

Taking log on both sides, we get

$\log x = \log(a^{\sin^{-1}t})^{1/2} \Rightarrow \log x = \frac{1}{2}\log a^{\sin^{-1}t}$

$\Rightarrow \log x = \frac{1}{2}\sin^{-1}t \log a \Rightarrow \log x = \frac{1}{2}\log a \sin^{-1}t$

On differentiating both sides w.r.t. t, we get

$\frac{1}{x}\cdot\frac{dx}{dt} = \frac{1}{2}\log a \frac{d}{dt}(\sin^{-1}t) = \frac{1}{2}\log a \frac{1}{\sqrt{1-t^2}}$

$\Rightarrow \quad \frac{dx}{dt} = \frac{x \log a}{2} \cdot \frac{1}{\sqrt{1-t^2}}$

Again, $y = \sqrt{a^{\cos^{-1}t}}$

Taking log on both sides, we get

$\log y = \log(a^{\cos^{-1}t})^{1/2}$

$\Rightarrow \quad \log y = \frac{1}{2}\log a^{\cos^{-1}t}$

$\Rightarrow \quad \log y = \frac{1}{2}\cos^{-1}t \log a$

On differentiating both sides w.r.t. t, we get

$\frac{1}{y}\cdot\frac{dy}{dt} = \frac{1}{2}\log a \cdot \left[\frac{-1}{\sqrt{1-t^2}}\right]$

$\Rightarrow \quad \frac{dy}{dt} = \frac{y \log a}{2} \cdot \frac{-1}{\sqrt{1-t^2}}$

Now, $\frac{dy}{dx} = \left(\frac{dy}{dt}\right)\cdot\left(\frac{dt}{dx}\right)$

$\Rightarrow \quad \frac{dy}{dx} = \frac{-y \log a}{2} \cdot \frac{1}{\sqrt{1-t^2}} \cdot \frac{2\cdot\sqrt{1-t^2}}{x \log a} \Rightarrow \frac{dy}{dx} = \frac{-y}{x}$

Hence proved.

Topic Practice 6

Directions (Q. Nos. 1-11) *Find* $\frac{dy}{dx}$.

1 $x = at^2$ and $y = 2at$ [NCERT]

2 $x = a\cos\theta$ and $y = a\sin\theta$

3 $x = a(\theta - \sin\theta)$ and $y = a(1 + \cos\theta)$. [NCERT]

4 $x = \frac{1-t^2}{1+t^2}$ and $y = \frac{2t}{1+t^2}$.

5 $x = e^{\theta}\left(\theta + \frac{1}{\theta}\right)$, $y = e^{-\theta}\left(\theta - \frac{1}{\theta}\right)$. [NCERT Exemplar]

6 $x = \frac{\sin^3 t}{\sqrt{\cos 2t}}$ and $y = \frac{\cos^3 t}{\sqrt{\cos 2t}}$. [NCERT]

7 $x = a(2\theta - \sin 2\theta)$ and $y = a(1 - \cos 2\theta)$, when $\theta = \frac{\pi}{3}$. [CBSE 2018]

8 If $x = a\sec^3\theta$ and $y = a\tan^3\theta$ at $\theta = \frac{\pi}{3}$. [NCERT Exemplar]

9 $x = ae^{\theta}(\sin\theta - \cos\theta)$ and $y = ae^{\theta}(\sin\theta + \cos\theta)$ at $\theta = \frac{\pi}{4}$. [All India 2014]

10 If $x = \cos t(3 - 2\cos^2 t)$ and $y = \sin t(3 - 2\sin^2 t)$, at $x = \frac{\pi}{4}$. [All India 2014]

11 $x = a\sin 2t(1+\cos 2t)$ and $y = b\cos 2t(1-\cos 2t)$, at $t = \frac{\pi}{4}$ and $t = \frac{\pi}{3}$. [Delhi 2016]

12 If $x = 2\cos\theta - \cos 2\theta$ and $y = 2\sin\theta - \sin 2\theta$, then prove that $\frac{dy}{dx} = \tan\left(\frac{3\theta}{2}\right)$. [All India 2016C, Delhi 2013C]

13 If $x = e^{\cos 2t}$ and $y = e^{\sin 2t}$, then prove that $\frac{dy}{dx} = \frac{-y\log x}{x\log y}$.

14 For $a > 0$, if $x = \left(t + \frac{1}{t}\right)^a$ and $y = a^{t+\frac{1}{t}}$, then prove that $\frac{dy}{dx} = \frac{a^{\left(t+\frac{1}{t}\right)}\log a}{\left(t+\frac{1}{t}\right)^{a-1}}$.

Hints & Solutions

1. Similar as Example 1. **Ans.** $\frac{1}{t}$

2. **Hint** $\frac{dx}{d\theta} = -a\sin\theta$, $\frac{dy}{d\theta} = a\cos\theta$ **Ans.** $\frac{dy}{dx} = -\cot\theta$

3. We have, $x = a(\theta - \sin\theta)$ and $y = a(1+\cos\theta)$
On differentiating x and y w.r.t. θ, we get
$$\frac{dx}{d\theta} = a(1-\cos\theta)$$
and $$\frac{dy}{d\theta} = a(0-\sin\theta) = -a\sin\theta$$
$$\text{Now, } \frac{dy}{dx} = \left(\frac{dy}{d\theta}\right)\times\left(\frac{d\theta}{dx}\right) = \frac{-a\sin\theta}{a(1-\cos\theta)} = \frac{\left(-2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\right)}{2\sin^2\frac{\theta}{2}}$$
$$\left[\because \sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2} \text{ and } 1-\cos\theta = 2\sin^2\frac{\theta}{2}\right]$$
$$= -\cot\left(\frac{\theta}{2}\right)$$

4. Similar as Example 5. **Ans.** $\frac{t^2-1}{2t}$

5. Similar as Example 5. **Ans.** $e^{-2\theta}\left(\frac{-\theta^3+\theta^2+\theta+1}{\theta^3+\theta^2+\theta-1}\right)$

6. Given, $x = \frac{\sin^3 t}{\sqrt{\cos 2t}}$, $y = \frac{\cos^3 t}{\sqrt{\cos 2t}}$.
On differentiating x and y w.r.t. t, we get
$$\frac{dx}{dt} = \frac{d}{dt}\left(\frac{\sin^3 t}{\sqrt{\cos 2t}}\right)$$
$$= \frac{\sqrt{\cos 2t}(3\sin^2 t\cos t) - \sin^3 t\left(\frac{-2\sin 2t}{2\sqrt{\cos 2t}}\right)}{(\sqrt{\cos 2t})^2}$$
[using quotient rule of derivative]
$$= \frac{3(\cos 2t)\sin^2 t\cos t + \sin 2t\sin^3 t}{\cos 2t\sqrt{\cos 2t}}$$
$$= \frac{3(1-2\sin^2 t)\sin^2 t\cos t + (2\sin t\cos t)\sin^3 t}{\cos 2t\sqrt{\cos 2t}}$$
$$[\because \cos 2t = 1-2\sin^2 t \text{ and } \sin 2t = 2\sin t\cos t]$$
$$= \frac{3\sin^2 t\cos t - 4\sin^4 t\cos t}{\cos 2t\sqrt{\cos 2t}}$$
and $$\frac{dy}{dt} = \frac{d}{dt}\left(\frac{\cos^3 t}{\sqrt{\cos 2t}}\right)$$
$$= \frac{\sqrt{\cos 2t}(-3\cos^2 t\sin t) - \cos^3 t\left(\frac{-2\sin 2t}{2\sqrt{\cos 2t}}\right)}{(\sqrt{\cos 2t})^2}$$
[using quotient rule of derivative]
$$= \frac{-3(\cos 2t)\cos^2 t\sin t + \sin 2t\cos^3 t}{\cos 2t\sqrt{\cos 2t}}$$
$$= \frac{-3(2\cos^2 t-1)\cos^2 t\sin t + \cos^3 t(2\sin t\cos t)}{\cos 2t\sqrt{\cos 2t}}$$
$$= \frac{3\cos^2 t\sin t - 4\cos^4 t\sin t}{\cos 2t\sqrt{\cos 2t}}$$
$$\text{Now, } \frac{dy}{dx} = \frac{dy}{dt}\times\frac{dt}{dx} = \frac{3\cos^2 t\sin t - 4\cos^4 t\sin t}{3\sin^2 t\cos t - 4\sin^4 t\cos t}$$
$$= \frac{\cos^2 t\sin t(3-4\cos^2 t)}{\sin^2 t\cos t(3-4\sin^2 t)} = \frac{\cos t(3-4\cos^2 t)}{\sin t(3-4\sin^2 t)}$$
$$= \frac{3\cos t - 4\cos^3 t}{3\sin t - 4\sin^3 t} = \frac{-\cos 3t}{\sin 3t} = -\cot 3t$$
$$[\because \cos 3t = 4\cos^3 t - 3\cos t \text{ and } \sin 3t = \sin t - 4\sin^3 t]$$

7. We have, $x = a(2\theta - \sin 2\theta)$ and $y = a(1-\cos 2\theta)$
Clearly, $$\frac{dy}{dx} = \frac{(dy/d\theta)}{(dx/d\theta)} \quad \ldots(i)$$
Here, $$\frac{dx}{d\theta} = a(2-\cos 2\theta\cdot 2) = 2a(1-\cos 2\theta)$$
and $$\frac{dy}{d\theta} = a(0+2\sin 2\theta) = 2a\sin 2\theta$$
From Eq. (i), we get
$$\frac{dy}{dx} = \frac{2a\sin 2\theta}{2a(1-\cos 2\theta)} = \frac{\sin 2\theta}{1-\cos 2\theta} = \frac{2\sin\theta\cos\theta}{2\sin^2\theta}$$
$$[\because \cos 2\theta = 1-2\sin^2\theta]$$
$$= \cot\theta$$
$$\text{Now, } \left(\frac{dy}{dx}\right)_{\theta=\pi/3} = \cot\frac{\pi}{3} = \frac{1}{\sqrt{3}}$$

8. Similar as Question 7. Ans. $\frac{\sqrt{3}}{2}$

9. We have, $x = ae^{\theta}(\sin\theta - \cos\theta) = a(e^{\theta}\sin\theta - e^{\theta}\cos\theta)$

On differentiating both sides w.r.t. θ, we get

$$\frac{dx}{d\theta} = a\frac{d}{d\theta}[e^{\theta}\sin\theta - e^{\theta}\cos\theta]$$

$$= a\left[\frac{d}{d\theta}(e^{\theta}\sin\theta) - \frac{d}{d\theta}(e^{\theta}\cos\theta)\right]$$

$$= a\left[e^{\theta}\frac{d}{d\theta}(\sin\theta) + \sin\theta\frac{d}{d\theta}(e^{\theta}) - e^{\theta}\frac{d}{d\theta}(\cos\theta) - \cos\theta\frac{d}{d\theta}(e^{\theta})\right]$$

$$= a[e^{\theta}\cos\theta + e^{\theta}\sin\theta - e^{\theta}(-\sin\theta) - e^{\theta}\cos\theta]$$

$$= a[e^{\theta}\cos\theta + e^{\theta}\sin\theta + e^{\theta}\sin\theta - e^{\theta}\cos\theta]$$

$$= a[2e^{\theta}\sin\theta] = 2ae^{\theta}\sin\theta \quad ...(i)$$

and $y = ae^{\theta}(\sin\theta + \cos\theta) = a(e^{\theta}\sin\theta + e^{\theta}\cos\theta)$

On differentiating both sides w.r.t. θ, we get

$$\frac{dy}{d\theta} = a\left[\frac{d}{d\theta}(e^{\theta}\sin\theta) + \frac{d}{d\theta}(e^{\theta}\cos\theta)\right]$$

$$= a\left[e^{\theta}\frac{d}{d\theta}(\sin\theta) + \sin\theta\frac{d}{d\theta}(e^{\theta}) + e^{\theta}\frac{d}{d\theta}(\cos\theta) + \cos\theta\frac{d}{d\theta}(e^{\theta})\right]$$

$$= a[e^{\theta}\cos\theta + e^{\theta}\sin\theta + e^{\theta}(-\sin\theta) + e^{\theta}\cos\theta]$$

$$= a[e^{\theta}\cos\theta + e^{\theta}\sin\theta - e^{\theta}\sin\theta + e^{\theta}\cos\theta]$$

$$\Rightarrow \frac{dy}{d\theta} = 2ae^{\theta}\cos\theta \quad ...(ii)$$

Now, $\frac{dy}{dx} = \frac{dy}{d\theta}\cdot\frac{d\theta}{dx} = \frac{2ae^{\theta}\cos\theta}{2ae^{\theta}\sin\theta}$ [from Eqs. (i) and (ii)]

$$= \cot\theta$$

At $\theta = \frac{\pi}{4}$, $\frac{dy}{dx} = \cot\frac{\pi}{4} \Rightarrow \frac{dy}{dx} = 1$ $\left[\because \cot\frac{\pi}{4} = 1\right]$

10. Similar as Question 7. Ans. $\frac{dy}{dx} = 1$

11. Given, $x = a\sin 2t(1 + \cos 2t)$
and $y = b\cos 2t(1 - \cos 2t)$

On differentiating both sides w.r.t. t, we get

$$\frac{dx}{dt} = a\left[\sin 2t\frac{d}{dt}(1 + \cos 2t) + (1 + \cos 2t)\frac{d}{dt}(\sin 2t)\right]$$

[by using product rule of derivative]

$$= a[\sin 2t\cdot(0 - 2\sin 2t) + (1 + \cos 2t)(2\cos 2t)]$$

$$= a[-2\sin^2 2t + 2\cos 2t + 2\cos^2 2t]$$

$$= a[2(\cos^2 2t - \sin^2 2t) + 2\cos 2t]$$

$$= a(2\cos 4t + 2\cos 2t)$$

$$= 2a(\cos 4t + \cos 2t) \quad [\because \cos^2 2\theta - \sin^2 2\theta = \cos 4\theta]$$

and $\frac{dy}{dt} = b\left[\cos 2t\frac{d}{dt}(1 - \cos 2t) + (1 - \cos 2t)\frac{d}{dt}(\cos 2t)\right]$

[by using product rule of derivative]

$$= b[\cos 2t\times(0 + 2\sin 2t) + (1 - \cos 2t)(-2\sin 2t)]$$

$$= b(2\sin 2t\cos 2t - 2\sin 2t + 2\sin 2t\cos 2t)$$

$$= b(4\sin 2t\cos 2t - 2\sin 2t)$$

$$= b(2\sin 4t - 2\sin 2t) \quad [\because 2\sin 2\theta\cos 2\theta = \sin 4\theta]$$

$$= 2b[\sin 4t - \sin 2t]$$

Now, $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2b[\sin 4t - \sin 2t]}{2a[\cos 4t + \cos 2t]}$

$$= \frac{b(\sin 4t - \sin 2t)}{a(\cos 4t + \cos 2t)}$$

At $t = \frac{\pi}{4}$, $\frac{dy}{dx} = \frac{b\left(\sin 4\times\frac{\pi}{4} - \sin 2\times\frac{\pi}{4}\right)}{a\left(\cos 4\times\frac{\pi}{4} + \cos 2\times\frac{\pi}{4}\right)}$

$$= \frac{b\left(\sin\pi - \sin\frac{\pi}{2}\right)}{a\left(\cos\pi + \cos\frac{\pi}{2}\right)}$$

$$= \frac{b(0-1)}{a(-1+0)} = \frac{b}{a}$$

At $t = \frac{\pi}{3}$, $\frac{dy}{dx} = \frac{b\left(\sin 4\times\frac{\pi}{3} - \sin 2\times\frac{\pi}{3}\right)}{a\left(\cos 4\times\frac{\pi}{3} + \cos 2\times\frac{\pi}{3}\right)}$

$$= \frac{b\left[\sin\left(\pi + \frac{\pi}{3}\right) - \sin\left(\pi - \frac{\pi}{3}\right)\right]}{a\left[\cos\left(\pi + \frac{\pi}{3}\right) + \cos\left(\pi - \frac{\pi}{3}\right)\right]}$$

$$= \frac{b\left(-\sin\frac{\pi}{3} - \sin\frac{\pi}{3}\right)}{a\left(-\cos\frac{\pi}{3} - \cos\frac{\pi}{3}\right)}$$

$$= \frac{b\left(-2\sin\frac{\pi}{3}\right)}{-a\times 2\cos\frac{\pi}{3}} = \frac{b}{a}\frac{\left(\frac{\sqrt{3}}{2}\right)}{\left(\frac{1}{2}\right)} = \frac{b\sqrt{3}}{a}$$

12. Similar as Example 2.

13. **Hint** $\frac{dx}{dt} = e^{\cos 2t}(-\sin 2t)(2)$

$$\frac{dy}{dt} = e^{\sin 2t}(\cos 2t)(2)$$

$\because$ $x = e^{\cos 2t} \Rightarrow \log x = \cos 2t$

$y = e^{\sin 2t} \Rightarrow \log y = \sin 2t$

$\therefore$ $\frac{dx}{dt} = -2x\log y, \frac{dy}{dt} = 2y\log x$

14. **Hint** $\frac{dx}{dt} = a\left(t + \frac{1}{t}\right)^{a-1}\left(1 - \frac{1}{t^2}\right)$

and $\frac{dy}{dt} = a^{\left(t + \frac{1}{t}\right)}\log a\left(1 - \frac{1}{t^2}\right)$

Differentiation of a Function w.r.t. Another Function

Suppose it is required to differentiate a function $f(x)$ with respect to another function $g(x)$.

Let $y = f(x)$ and $z = g(x)$.

Here, both functions are different but both are in same variable x. Then, to find the derivative of $f(x)$ with respect to $g(x)$ or derivative of $g(x)$ with respect to $f(x)$ i.e. to find $\frac{dy}{dz}$ or $\frac{dz}{dy}$.

First differentiate both functions $f(x)$ and $g(x)$ with respect to x separately and then put these values in the following formulae

$$\frac{dy}{dz} = \frac{dy/dx}{dz/dx} \text{ or } \frac{dz}{dy} = \frac{dz/dx}{dy/dx}$$

EXAMPLE 1. Differentiate x^6 w.r.t. $\frac{1}{\sqrt{x}}$.

Sol. Let $f = x^6$ and $g = \frac{1}{\sqrt{x}} = x^{-1/2}$

$$\Rightarrow \frac{df}{dx} = 6x^5 \text{ and } \frac{dg}{dx} = -\frac{1}{2}x^{-\frac{1}{2}-1} = -\frac{1}{2}x^{-\frac{3}{2}}$$

$$\because \frac{df}{dg} = \frac{df/dx}{dg/dx} = \frac{6x^5}{-\frac{1}{2}x^{-\frac{3}{2}}} = -12\,x^{5+\frac{3}{2}} = -12\,x^{\frac{13}{2}}$$

EXAMPLE 2. Write the derivative of $\sin x$ w.r.t. $\cos x$. [Delhi 2014C]

Sol. Let $y = \sin x$ and $z = \cos x$

On differentiating y and z w.r.t. x, we get

$$\frac{dy}{dx} = \cos x \text{ and } \frac{dz}{dx} = -\sin x$$

Now, $$\frac{dy}{dz} = \frac{dy/dx}{dz/dx} = \frac{\cos x}{-\sin x} = -\tan x$$

EXAMPLE 3. Differentiate $\sin^2 x$ w.r.t. $e^{\cos x}$. [NCERT]

Sol. Let $y = \sin^2 x$ and $z = e^{\cos x}$.

On differentiating y and z w.r.t. x, we get

$$\frac{dy}{dx} = 2\sin x \cos x$$

and $$\frac{dz}{dx} = e^{\cos x}(-\sin x) = -(\sin x)\,e^{\cos x}$$

Now, $$\frac{dy}{dz} = \frac{dy/dx}{dz/dx} = \frac{2\sin x \cos x}{-(\sin x)\,e^{\cos x}} = \frac{-2\cos x}{e^{\cos x}}$$

EXAMPLE 4. Find derivative of $\sin^{-1} x$ w.r.t. $\tan^{-1} x$.

Sol. Let $f(x) = \sin^{-1} x$ and $g(x) = \tan^{-1} x$

On differentiating $f(x)$ and $g(x)$ w.r.t. x, we get

$$\frac{df}{dx} = \frac{1}{\sqrt{1-x^2}} \text{ and } \frac{dg}{dx} = \frac{1}{1+x^2}$$

Now, $$\frac{df}{dg} = \frac{df}{dx} \times \frac{dx}{dg} = \frac{1}{\sqrt{1-x^2}} \times (1+x^2) = \frac{1+x^2}{\sqrt{1-x^2}}$$

EXAMPLE 5. Differentiate $\tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$ w.r.t. $\cos^{-1}(2x\sqrt{1-x^2})$, when $x \neq 0$. [Delhi 2014]

Sol. Let $u = \tan^{-1}\left[\frac{\sqrt{1-x^2}}{x}\right]$. Put $x = \cos\theta \Rightarrow \theta = \cos^{-1} x$

$$\therefore u = \tan^{-1}\left[\frac{\sqrt{1-\cos^2\theta}}{\cos\theta}\right] = \tan^{-1}\left[\frac{\sqrt{\sin^2\theta}}{\cos\theta}\right]$$

$$= \tan^{-1}\left[\frac{\sin\theta}{\cos\theta}\right] = \tan^{-1}[\tan\theta] = \theta$$

$$\Rightarrow u = \cos^{-1} x$$

On differentiating both sides w.r.t. x, we get

$$\frac{du}{dx} = -\frac{1}{\sqrt{1-x^2}}$$

Again, let $v = \cos^{-1}(2x\sqrt{1-x^2})$

Put $x = \cos\theta \Rightarrow \theta = \cos^{-1} x$

$$\therefore v = \cos^{-1}[2\cos\theta\sqrt{1-\cos^2\theta}]$$

$$= \cos^{-1}[2\cos\theta\sin\theta]$$

$$[\because \cos^2\theta + \sin^2\theta = 1 \Rightarrow \sin\theta = \sqrt{1-\cos^2\theta}]$$

$$= \cos^{-1}[\sin 2\theta]$$

$$= \cos^{-1}\left[\cos\left(\frac{\pi}{2} - 2\theta\right)\right] \quad \left[\because \sin\theta = \cos\left(\frac{\pi}{2} - \theta\right)\right]$$

$$= \frac{\pi}{2} - 2\theta = \frac{\pi}{2} - 2\cos^{-1} x$$

On differentiating both sides, w.r.t. x, we get

$$\frac{dv}{dx} = \frac{2}{\sqrt{1-x^2}}$$

Now, $$\frac{du}{dv} = \left(\frac{du}{dx}\right) \times \left(\frac{dx}{dv}\right) = -\frac{1}{\sqrt{1-x^2}} \times \frac{\sqrt{1-x^2}}{2} = -\frac{1}{2}$$

EXAMPLE 6. Find the derivative of $x^{\log x}$ w.r.t. $\log x$. [CBSE 2023, 20]

Sol. Let $y = x^{\log x}$

Taking log on both sides, we get

$$\log y = \log x \cdot \log x = (\log x)^2$$

On differentiating both sides w.r.t. x, we get

$$\frac{1}{y}\frac{dy}{dx} = 2\log x \frac{d}{dx}(\log x)$$

$$\frac{1}{y}\frac{dy}{dx} = 2\log x\left(\frac{1}{x}\right)$$

$$\Rightarrow \quad \frac{dy}{dx} = y\left(\frac{2\log x}{x}\right)$$

$$\Rightarrow \quad \frac{dy}{dx} = x^{\log x}\left(\frac{2\log x}{x}\right)$$

$$= x^{\log x - 1}(2\log x)$$

Again, let $z = \log x$

On differentiating both sides w.r.t. x, we get

$$\frac{dz}{dx} = \frac{1}{x}$$

Now, $$\frac{dy}{dz} = \frac{\frac{dy}{dx}}{\frac{dz}{dx}} = \frac{x^{\log x - 1}(2\log x)}{\frac{1}{x}}$$

$$= x^{\log x}(2\log x)$$

Differentiation of Infinite Series

When the value of y is given as an infinite series, then the process to find the derivative of such infinite series is called **differentiation of infinite series.**

In such cases, we use the fact that, if one term is deleted from an infinite series, it remains unaffected, so we can replace all terms except first term by y. Thus, we convert it into a finite series or function. Then, we differentiate it to find the required value.

EXAMPLE 7. If $y = x^{x^{x^{x^{\cdots}}}}$, then find $\frac{dy}{dx}$.

Sol. We have, $y = x^{x^{x^{x^{\cdots}}}}$

Then, $y = x^y$

[we know that by deleting a single term from an infinite series, it remains unaffected.]

$$\Rightarrow \quad y = e^{y\log x} \qquad [\because e^{\log_e x} = x]$$

$$\Rightarrow \quad \frac{dy}{dx} = e^{y\log x}\frac{d}{dx}(y\log x)$$

$$\Rightarrow \quad \frac{dy}{dx} = x^y\left(\frac{dy}{dx}\log x + \frac{y}{x}\right)$$

$$\Rightarrow \quad \frac{dy}{dx} = y\left(\frac{dy}{dx}\log x + \frac{y}{x}\right) \quad [\because y = x^y]$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{y^2}{x(1 - y\log x)}$$

EXAMPLE 8. If $y = \sqrt{\log x + \sqrt{\log x + \sqrt{\log x + \ldots + \infty}}}$, then find $\frac{dy}{dx}$.

Sol. We have, $y = \sqrt{\log x + \sqrt{\log x + \sqrt{\log x + \ldots + \infty}}}$

Then, $y = \sqrt{\log x + y}$

$$\Rightarrow \quad y^2 = \log x + y \qquad \text{[squaring both sides]}$$

On differentiating both sides w.r.t. x, we get

$$2y\cdot\frac{dy}{dx} = \frac{1}{x} + \frac{dy}{dx}$$

$$\Rightarrow 2y\frac{dy}{dx} - \frac{dy}{dx} = \frac{1}{x}$$

$$\Rightarrow \frac{dy}{dx}(2y - 1) = \frac{1}{x}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{1}{x(2y - 1)}$$

EXAMPLE 9. If $y = (\sin x)^{(\sin x)^{(\sin x)\ldots\infty}}$, then prove that $\frac{dy}{dx} = \frac{y^2\cos x}{1 - y\log\sin x}$.

Sol. We have, $y = (\sin x)^{(\sin x)^{(\sin x)\ldots\infty}}$.

Then, $y = (\sin x)^y$

Taking log on both sides, we get

$$\log y = y\log\sin x$$

On differentiating both sides w.r.t. x, we get

$$\frac{1}{y}\frac{dy}{dx} = y\frac{d}{dx}(\log\sin x) + \log\sin x\frac{d}{dx}(y)$$

$$\Rightarrow \quad \frac{1}{y}\frac{dy}{dx} = y\cdot\frac{1}{\sin x}\cdot\cos x + \log\sin x\cdot\frac{dy}{dx}$$

$$\Rightarrow \quad \frac{dy}{dx}\left(\frac{1}{y} - \log\sin x\right) = y\cot x$$

$$\Rightarrow \quad \frac{dy}{dx}\left(\frac{1 - y\log\sin x}{y}\right) = y\cot x$$

$$\therefore \quad \frac{dy}{dx} = \frac{y^2\cot x}{1 - y\log\sin x}$$

Hence proved.

Topic Practice 7

1 Differentiate x^4 with respect to x^3.

2 Differentiate $\tan x$ with respect to $\sin x$.

3 Differentiate $\sin x^2$ w.r.t. x^2.

4 Differentiate $b \tan\theta$ w.r.t. $a\sec\theta$.

5 Differentiate $\sin x^2$ w.r.t. x^2.

6 Differentiate $\tan e^x$ w.r.t. e^x.

7 Differentiate $\cos^{-1}(2x^2 - 1)$ w.r.t. $\cos^{-1} x$.

8 Find the derivative of $\tan^{-1}\left[\frac{\sqrt{1+x^2}-1}{x}\right]$ w.r.t. $\tan^{-1}\frac{2x}{1-x^2}$.

9 Find the derivative of $\sec^{-1}\left(\frac{1}{2x^2-1}\right)$ w.r.t. $\sqrt{1-x^2}$ at $x = \frac{1}{2}$.

10 Differentiate $\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$ w.r.t. $\tan^{-1} x$, when $x \neq 0$. **[NCERT Exemplar]**

11 Differentiate $\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$ w.r.t. $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$, when $x \neq 0$. **[Delhi 2016, 2014]**

12 Differentiate $\cos^{-1}(4x^3 - 3x)$ w.r.t. $\tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$, if $\frac{1}{2} < x < 1$.

13 Differentiate $\sin^{-1}(2ax\sqrt{1-a^2x^2})$ w.r.t. $\sqrt{1-a^2x^2}$, if $-\frac{1}{\sqrt{2}} < ax < \frac{1}{\sqrt{2}}$.

14 If $y = \sqrt{x + \sqrt{x + \sqrt{x + \ldots \text{ to } \infty}}}$, then prove that $\frac{dy}{dx} = \frac{1}{2y-1}$.

15 If $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \ldots + \infty}}}$, then prove that $\frac{dy}{dx} = \frac{\cos x}{2y-1}$.

16 If $y = (\cos x)^{(\cos x)^{(\cos x)\ldots\infty}}$, then prove that $\frac{dy}{dx} = \frac{y^2 \tan x}{(y \log\cos x - 1)}$.

17 If $y = e^{x + e^{x + e^{x + e^{x + \ldots\infty}}}}$, then prove that $\frac{dy}{dx} = \frac{y}{1-y}$.

18 If $y = x + \cfrac{1}{x + \cfrac{1}{x + \cfrac{1}{x + \ldots + \infty}}}$, then prove that $\frac{dy}{dx} = \frac{y}{2y-x}$.

19 If $y = a^{x^{x^{x^{\cdots}}}}$, then prove that $\frac{dy}{dx} = \frac{y^2(\log y)}{x[1 - y(\log x)(\log y)]}$.

Hints & Solutions

1. Similar as Example 1. **Ans.** $\frac{4}{3}x$

2. Similar as Example 2. **Ans.** $\sec^3 x$

3. Similar as Example 2. **Ans.** $\cos x^2$

4. Let $x = b\tan\theta$ and $y = a\sec\theta$

   On differentiating w.r.t. θ, we get

   $$\frac{dx}{d\theta} = b\sec^2\theta \text{ and } \frac{dy}{d\theta} = a\sec\theta\tan\theta$$

   $$\therefore \frac{dx}{dy} = \frac{dx/d\theta}{dy/d\theta} = \frac{b\sec^2\theta}{a\sec\theta\tan\theta} = \frac{b}{a}\left(\frac{\sec\theta}{\tan\theta}\right) = \frac{b}{a}\operatorname{cosec}\theta$$

5. Let $u = \sin x^2$ and $v = x^2$

   On differentiating u and v w.r.t. x, we get

   $$\frac{du}{dx} = \cos x^2 \cdot \frac{d}{dx}(x^2) = 2x \cdot \cos x^2 \text{ and } \frac{dv}{dx} = 2x$$

   Now, $\frac{du}{dv} = \frac{\left(\frac{du}{dx}\right)}{\left(\frac{dv}{dx}\right)} = \left(\frac{du}{dx}\right) \times \left(\frac{dx}{dv}\right) = \frac{\cos x^2 \times 2x}{2x}$

   $\therefore \frac{du}{dv} = \cos x^2$

6. Similar as Question 5. **Ans.** $\sec^2 e^x$

7. Let $u = \cos^{-1}(2x^2 - 1)$ and $v = \cos^{-1} x$

   $$\therefore \frac{du}{dx} = -\frac{1}{\sqrt{1-(2x^2-1)^2}} \cdot 4x = \frac{-4x}{\sqrt{1-(4x^4+1-4x^2)}}$$

   $$= \frac{-4x}{\sqrt{-4x^4+4x^2}} = \frac{-4x}{\sqrt{4x^2(1-x^2)}} = \frac{-2}{\sqrt{1-x^2}}$$

   and $\frac{dv}{dx} = \frac{-1}{\sqrt{1-x^2}}$

$$\therefore \quad \frac{du}{dv} = \frac{du/dx}{dv/dx} = \frac{-2/\sqrt{1-x^2}}{-1/\sqrt{1-x^2}} = 2$$

8. Let $u = \tan^{-1}\left[\frac{\sqrt{1+x^2}-1}{x}\right]$.

Put $x = \tan\theta \Rightarrow \theta = \tan^{-1}x$

$$\therefore \quad u = \tan^{-1}\left[\frac{\sqrt{1+\tan^2\theta}-1}{\tan\theta}\right] = \tan^{-1}\left[\frac{\sec\theta - 1}{\tan\theta}\right]$$

$[\because \sec^2\theta = 1+\tan^2\theta]$

$$= \tan^{-1}\left[\frac{\frac{1}{\cos\theta}-1}{\frac{\sin\theta}{\cos\theta}}\right] = \tan^{-1}\left[\frac{1-\cos\theta}{\cos\theta}\times\frac{\cos\theta}{\sin\theta}\right]$$

$$= \tan^{-1}\left[\frac{1-\cos\theta}{\sin\theta}\right] = \tan^{-1}\left[\frac{2\sin^2\frac{\theta}{2}}{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}\right]$$

$\left[\because \sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2} \text{ and } 1-\cos\theta = 2\sin^2\frac{\theta}{2}\right]$

$$= \tan^{-1}\left[\tan\frac{\theta}{2}\right] = \frac{\theta}{2} \qquad \left[\because \tan^{-1}\left(\tan\frac{\theta}{2}\right) = \frac{\theta}{2}\right]$$

$$\Rightarrow u = \frac{1}{2}\tan^{-1}x \qquad [\because \theta = \tan^{-1}x]$$

On differentiating both sides w.r.t. x, we get

$$\frac{du}{dx} = \frac{1}{2}\times\frac{1}{1+x^2} = \frac{1}{2(1+x^2)}$$

Again, let $v = \tan^{-1}\left(\frac{2x}{1-x^2}\right) = 2\tan^{-1}x$

On differentiating both sides w.r.t. x, we get

$$\frac{dv}{dx} = 2\times\frac{1}{1+x^2}$$

Now, $\quad \frac{du}{dv} = \frac{du}{dx}\times\frac{dx}{dv} = \frac{1}{2(1+x^2)}\times\frac{(1+x^2)}{2} = \frac{1}{4}$

9. **Hint** Put $x = \cos\theta$ in first function and then solve as Question 8. **Ans.** 4

10. Similar as Question 8. **Ans.** $\frac{1}{2}$

11. Similar as Question 8. **Ans.** $\frac{1}{4}$

12. **Hint** Put $x = \cos\theta$ in both functions and then solve as Question 8. **Ans.** 3

13. **Hint** Put $ax = \sin\theta$ in first function and then solve as Question 8. **Ans.** $-\frac{2}{ax}$

14. **Hint** $y = \sqrt{x+y} \Rightarrow y^2 = x+y \Rightarrow 2y\frac{dy}{dx} = 1+\frac{dy}{dx}$

15. We have, $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \ldots + \infty}}}$

$$\therefore \quad y = \sqrt{\sin x + y}$$

$$\Rightarrow \quad y^2 = \sin x + y \qquad \text{[squaring both sides]}$$

On differentiating both sides w.r.t. x, we get

$$2y\frac{dy}{dx} = \cos x + \frac{dy}{dx} \Rightarrow 2y\frac{dy}{dx} - \frac{dy}{dx} = \cos x$$

$$\Rightarrow \quad \frac{dy}{dx}(2y-1) = \cos x \Rightarrow \frac{dy}{dx} = \frac{\cos x}{2y-1}$$

16. Given that, $y = (\cos x)^{(\cos x)^{(\cos x)\ldots\infty}} \Rightarrow y = (\cos x)^y$

Taking log on both sides, we get

$$\log y = y\log(\cos x)$$

Now, differentiating w.r.t. x, we get

$$\frac{1}{y}\cdot\frac{dy}{dx} = y\cdot\frac{1}{\cos x}(-\sin x) + \log(\cos x)\cdot\frac{dy}{dx}$$

$$\Rightarrow \quad \frac{dy}{dx} = y\left\{-y\tan x + \log\cos x\cdot\frac{dy}{dx}\right\}$$

$$\Rightarrow (1 - y\log\cos x)\frac{dy}{dx} = -y^2\tan x$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{y^2\tan x}{(y\log\cos x - 1)}$$

17. We have, $y = e^{x+e^{x+e^{x+\ldots\infty}}}$. Then, $y = e^{x+y}$

Taking log on both sides, we get

$$\log y = \log e^{(x+y)}$$

$$\Rightarrow \quad \log y = (x+y)\log e \Rightarrow \log y = (x+y) \qquad [\because \log e = 1]$$

On differentiating both sides w.r.t. x, we get

$$\frac{1}{y}\frac{dy}{dx} = 1+\frac{dy}{dx}$$

$$\Rightarrow \quad \frac{1}{y}\frac{dy}{dx} - \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx}\left(\frac{1}{y}-1\right) = 1$$

$$\Rightarrow \quad \frac{dy}{dx}\left(\frac{1-y}{y}\right) = 1 \Rightarrow \frac{dy}{dx} = \frac{y}{1-y}$$

18. We have, $y = x + \cfrac{1}{x + \cfrac{1}{x + \cfrac{1}{x+\ldots+\infty}}}$

$$\therefore \quad y = x + \frac{1}{y} \Rightarrow y = \frac{xy+1}{y} \Rightarrow y^2 = xy+1$$

On differentiating both sides w.r.t. x, we get

$$2y\frac{dy}{dx} = x\frac{dy}{dx} + y\cdot 1 + 0$$

[by product rule of derivative]

$$\Rightarrow \quad 2y\frac{dy}{dx} - x\frac{dy}{dx} = y$$

$$\Rightarrow \quad (2y-x)\frac{dy}{dx} = y \Rightarrow \frac{dy}{dx} = \frac{y}{2y-x}$$

19. **Hint** $y = a^{x^y} \Rightarrow \log y = x^y\log a$

$$\Rightarrow \quad \log\log y = \log x^y + \log\log a$$

Now, find $\frac{dy}{dx}$ by differentiating the above expression.

Second Order Derivative

Second order derivative is the derivative of the first order derivative.

Let $y = f(x)$ ...(i)

Then, $\frac{dy}{dx} = f'(x)$ is called the **first derivative** of y or $f(x)$. It can also be written as y' or y_1.

Again, differentiating Eq. (i) w.r.t. x, we get

$$\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}[f'(x)] \Rightarrow \frac{d^2y}{dx^2} = f''(x)$$

where, $\frac{d}{dx}\left(\frac{dy}{dx}\right)$ is called the **second order derivative** of y w.r.t. x and is denoted by $\frac{d^2y}{dx^2}$ or $f''(x)$ or y'' or D^2y or y_2.

Other higher order derivatives may be defined similarly.

EXAMPLE 1. Find the second order derivative of

(i) x^{20} (ii) $\tan^{-1} x$

(iii) $\sin(\log x)$

Sol. (i) Let $y = x^{20}$

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = 20x^{19}$$

Again, differentiating both sides w.r.t. x, we get

$$\frac{d}{dx}\left(\frac{dy}{dx}\right) = (20 \times 19)\, x^{18}$$

$$\Rightarrow \quad \frac{d^2y}{dx^2} = 380x^{18}$$

(ii) Let $y = \tan^{-1} x$

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = \frac{1}{1+x^2} = (1+x^2)^{-1}$$

Again, differentiating both sides w.r.t. x, we get

$$\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}(1+x^2)^{-1}$$

$$\Rightarrow \quad \frac{d^2y}{dx^2} = (-1)(1+x^2)^{-2}\frac{d}{dx}(1+x^2)$$

$$= (-1)(1+x^2)^{-2}(2x) = \frac{-2x}{(1+x^2)^2}$$

(iii) Let $y = \sin(\log x)$

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = \cos(\log x)\frac{d}{dx}(\log x)$$

$$\Rightarrow \quad \frac{dy}{dx} = \cos(\log x)\cdot\frac{1}{x}$$

$$\Rightarrow \quad x\frac{dy}{dx} = \cos(\log x) \qquad ...(i)$$

Again, differentiating both sides w.r.t. x, we get

$$x\cdot\frac{d^2y}{dx^2} + \frac{dy}{dx} = -\frac{\sin(\log x)}{x}$$

$$\Rightarrow \quad x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} = -\sin(\log x)$$

$$\Rightarrow \quad x^2\frac{d^2y}{dx^2} + \cos(\log x) = -\sin(\log x) \text{ [from Eq. (i)]}$$

$$\Rightarrow \quad x^2\frac{d^2y}{dx^2} = -\sin(\log x) - \cos(\log x)$$

$$\Rightarrow \quad \frac{d^2y}{dx^2} = -\frac{1}{x^2}[\sin(\log x) + \cos(\log x)]$$

EXAMPLE 2. If $y = (\tan x + \sec x)$, then find $\frac{d^2y}{dx^2}$.

[CBSE 2023]

Sol. We have, $y = (\tan x + \sec x)$

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = \sec^2 x + \sec x \tan x$$

$$= \frac{1}{\cos^2 x} + \frac{1}{\cos x}\cdot\frac{\sin x}{\cos x}$$

$$= \frac{1+\sin x}{\cos^2 x} = \frac{1+\sin x}{1-\sin^2 x}$$

$$= \frac{1}{1-\sin x}$$

Again, differentiating both sides w.r.t. x, we get

$$\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}\left(\frac{1}{1-\sin x}\right)$$

$$= \frac{d}{dx}(1-\sin x)^{-1}$$

$$= (-1)(1-\sin x)^{-2}\frac{d}{dx}(1-\sin x)$$

$$= (-1)(1-\sin x)^{-2}(-\cos x)$$

$$\Rightarrow \quad \frac{d^2y}{dx^2} = \frac{\cos x}{(1-\sin x)^2}$$

EXAMPLE 3. If $y = e^x \sin 5x$, then find $\frac{d^2y}{dx^2}$. [NCERT]

Sol. We have, $y = e^x \sin 5x$.

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = e^x \cdot \frac{d}{dx}\sin 5x + \sin 5x \cdot \frac{d}{dx}e^x \quad \text{[by product rule]}$$

$$= e^x \cdot \cos 5x \cdot \frac{d}{dx}(5x) + \sin 5x \cdot e^x \quad \text{[by chain rule]}$$

$$= e^x \cdot \cos 5x \cdot 5 + e^x \sin 5x$$

$$\Rightarrow \quad \frac{dy}{dx} = e^x (5\cos 5x + \sin 5x)$$

Again, differentiating both sides w.r.t. x, we get

$$\frac{d^2y}{dx^2} = e^x \cdot \frac{d}{dx}[5\cos 5x + \sin 5x] + [5\cos 5x + \sin 5x]\frac{d}{dx}\cdot(e^x) \quad \text{[by product rule]}$$

$$= e^x \left[5(-\sin 5x)\cdot\frac{d}{dx}(5x) + \cos 5x \cdot \frac{d}{dx}(5x)\right] + [5\cos 5x + \sin 5x]\cdot e^x \quad \text{[by chain rule]}$$

$$= e^x [5(-\sin 5x)5 + \cos 5x\,(5)] + [5\cos 5x + \sin 5x]e^x$$

$$= e^x [-25\sin 5x + 5\cos 5x + 5\cos 5x + \sin 5x]$$

$$= e^x [10\cos 5x - 24\sin 5x]$$

$$\therefore \frac{d^2y}{dx^2} = 2e^x [5\cos 5x - 12\sin 5x]$$

EXAMPLE 4. Find $\frac{d^2y}{dx^2}$, if $x = t^2$, $y = 2t$.

Sol. We have, $x = t^2$ ...(i)

and $y = 2t$ ...(ii)

Here, t is the parameter.

On differentiating Eqs (i) and (ii), respectively w.r.t. t we get

$$\frac{dx}{dt} = 2t \quad \text{...(iii)}$$

and $$\frac{dy}{dt} = 2 \quad \text{...(iv)}$$

On putting the values from Eqs. (iii) and (iv) in the formula $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$, we get

$$\frac{dy}{dx} = \frac{2}{2t} = \frac{1}{t}$$

Now, Differentiating both sides w.r.t. x, we get

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{1}{t}\right) = \frac{d}{dt}\left(\frac{1}{t}\right)\frac{dt}{dx} = -\frac{1}{t^2}\frac{dt}{dx}$$

$$= -\frac{1}{t^2}\frac{1}{2t} \quad \text{[from Eq. (iii)]}$$

$$= -\frac{1}{2t^3}$$

EXAMPLE 5. If $x = a(\theta + \sin\theta)$ and $y = a(1 - \cos\theta)$, then find $\frac{d^2y}{dx^2}$ at $\theta = \frac{\pi}{2}$. [All India 2011]

Sol. We have, $x = a(\theta + \sin\theta)$...(i)

and $y = a(1 - \cos\theta)$ ...(ii)

Here, θ is the parameter.

On differentiating Eqs. (i) and (ii), respectively w.r.t. θ, we get

$$\frac{dx}{d\theta} = \frac{d}{d\theta}[a(\theta + \sin\theta)] = a(1 + \cos\theta) \quad \text{...(iii)}$$

and $$\frac{dy}{d\theta} = \frac{d}{d\theta}[a(1 - \cos\theta)] = a(0 + \sin\theta) = a\sin\theta \quad \text{...(iv)}$$

On putting the values from Eqs. (iii) and (iv) in the formula $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$, we get

$$\frac{dy}{dx} = \frac{a\sin\theta}{a(1 + \cos\theta)}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{\sin\theta}{1 + \cos\theta} = \frac{2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)}{2\cos^2\left(\frac{\theta}{2}\right)} = \tan\frac{\theta}{2}$$

Now, differentiating both sides w.r.t. x, we get

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\tan\frac{\theta}{2}\right) = \frac{d}{d\theta}\left(\tan\frac{\theta}{2}\right)\cdot\frac{d\theta}{dx}$$

$$= \frac{1}{2}\sec^2\frac{\theta}{2}\cdot\frac{1}{a(1 + \cos\theta)} \quad \text{[from Eq. (iii)]}$$

$$= \frac{\sec^2\frac{\theta}{2}}{2a(1 + \cos\theta)}$$

Now, at $\theta = \frac{\pi}{2}$, $\frac{d^2y}{dx^2} = \frac{\sec^2\frac{\pi}{4}}{2a\left(1 + \cos\frac{\pi}{2}\right)} = \frac{2}{2a(1+0)} = \frac{1}{a}$

EXAMPLE 6. If $y = x^x$, then prove that

$$\frac{d^2y}{dx^2} - \frac{1}{y}\left(\frac{dy}{dx}\right)^2 - \frac{y}{x} = 0.$$ [CBSE 2023, Delhi 2016, 2014]

Sol. We have, $y = x^x$

Taking log on both sides, we get

$$\log y = \log x^x \Rightarrow \log y = x\log x$$

On differentiating both sides w.r.t. x, we get

$$\frac{1}{y}\frac{dy}{dx} = x\frac{d}{dx}(\log x) + \log x\frac{d}{dx}(x)$$

$$\Rightarrow \quad \frac{1}{y}\frac{dy}{dx} = x \times \frac{1}{x} + \log x \Rightarrow \frac{dy}{dx} = y(1 + \log x) \quad \text{...(i)}$$

Again, differentiating both sides of Eq. (i) w.r.t. x, we get

$$\frac{d^2y}{dx^2} = y\frac{d}{dx}(1+\log x) + (1+\log x)\frac{dy}{dx}$$

$$= y \times \frac{1}{x} + (1+\log x)\frac{dy}{dx}$$

$$\Rightarrow \quad \frac{d^2y}{dx^2} = \frac{y}{x} + (1+\log x)\frac{dy}{dx} \quad \text{...(ii)}$$

Now, $\text{LHS} = \frac{d^2y}{dx^2} - \frac{1}{y}\left(\frac{dy}{dx}\right)^2 - \frac{y}{x}$

On putting the values of $\frac{d^2y}{dx^2}$ and $\frac{dy}{dx}$ from Eqs. (i) and (ii), we get

$$\text{LHS} = \frac{y}{x} + (1+\log x)\, y(1+\log x) - \frac{1}{y}[y^2(1+\log x)^2] - \frac{y}{x}$$

$$= y(1+\log x)^2 - y(1+\log x)^2 = 0 = \text{RHS}$$

Hence proved.

EXAMPLE 7. If $y = \sin^{-1} x$, then show that

$$(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} = 0.$$

Sol. We have, $y = \sin^{-1} x$

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow \quad \sqrt{1-x^2}\frac{dy}{dx} = 1$$

On squaring both sides, we get

$$(1-x^2)\left(\frac{dy}{dx}\right)^2 = 1$$

Again, differentiating both sides w.r.t. x, we get

$$\frac{d}{dx}\left[(1-x^2)\left(\frac{dy}{dx}\right)^2\right] = \frac{d}{dx}(1) \quad \text{...(i)}$$

$$\Rightarrow (1-x^2)\frac{d}{dx}\left(\frac{dy}{dx}\right)^2 + \left(\frac{dy}{dx}\right)^2\frac{d}{dx}(1-x^2) = 0$$

$$\Rightarrow \quad (1-x^2)2\frac{dy}{dx}\cdot\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2(0-2x) = 0$$

$$\Rightarrow \quad \frac{2dy}{dx}\left[(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx}\right] = 0$$

$$\Rightarrow \quad (1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} = 0$$

Hence proved.

EXAMPLE 8. If $y = Pe^{ax} + Qe^{bx}$, then show that

$$\frac{d^2y}{dx^2} - (a+b)\frac{dy}{dx} + aby = 0.$$

[All India 2015C]

Sol. We have, $y = Pe^{ax} + Qe^{bx}$ ...(i)

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = P\frac{d}{dx}(e^{ax}) + Q\frac{d}{dx}(e^{bx})$$

$$\Rightarrow \quad \frac{dy}{dx} = Pae^{ax} + Qbe^{bx} \quad \text{...(ii)}$$

Again, differentiating both sides w.r.t. x, we get

$$\frac{d^2y}{dx^2} = Pa\frac{d}{dx}(e^{ax}) + bQ\frac{d}{dx}(e^{bx})$$

$$= Pa(ae^{ax}) + bQ(be^{bx}) = a^2Pe^{ax} + b^2Qe^{bx} \quad \text{...(iii)}$$

Now, $\text{LHS} = \frac{d^2y}{dx^2} - (a+b)\frac{dy}{dx} + aby$

On putting the values from Eqs. (i), (ii) and (iii), we get

$$\text{LHS} = a^2Pe^{ax} + b^2Qe^{bx} - (a+b)(aPe^{ax} + bQe^{bx}) + ab(Pe^{ax} + Qe^{bx})$$

$$= a^2Pe^{ax} + b^2Qe^{bx} - a^2Pe^{ax} - abQe^{bx} - abPe^{ax} - b^2Qe^{bx} + abPe^{ax} + abQe^{bx}$$

$$= 0 = \text{RHS}$$

EXAMPLE 9. If $x = a\cos\theta + b\sin\theta$ and $y = a\sin\theta - b\cos\theta$, then show that

$$y^2\frac{d^2y}{dx^2} - x\frac{dy}{dx} + y = 0.$$

[Delhi 2015]

Sol. Given, $x = a\cos\theta + b\sin\theta$ ...(i)

and $y = a\sin\theta - b\cos\theta$ ...(ii)

Here, θ is the parameter.

On differentiating Eqs. (i) and (ii), respectively w.r.t. θ, we get

$$\frac{dx}{d\theta} = \frac{d}{d\theta}(a\cos\theta + b\sin\theta)$$

$$\Rightarrow \quad \frac{dx}{d\theta} = -a\sin\theta + b\cos\theta$$

and $$\frac{dy}{d\theta} = \frac{d}{d\theta}(a\sin\theta - b\cos\theta)$$

$$\Rightarrow \quad \frac{dy}{d\theta} = a\cos\theta + b\sin\theta$$

Now, $$\frac{dy}{dx} = \frac{(dy/d\theta)}{(dx/d\theta)} = \frac{a\cos\theta + b\sin\theta}{-(a\sin\theta - b\cos\theta)} = -\frac{x}{y}$$

[from Eqs. (i) and (ii)]

Again, differentiating both sides w.r.t. x, we get

$$\frac{d^2y}{dx^2} = -\left[\frac{y\frac{d}{dx}(x) - x\frac{dy}{dx}}{y^2}\right]$$

$$\Rightarrow y^2\frac{d^2y}{dx^2} = -y + x\frac{dy}{dx} \Rightarrow y^2\frac{d^2y}{dx^2} - x\frac{dy}{dx} + y = 0$$

Topic Practice 8

1 Find the second order derivative of the following.

(i) $x^3 - x^2 + 2$ (ii) $x^{10} + x^9$

(iii) e^{ax} (iv) 2^x

(v) $\log x$ (vi) $\tan x$

(vii) $x \cos x$ (viii) $\log(\log x)$

(ix) $x^3 \log x$ (x) $e^{6x} \cos 3x$ [NCERT]

(xi) x^x (xii) $\tan^{-1} x$

(xiii) $x^4 + \cot x$

2 A curve has the equation $y = \frac{x+6}{2x-3}$, when $x \neq \frac{3}{2}$.

(i) Show that the derivative of the curve can be expressed as $\frac{k}{(2x-3)^2}$, where k is a constant.

(ii) Find the value of x for which $\frac{d^2y}{dx^2} = \frac{12}{25}$.

3 If $y = A \sin x + B \cos x$, then prove that $\frac{d^2y}{dx^2} + y = 0$. [NCERT]

4 If $y = 5\cos x - 3\sin x$, then prove that $\frac{d^2y}{dx^2} + y = 0$. [NCERT]

5 If $y = 3e^{2x} + 2e^{3x}$, then prove that $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$. [NCERT]

6 If $y = \text{cosec}\, x + \cot x$, then prove that $(\sin x)\frac{d^2y}{dx^2} - y^2 = 0$.

7 If $x \cos(a+y) = \cos y$, then prove that $\frac{dy}{dx} = \frac{\cos^2(a+y)}{\sin a}$. Hence, show that $\sin a \frac{d^2y}{dx^2} + \sin 2(a+y)\frac{dy}{dx} = 0$. [All India 2016]

8 If $e^y(x+1) = 1$, then prove that $\frac{d^2y}{dx^2} = \left(\frac{dy}{dx}\right)^2$. [All India 2017]

9 If $y = \sin(\sin x)$, then prove that $\frac{d^2y}{dx^2} + \tan x \frac{dy}{dx} + y\cos^2 x = 0$. [CBSE 2018]

10 If $y = e^{a\cos^{-1} 3x}$, then prove that $(1-9x^2)\frac{d^2y}{dx^2} - 9x\frac{dy}{dx} - 9a^2y = 0$. [All India 2016]

11 $x = \tan\left(\frac{1}{a}\log y\right)$, then prove that $(1+x^2)\frac{d^2y}{dx^2} + (2x-a)\frac{dy}{dx} = 0$. [All India 2011]

12 If $y = x^3 \log\left(\frac{1}{x}\right)$, then prove that $x\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 3x^2 = 0$.

13 If $y = e^{ax} \cdot \cos bx$, then prove that $\frac{d^2y}{dx^2} - 2a\frac{dy}{dx} + (a^2+b^2)y = 0$.

14 If $y = x\log\left(\frac{x}{a+bx}\right)$, then prove that $x^3\frac{d^2y}{dx^2} = \left(x\frac{dy}{dx} - y\right)^2$. [Delhi 2013]

15 If $y = \sqrt{x+1} - \sqrt{x-1}$, then prove that $(x^2-1)\frac{d^2y}{dx^2} + x\frac{dy}{dx} - \frac{1}{4}y = 0$.

16 If $y = 500e^{7x} + 600e^{-7x}$, then prove that $\frac{d^2y}{dx^2} = 49y$. [NCERT]

17 If $y = \log(1+\cos x)$, then prove that $\frac{d^3y}{dx^3} + \frac{d^2y}{dx^2}\cdot\frac{dy}{dx} = 0$.

18 If $y = Ae^{-kt}\cos(pt+c)$, then prove that $\frac{d^2y}{dt^2} + 2k\frac{dy}{dt} + n^2y = 0$, where $n^2 = p^2 + k^2$.

19 If $y = (\sec^{-1}x)^2$, then prove that $x^2(x^2-1)\frac{d^2y}{dx^2} + (2x^3 - x)\frac{dy}{dx} = 2$. [All India 2017C]

20 If $y = \log[x + \sqrt{x^2+1}]$, then prove that $(x^2+1)\frac{d^2y}{dx^2} + x\frac{dy}{dx} = 0$. [Foreign 2011]

21 If $y = \left(x + \sqrt{1+x^2}\right)^n$, then prove that $(1+x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx} = n^2y$. [Foreign 2015]

22 If $x^m y^n = (x+y)^{m+n}$, then prove that $\frac{d^2y}{dx^2} = 0$. [Delhi 2017]

23 If $y = 3\cos(\log x) + 4\sin(\log x)$, then prove that $x^2y_2 + xy_1 + y = 0$. [NCERT]

24 If $(x-a)^2+(y-b)^2=c^2$, then prove that $\dfrac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}}$ is a constant and independent of a and b. [NCERT]

25 If $x=\sin t$ and $y=\sin pt$, then prove that $(1-x^2)\dfrac{d^2y}{dx^2}-x\dfrac{dy}{dx}+p^2y=0$. [NCERT Exemplar]

26 If $y=\cos^{-1}x$, then find the value of $\dfrac{d^2y}{dx^2}$ in terms of y.

27 If $y=\tan^{-1}x$, then find $\dfrac{d^2y}{dx^2}$ in terms of y.

28 If $x=3\cos t-2\cos^3 t$ and $y=3\sin t-2\sin^3 t$, then find $\dfrac{d^2y}{dx^2}$. [Delhi 2017C]

29 If $x=a\sec^3\theta$, $y=a\tan^3\theta$, then find $\dfrac{d^2y}{dx^2}$ at $\theta=\dfrac{\pi}{4}$. [Delhi 2015C]

30 If $x=a(\cos t+t\sin t)$ and $y=a(\sin t-t\cos t)$, then find $\dfrac{d^2y}{dx^2}$ at $t=\dfrac{\pi}{4}$. [Delhi 2017C, 2014C]

31 If $y=a\left(\cos t+\log\tan\dfrac{t}{2}\right)$, $x=a\sin t$, then evaluate $\dfrac{d^2y}{dx^2}$ at $t=\dfrac{\pi}{3}$. [Delhi 2013]

Hints & Solutions

1. (i) Similar as Example 1 (i). **Ans.** $6x-2$

(ii) Similar as Example 1 (i). **Ans.** $18x^7(5x+4)$

(iii) Let $y=e^{ax}$

Then, $\dfrac{dy}{dx}=e^{ax}\dfrac{d}{dx}(ax)=ae^{ax}$

$\therefore \quad \dfrac{d^2y}{dx^2}=a\cdot e^{ax}(a)=a^2\cdot e^{ax}$

(iv) Let $y=2^x$

Then, $\dfrac{dy}{dx}=2^x\log_e 2$

$\therefore \quad \dfrac{d^2y}{dx^2}=2^x(\log 2)^2$

(v) Let $y=\log x$

Then, $\dfrac{dy}{dx}=\dfrac{1}{x}=x^{-1}$

$\therefore \quad \dfrac{d^2y}{dx^2}=\dfrac{d}{dx}(x^{-1})=(-1)x^{-2}=\dfrac{-1}{x^2}$

(vi) Let $y=\tan x$

Then, $\dfrac{dy}{dx}=\sec^2 x$

$\therefore \quad \dfrac{d^2y}{dx^2}=2\sec x\cdot(\sec x\tan x)=2\sec^2 x\tan x$

(vii) Let $y=x\cos x$

Then, $\dfrac{dy}{dx}=x\dfrac{d}{dx}(\cos x)+(\cos x)\dfrac{d}{dx}(x)$

$=x(-\sin x)+\cos x=-x\sin x+\cos x$

$\therefore \quad \dfrac{d^2y}{dx^2}=-[x\cos x+\sin x]-\sin x=-x\cos x-2\sin x$

(viii) **Hint** $\dfrac{dy}{dx}=\dfrac{1}{\log x}\cdot\dfrac{1}{x}=(x\cdot\log x)^{-1}$

and $\dfrac{d^2y}{dx^2}=(-1)(x\log x)^{-2}\left(x\cdot\dfrac{1}{x}+\log x\right)$

Ans. $\dfrac{1+\log x}{(x\log x)^2}$

(ix) Let $y=x^3\log x$

On differentiating both sides w.r.t. x, we get

$\dfrac{dy}{dx}=\dfrac{d}{dx}(x^3\log x)=x^3\dfrac{d}{dx}(\log x)+\log x\dfrac{d}{dx}(x^3)$

[by product rule of derivative]

$=x^3\left(\dfrac{1}{x}\right)+(\log x)(3x^2)=x^2(1+3\log x)$

Again, differentiating both sides w.r.t. x, we get

$\dfrac{d^2y}{dx^2}=x^2\dfrac{d}{dx}(1+3\log x)+\dfrac{d}{dx}(x^2)\cdot(1+3\log x)$

$=x^2\left(0+\dfrac{3}{x}\right)+(1+3\log x)(2x)$

$=3x+2x(1+3\log x)$

$=x(5+6\log x)$

(x) Similar as Example 3. **Ans.** $9e^{3x}(3\cos 3x-4\sin 3x)$

(xi) Similar as Example 6. **Ans.** $x^x\left\{(1+\log x)^2+\dfrac{1}{x}\right\}$

(xii) We have, $f(x)=\tan^{-1}x$

On differentiating both sides w.r.t. x, we get

$f'(x)=\dfrac{1}{1+x^2}=(1+x^2)^{-1}$